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Week 13

KI1101 General Chemistry 1

Chemical Bonding Theory

Molecular Structures

- Molecules containing 3 or more atoms may have many different shapes
- Almost all are “3-dimensional”
- Shapes are made from **five basic geometrical structures**
- Shapes classified according to number of electron **domains** they contain around central atom

VSPER Theory

- **Valence shell electron pair repulsion**
- Simple and useful model of electron domains
- Two (2) types of electron domains
  - **Bonding domains**
    - Electron pairs involved in bonds between 2 atoms
  - **Nonbonding domains**
    - Electron pairs associated with single atom
- All electrons in single, double, or triple bond considered to be in the **same** bonding domain

VSPER Theory

- Simple theory for predicting shapes of molecules

Fact:

- Negative electrons repel each other very strongly.

Result:

- Electron pairs arrange themselves to be as far apart as possible.
- Minimizes repulsions

Result:

- Electron pairs arrange to have lowest possible potential energy

VSPER Theory

Assumes:

- Bonds are shared pairs of electrons
  - Covalent bonds
- Central Atom will have 2, 3, 4, 5, or 6 pairs of electrons in its valence shell.
- Model includes central atoms with:
  - Incomplete octet
  - Complete octet
  - Extended octet

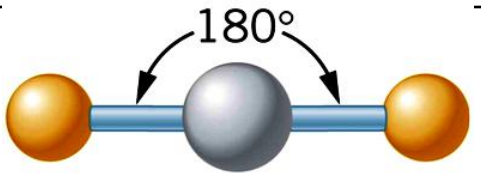
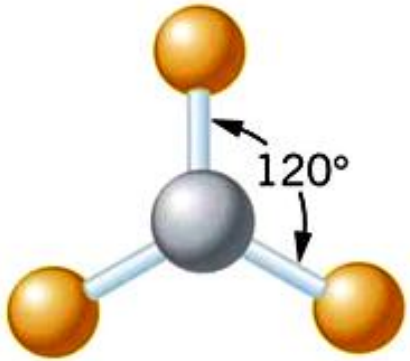
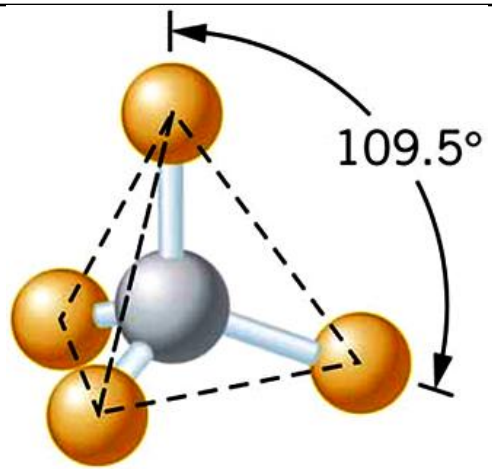
First look at cases where

- **All electron pairs around central atom are bonding pairs**

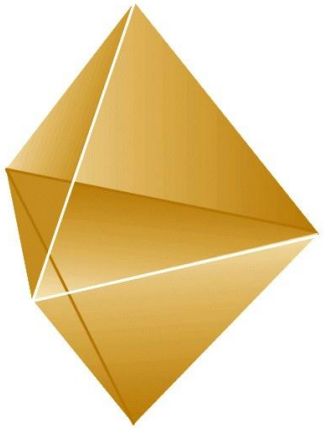
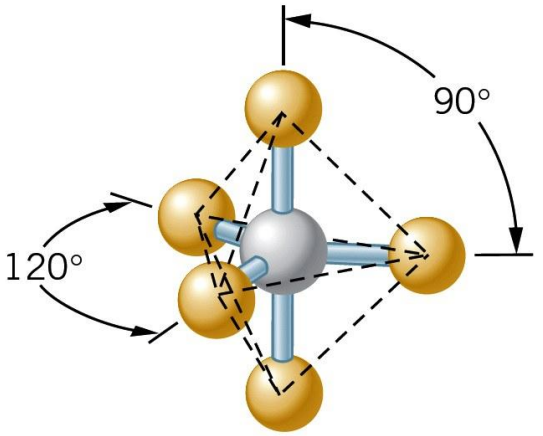
VSPER Theory

- Nonbonding domain contains
  - Lone pair (unshared pair of electrons) or
  - Unpaired electron for molecule with odd number of valence electrons
- VSEPR model is based on the notion that **electron domains keep as far away as possible from one another**
- Shapes expected for different numbers of electron domains around central atom may be summarized:

Five Basic Electron Domains

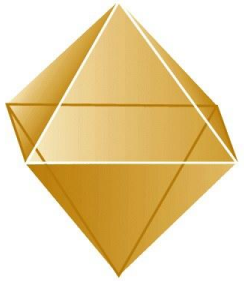
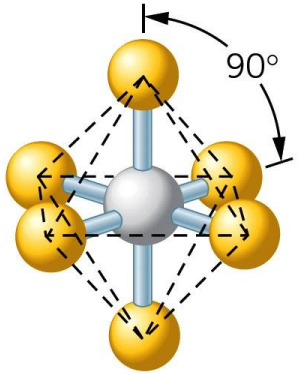
Electron Domains	Shape	Electron Pair Geometry
2		**linear**
3		**trigonal planar**
4		**tetrahedral**

Five Basic Electron Domains

Electron Domains	Shape	Electron Pair Geometry
5	 A trigonal bipyramid  A trigonal bipyramidal molecule	**trigonal bipyramidal** has equatorial and axial positions.

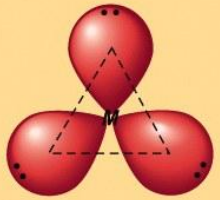
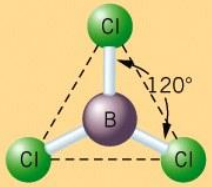
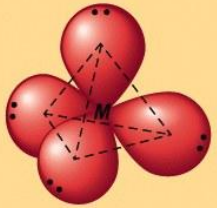
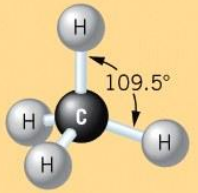
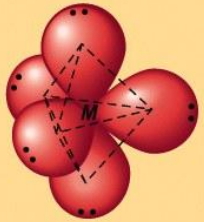
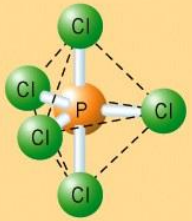
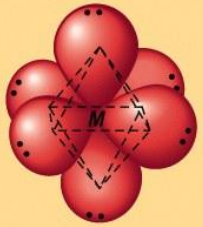
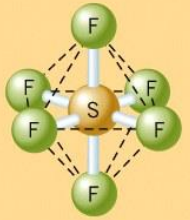
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Five Basic Electron Domains

Electron Domains	Shape	Electron Pair Geometry
6	 An octahedron  An octahedral molecule	**octahedral** has equatorial and axial positions

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Basic Electron Pair Geometries

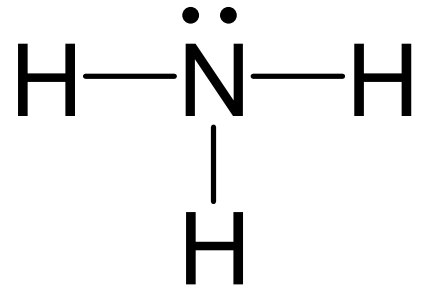
Number of Domains	Shape		Example	
2		Linear	$BeCl_2$	
3		Planar triangular	BCl_3	
4		Tetrahedral (A tetrahedron is pyramid shaped. It has four triangular faces and four corners.)	CH_4	
5		Trigonal bipyramidal (This figure consists of two three-sided pyramids joined by sharing a common face–the triangular plane through the center.)	PCl_5	
6		Octahedral (An octahedron is an eight-sided figure with *six* corners. It consists of two square pyramids that share a common square base.)	SF_6	

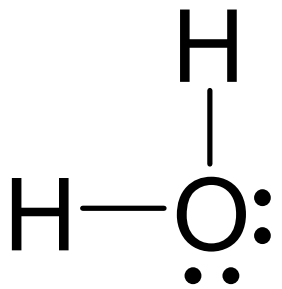
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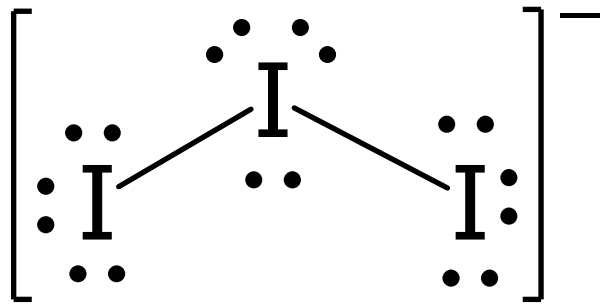
Learning Check

Identify, for each of the following:

1. Number of electron domains
2. Electron pair geometry







Your Turn!

How many electron domains are there around the central atom in SF_4O? What is the electron pair geometry for the compound?

A. 4, tetrahedron

B. 5, pentagon

C. 5, trigonal bipyramid

D. 4, square pyramid

E. 6, octahedron

VSPER Theory

- **What if one or more bonds are replaced by lone pairs?**
  - **Lone pairs**
    - Take up space around central atom
    - Effect overall geometry
    - Must be counted as electron domains
- **What if there are one or more multiple bonds?**
  - **Multiple bonds** (double and triple)
  - For purposes of molecular geometry
  - Treat as single electron domain
  - Same as single bonds

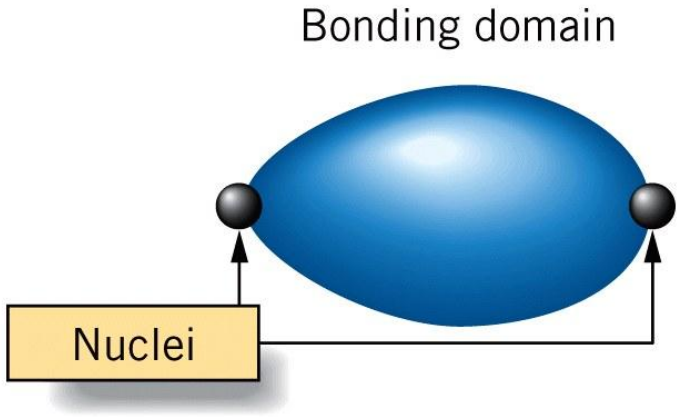
Relative Sizes of Electron Domain

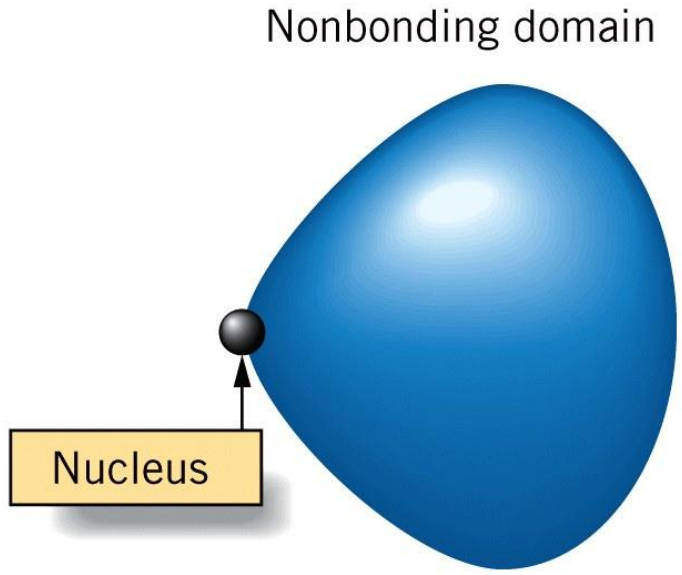
Bonding domains

- More oval in shape
- Electron density focused between 2 positive nuclei.

Nonbonding domains

- More bell or balloon shaped
- Electron density only has positive nuclei at one end





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Lewis Structure – Steps to Follow

1. **Draw Lewis Structure of Molecule**
   - Don't need to compute formal charge
   - If several resonance structures exist, pick only one

2. **Count e^- pair domains**
   - Lone pairs and bond pairs around central atom
   - Multiple bonds count as one set (or one effective pair)

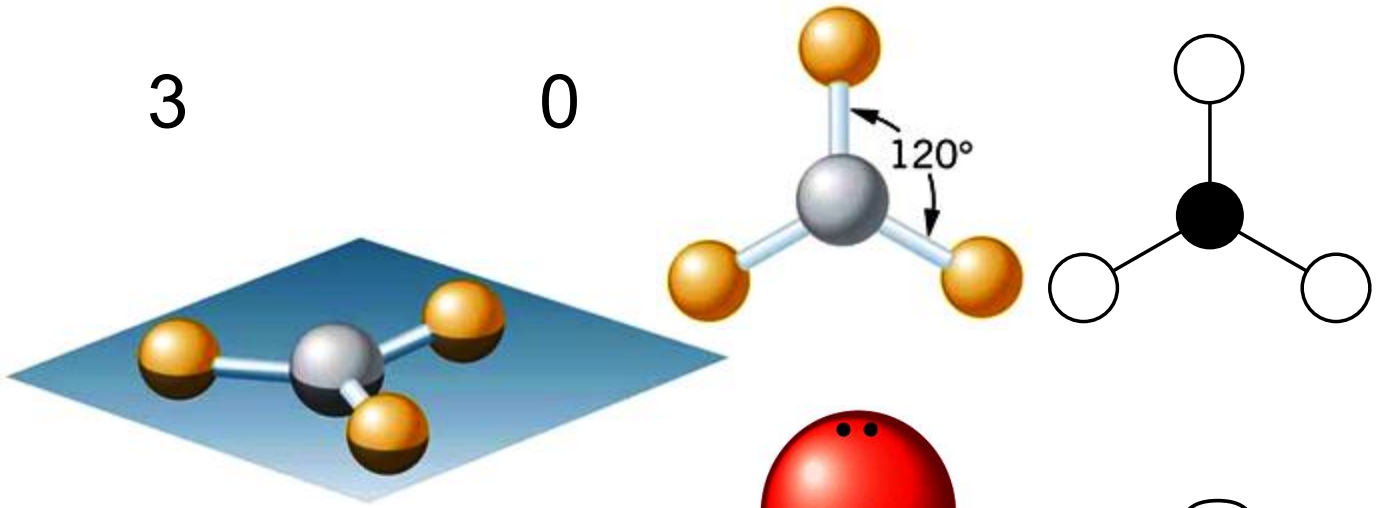
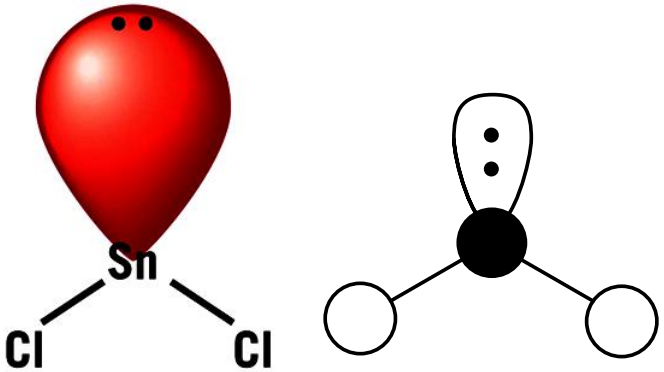
Lewis Structure – Steps to Follow

3. Arrange e^- pair domains to minimize repulsions

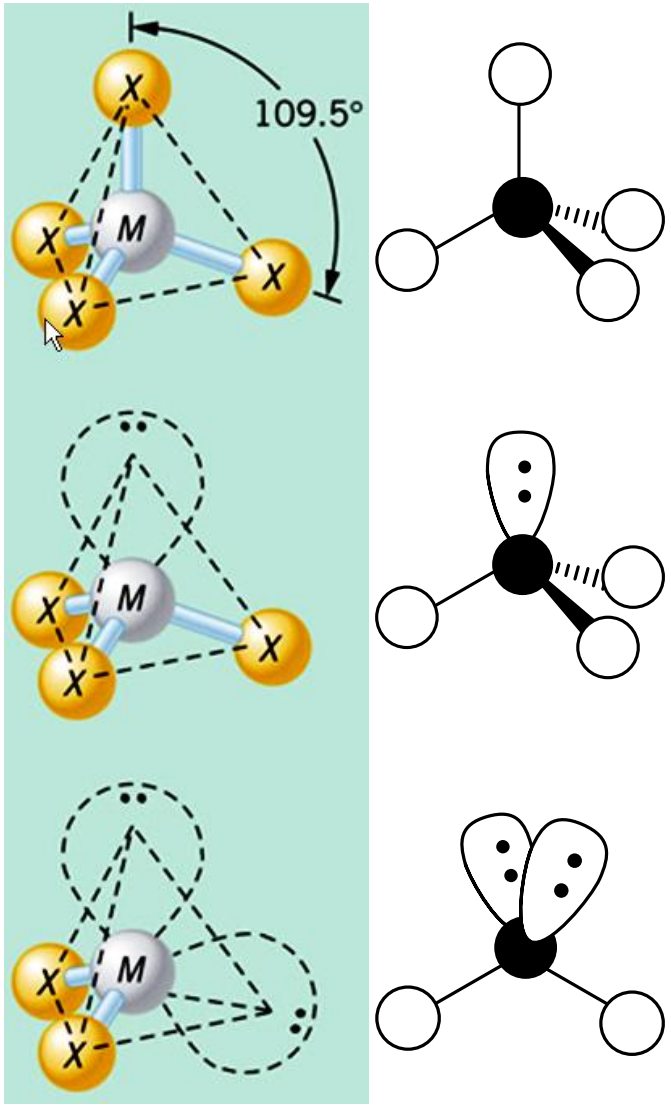
- Lone pairs
  - Require more space than bonding pairs
  - May slightly distort bond angles from those predicted.
  - In AX_5 lone pairs are equatorial
  - In AX_6 lone pairs are axial

4. Name molecular structure by position of atoms—only bonding e^-s

Structures Based on 3 e^- Domains

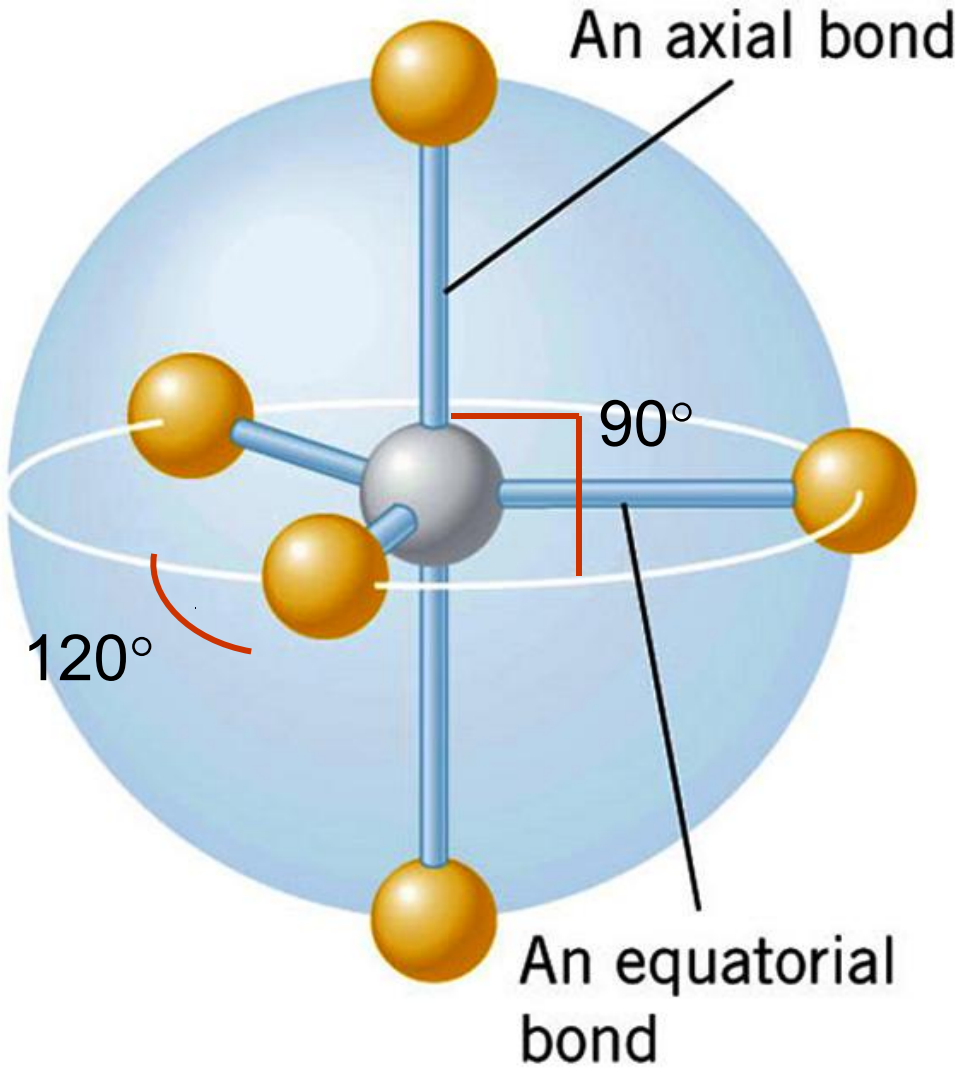
Number of Bonding Domains	Number of Nonbonding Domains	Structure	Molecular Structure
3	0		Trigonal Planar (Ex. BCl_3) All bond angles 120°
2	1		Nonlinear Bent or V-shaped (Ex. SO_2) Bond <120°

Structures Based on 4 e^- Domains

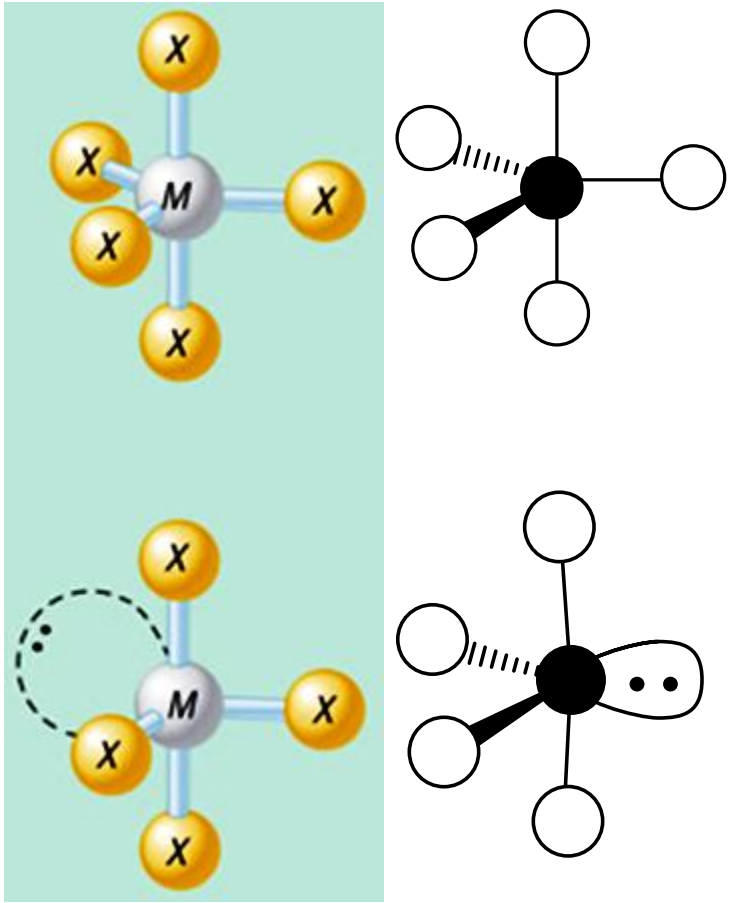
Number of Bonding Domains	Number of Nonbonding Domains	Structure	Molecular Structure
4	0		Tetrahedral (Ex. CH_4) All bond angles 109.5 °
3	1		Trigonal pyramidal (Ex. NH_3) Bond angle < 109.5°
2	2		Nonlinear, bent (Ex. H_2O) Bond angle <109.5°

Trigonal Bipyramidal

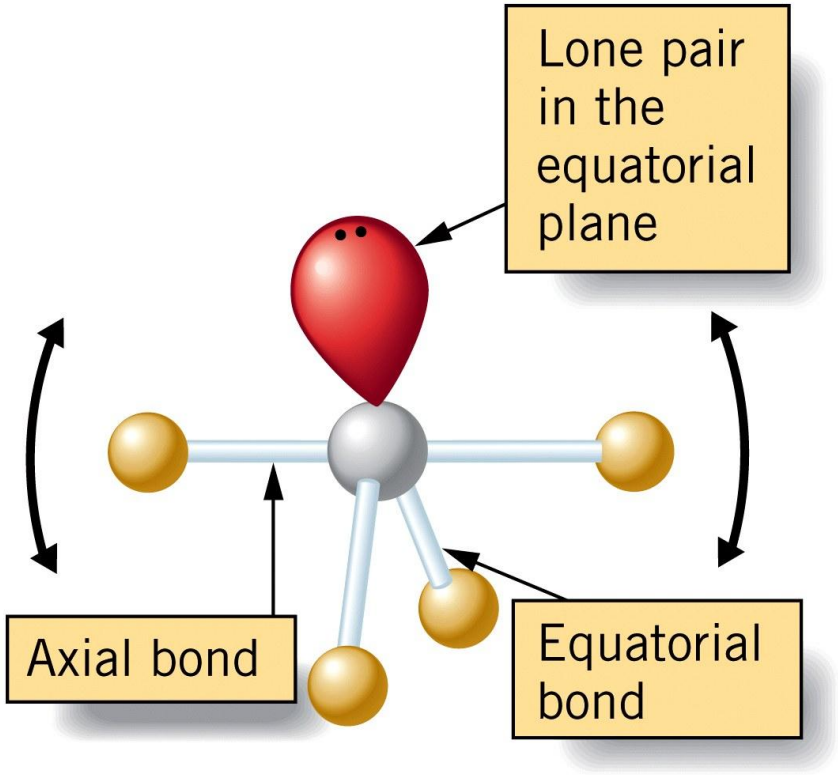
- 2 atoms in axial position
  - 90° to atoms in equatorial plane
- 3 atoms in equatorial position
  - 120° bond angle to atoms in axial position
  - More room here
  - Substitute here first



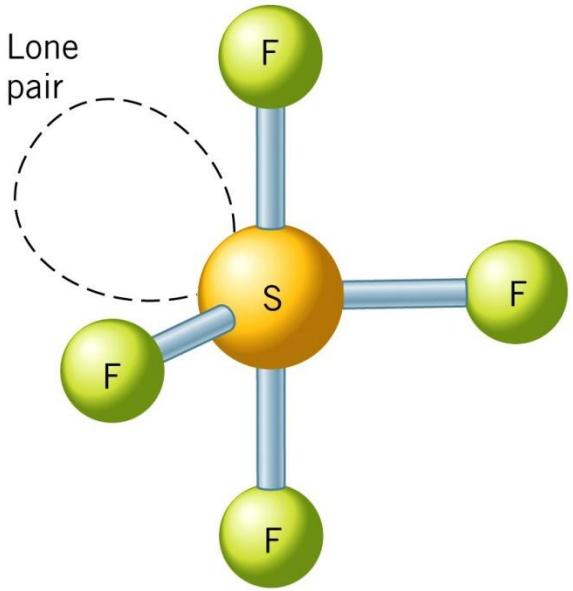
Structures Based on 5 e⁻ Domains

Number of Bonding Domains	Number of Nonbonding Domains	Structure	Molecular Structure
5	0		Trigonal bipyramidal (Ex. PF_5) Ax-eq bond angles 90° Eq-eq 120°
4	1		Distorted Tetrahedron, Sawhorse, or Seesaw (Ex. SF_4) Ax-eq bond angles < 90°

Where Do Lone Pairs Go?



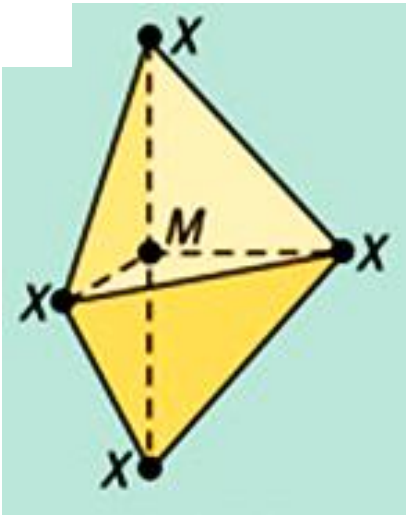
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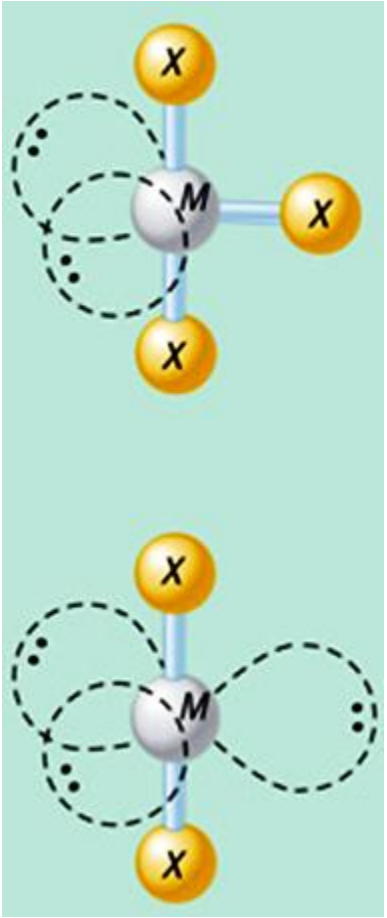
The structure of SF_4

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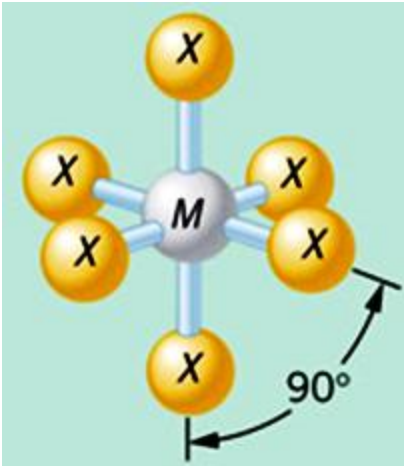
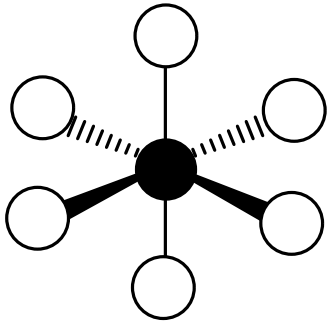
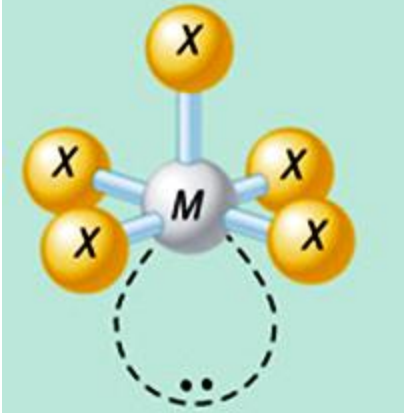
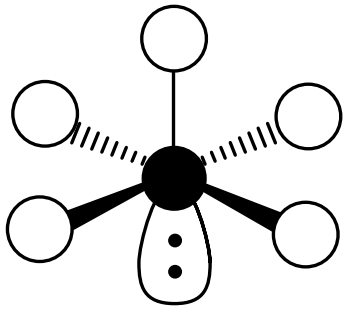
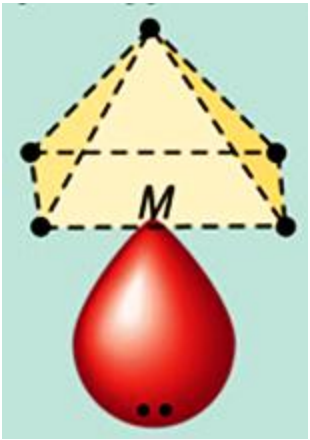
- Lone pair takes up more space
- Goes in equatorial plane
- Pushes bonding pairs out of way
- **Result:** distorted tetrahedron



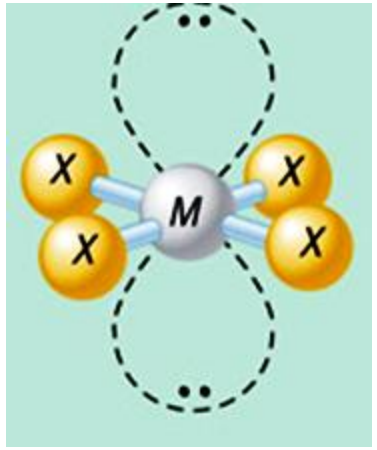
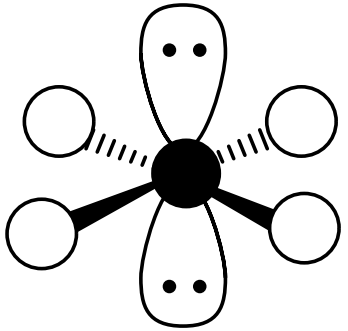
Structures Based on 5 e⁻ Domains

Number of Bonding Domains	Number of Nonbonding Domains	Structure	Molecular Structure
3	2		T-shaped (Ex. ClF_3) Bond angles 90°
2	3		Linear (Ex. I_3^-) Bond angles 180°

Structures Based on 6 e⁻ Domains

Number of Bonding Domains	Number of Nonbonding Domains	Structure	Molecular Structure
6	0		 Octahedral (Ex. SF_6)
5	1		 Square Pyramidal (Ex. BrF_5) 

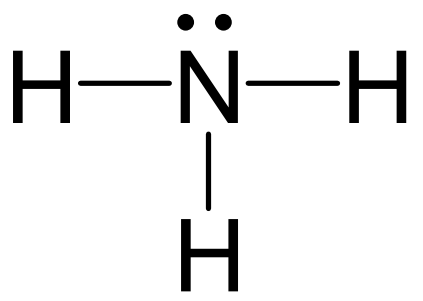
Structures Based on 6 e^- Domains

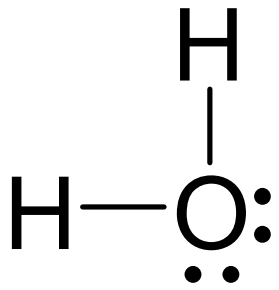
Number of Bonding Domains	Number of Nonbonding Domains	Structure	Molecular Structure
4	2		 Square Planar (Ex. XeF_4)

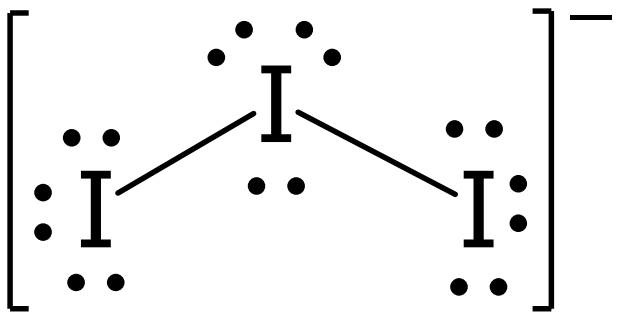
Learning Check

Identify for each of the following:

1. Number of bonding versus nonbonding domains
2. Molecular Geometry/Molecular structure







Your Turn!

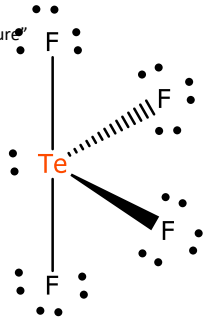
Here are four possible Lewis Structures for TeF_4

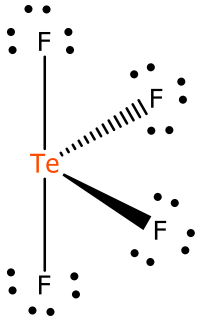
A. B.

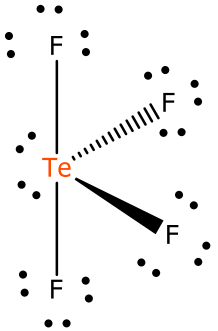
C. D.

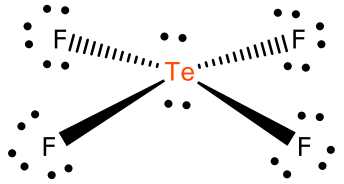
Which is the "best Lewis Structure"

A









Your Turn!

For the species, ICl_5 , answer the following:

1. How many bonding domains exist?

2. How many non-bonding domains exist?

3. What is the electron domain geometry?

4. What is the molecular geometry?

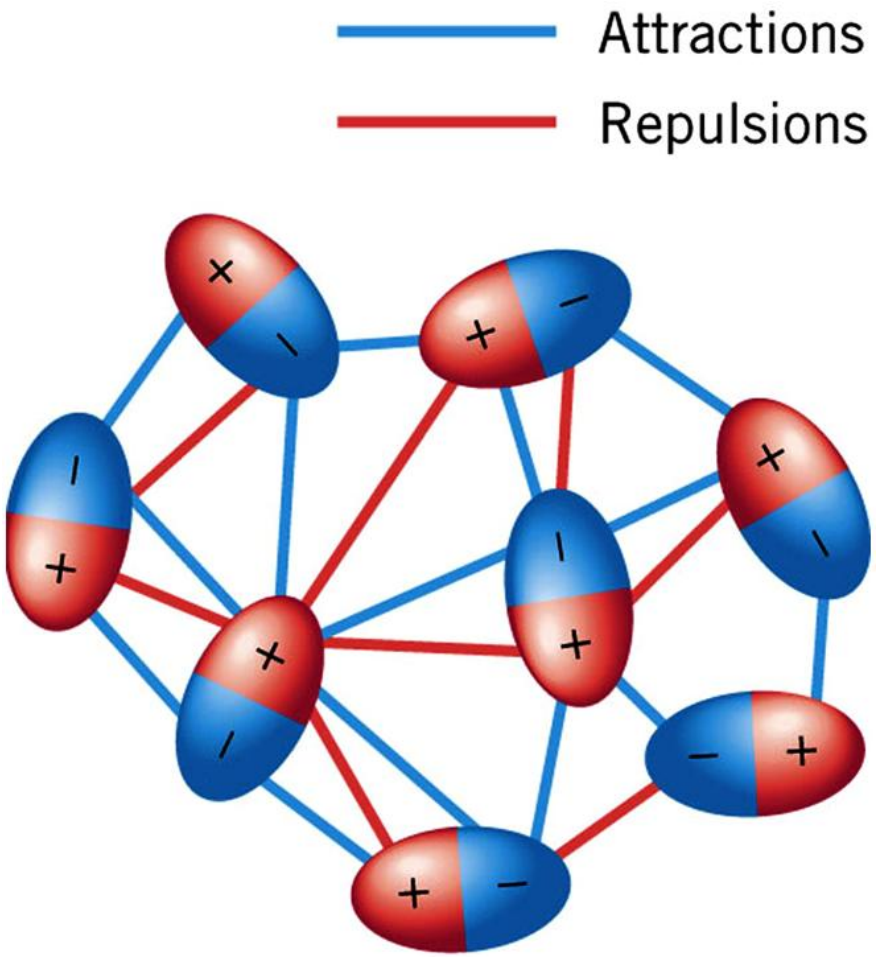
A. 1, 4, tetrahedron, trigonal bipyramid

B. 4, 1, tetrahedron, trigonal planar

C. 4, 1, trigonal bipyramid, distorted tetrahedron

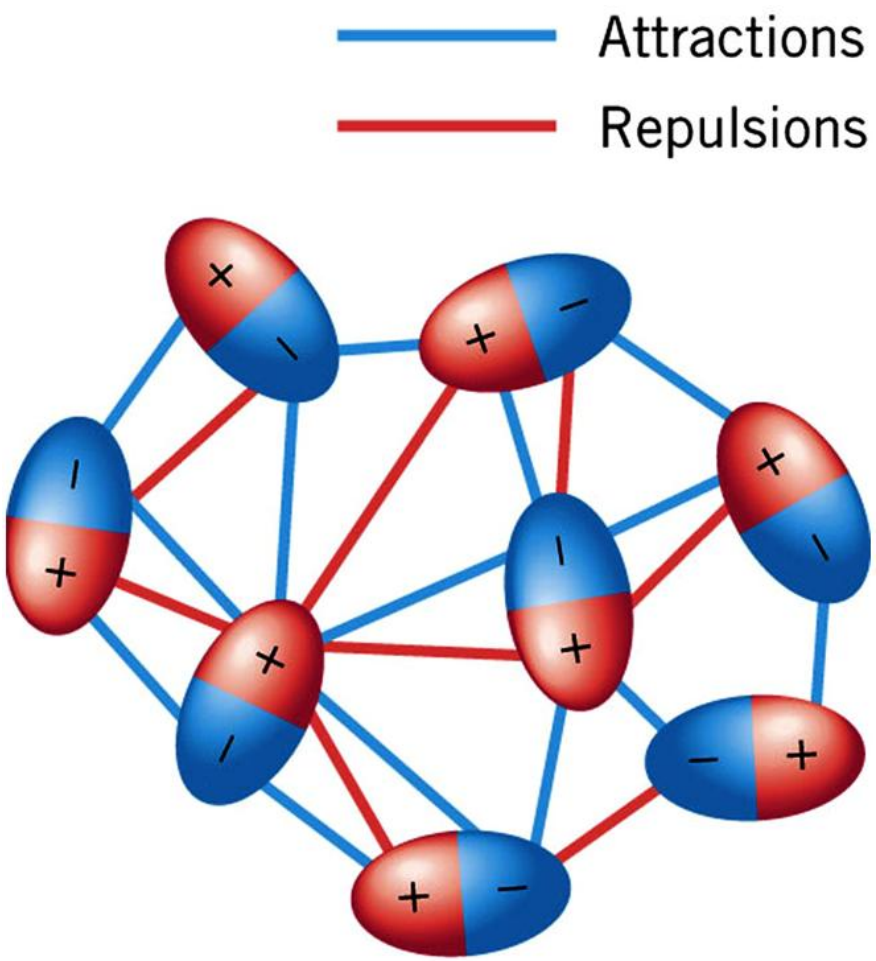
D. 5, 1, octahedron, square pyramid

Polar Molecules



- Have net dipole moment
  - Negative end
  - Positive end
- Polar molecules attract each other.
  - + end of polar molecule attracted to – end of next molecule.
  - Strength of this attraction depends on molecule's dipole moment
  - Dipole moment can be determined experimentally

Polar Molecules



- Polarity of molecule can be predicted by taking **vector** sum of bond dipoles
- Bond dipoles are usually shown as crossed arrows, where arrowhead indicates negative end

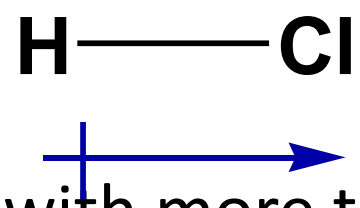
Molecular Shape and Molecular Polarity

- Many physical properties (mp, bp) affected by molecule **polar**ity
- For molecule to be polar:
  - Must have polar bonds
- Many molecules with polar bonds are nonpolar
  - Possible because certain arrangements of bond dipoles cancel
  - Nonpolar molecules even though they contain polar bonds

Why Nonpolar Molecules can have polar bonds

- Reason depends on ***Molecular Shape!***
- **Diatomics**: just consider 2 atoms
  - Calculate $\Delta\chi$

χ 2.1 2.9 $\Delta\chi = 0.8$



- For molecules with more than 2 atoms, must consider the ***combined effects*** of **all** polar bonds

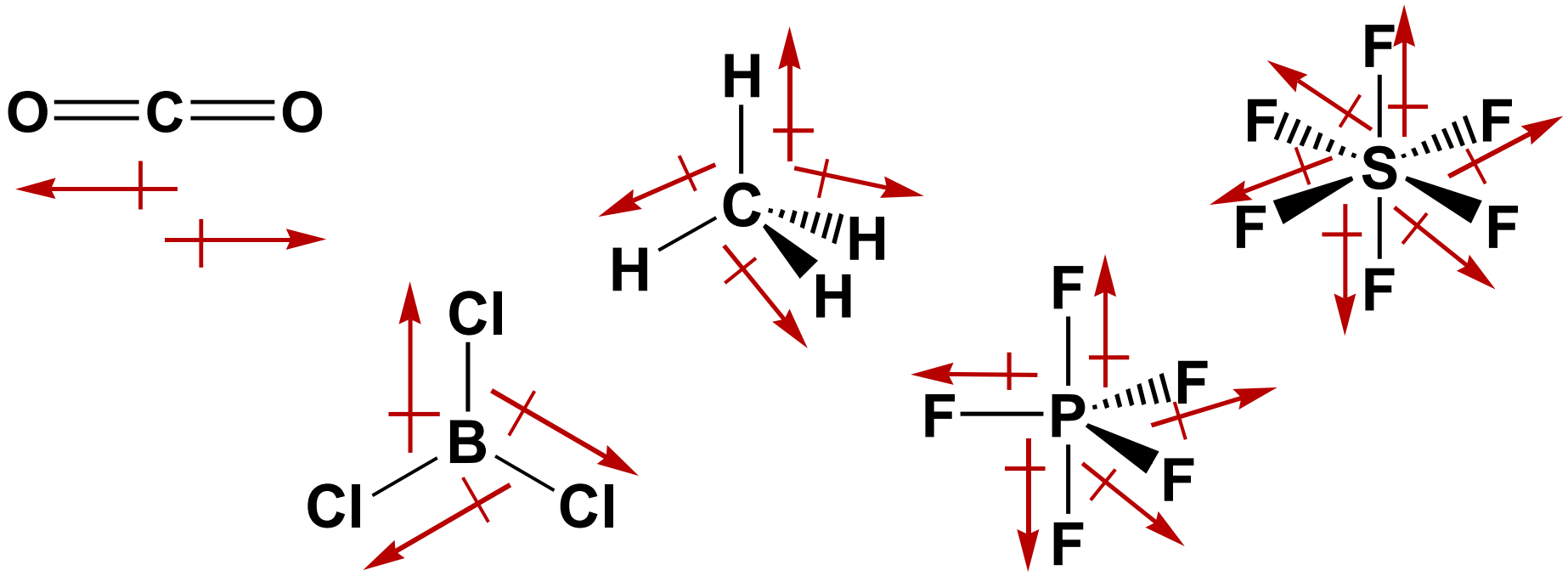
Polar Molecules are asymmetric

To determine polarity of molecule:

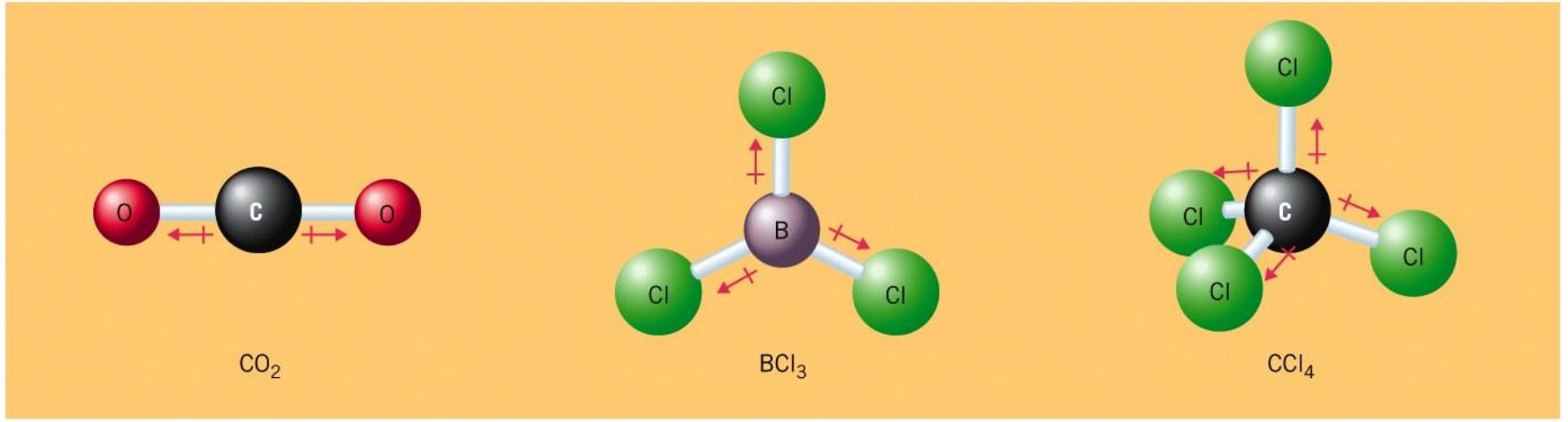
1. Draw structure using proper molecular geometry
2. Draw bond dipoles
3. If they cancel, molecule is non-polar
4. If molecule has uneven dipole distribution, it is polar

Molecular Polarity

1. **Molecule is nonpolar if**
   - All **e⁻** pairs around central atom are bonding pairs and
   - All terminal groups (atoms) are same
   - Then individual bond dipoles cancel



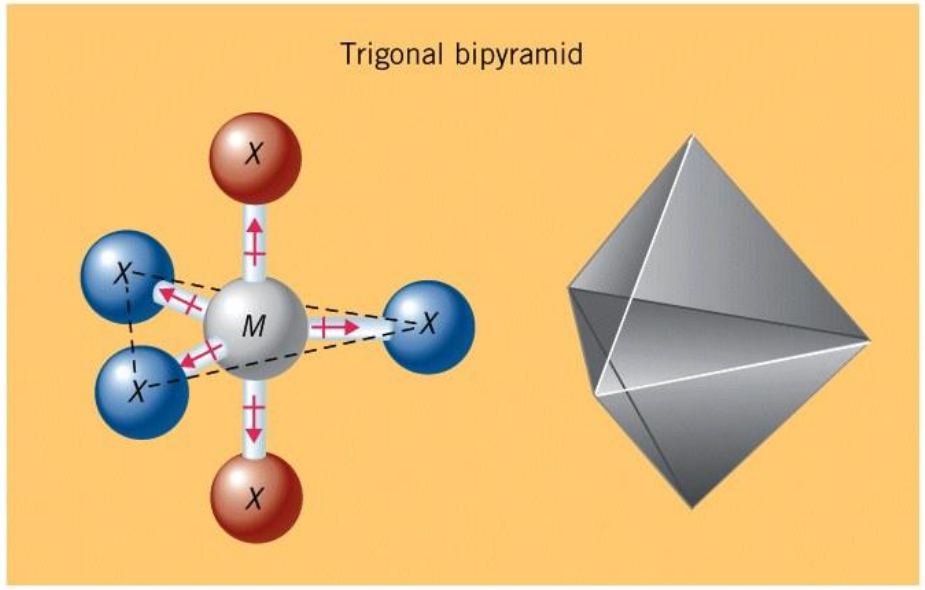
Molecular Polarity



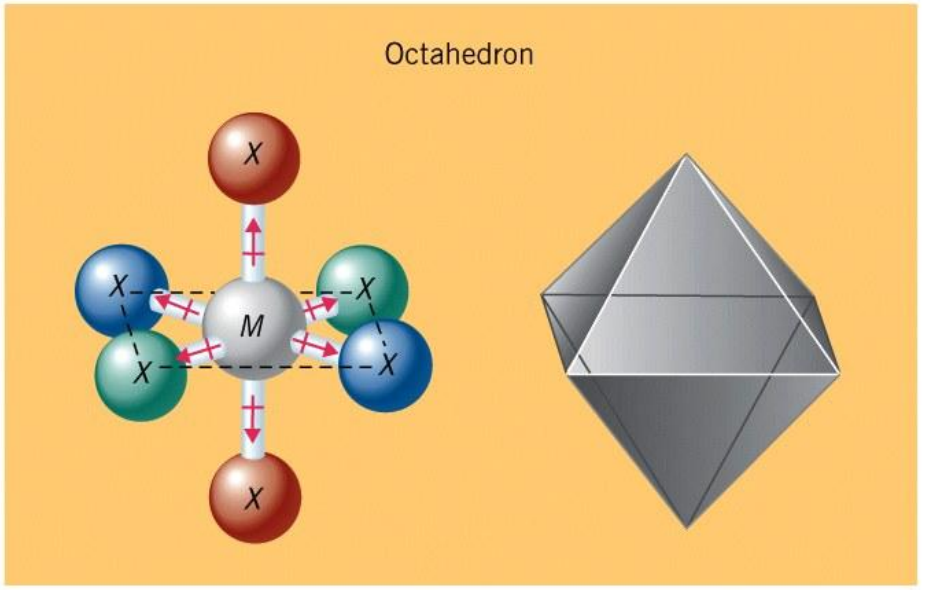
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- **Symmetric** molecules
  - Nonpolar because bond dipoles cancel
- All of **basic shapes** are **symmetric** when **all domains and groups** attached to them **are identical**

Cancellation of Bond Dipoles in Symmetric Trigonal Bipyramidal and Octahedral Molecules



(*a*)

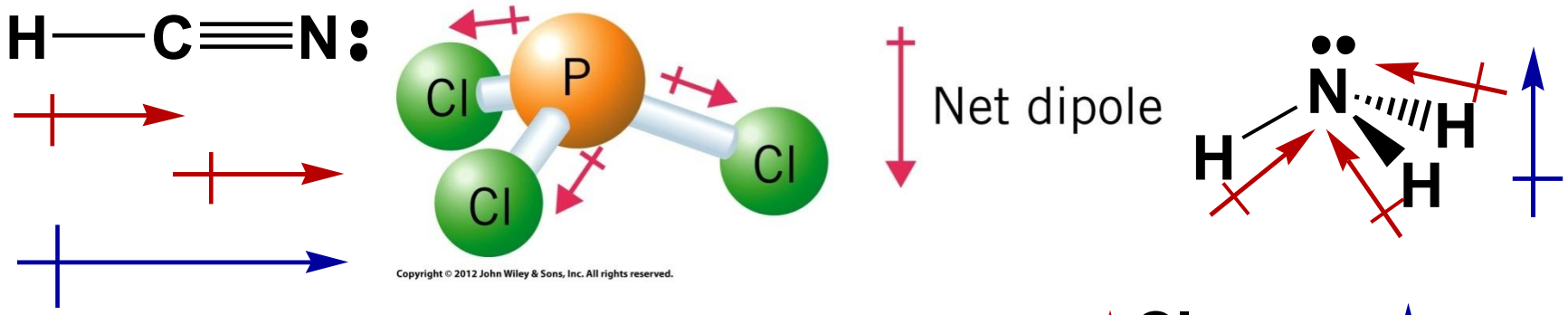


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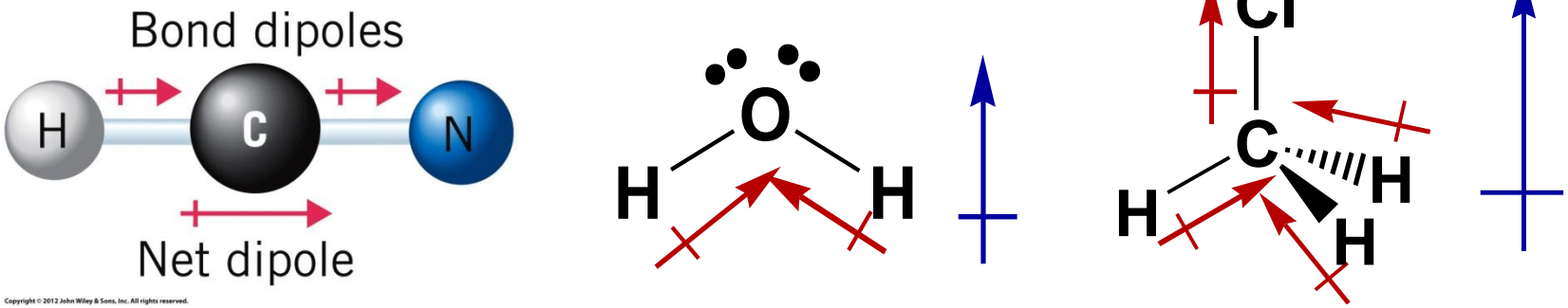
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Molecular Polarity

2. Molecule is usually **polar** if
   - All **atoms** attached to central atom are **NOT same**
   - Or,
   - There are **1 or more lone pairs** on **central atom**

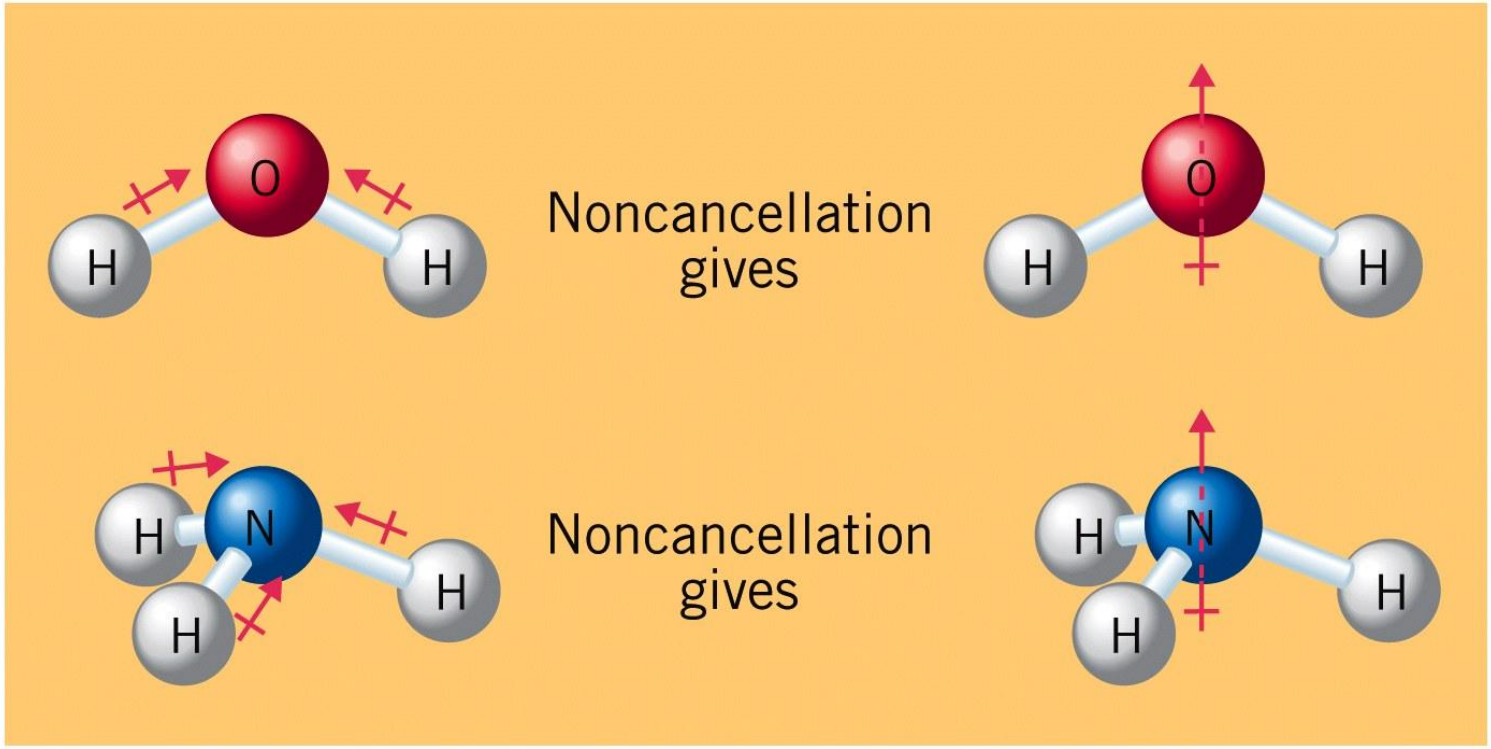


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Molecular Polarity

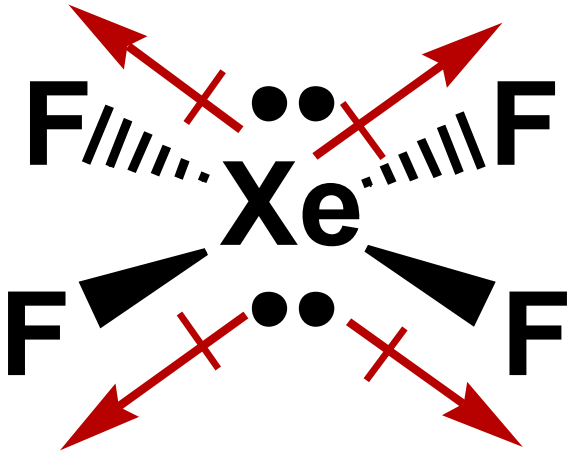


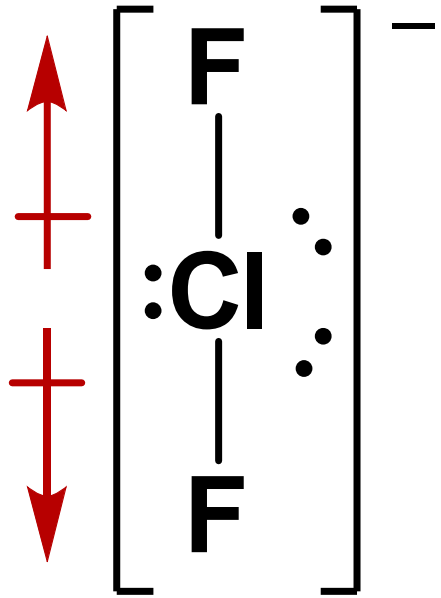
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- Water and ammonia both have non-bonding domains
- Bond dipoles do not cancel
- Molecules are polar

Molecular Polarity

- Following exceptions to rule 2 are **nonpolar**
- Nonbonding domains (lone pairs) are symmetrically placed around central atom





Your Turn!

Which of the following molecules is polar?

A. $BClF_2$

B. OF_2

C. NH_4^+

D. NO_3^-

E. C_2H_2

Modern Atomic Theory of Bonding

- Based on Wave Mechanics gave us
  - E's and shapes of orbitals
  - 4 quantum numbers
  - Heisenberg Uncertainty Principle (HUP)
    - $\mathbf{e}^-$ probabilities
  - Pauli Exclusion Principle

Problems with VSEPR

- Lewis Structures, VSEPR tell us nothing about
  - Why covalent bonds are formed.
  - How **e**$^{-}$'s manage to be shared between atoms.
  - How **e**$^{-}$ pairs in valence shell manage to avoid each other.

Modern Atomic Theory of Bonding

- Applied to molecules considers:
  - How orbitals on atoms come together to form covalent bond.
  - How these atomic orbitals interact with each other
  - How **e**$^{-}$'s are shared
- Two different theories have evolved
  - Complimentary
  - 2 ways to explain the same thing
  - Each useful for explaining different things

Valence Bond Theory

- Individual atoms, each with own orbitals and **e**⁻s come together to form bonds
- Worries about how atomic orbitals rearrange to form most efficient overlap for bonding

Molecular Orbital Theory (MOT)

- Views molecule as collection of + charged nuclei surrounded by **e**⁻s occupying a set of molecular orbitals
- Doesn't worry about how atoms come together to form molecule

Both Theories

- Try to explain structures and shapes of molecules, strengths of chemical bonds, bond orders, etc.
- Can be extended and refined to give same results

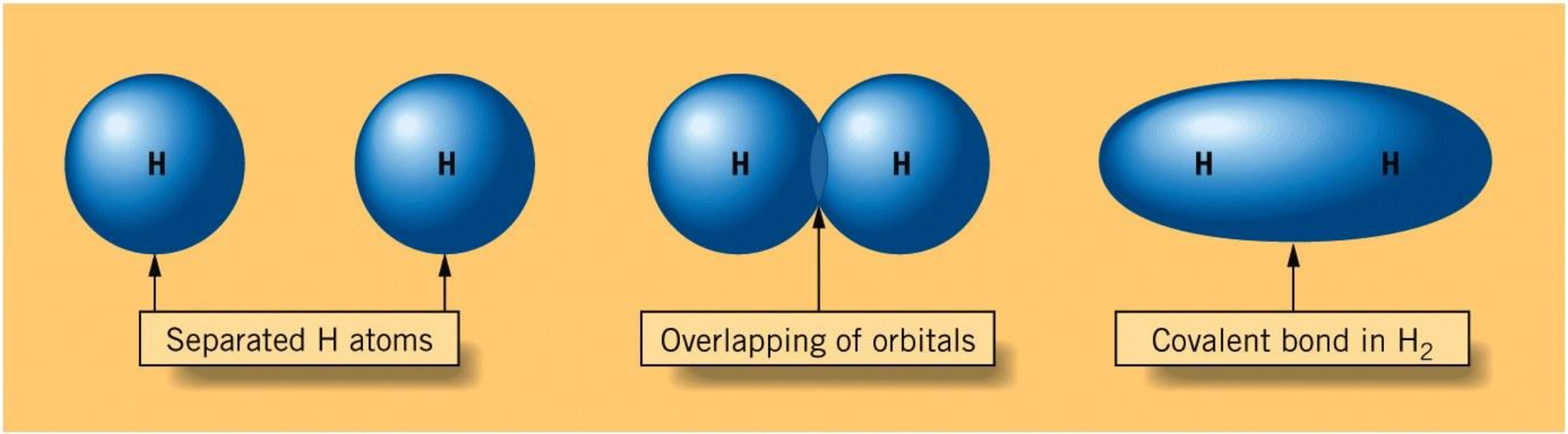
Valence Bond Theory (VBT)

- Bond between 2 atoms formed when pair of **e**$^-$s with paired (opposite) spins is shared by 2 overlapping atomic orbitals
- 1 AO from each atom

Valence Bond Theory – H_2

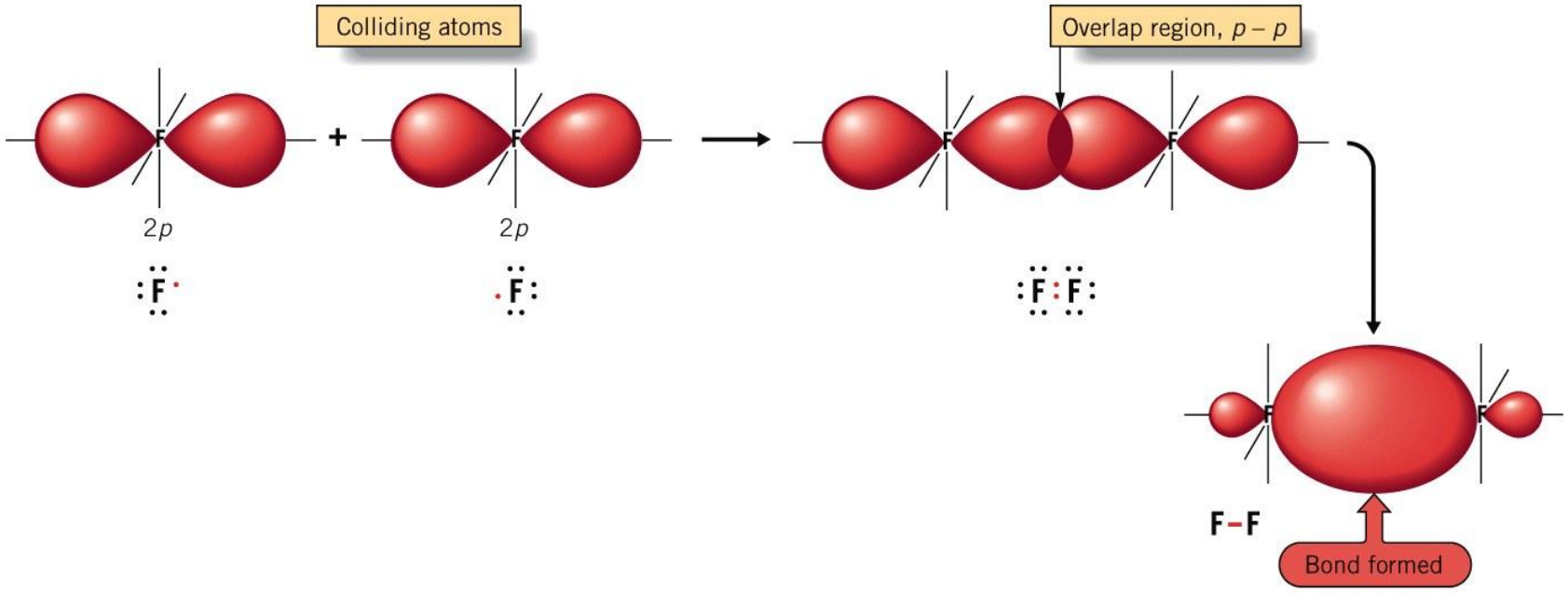
- H_2 bonds form because **1*s*** atomic valence orbital from each H atom overlaps





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Valence Bond Theory – F_2

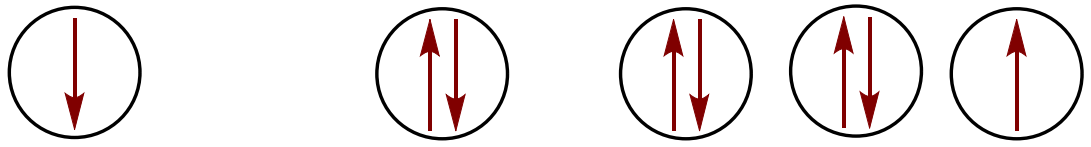


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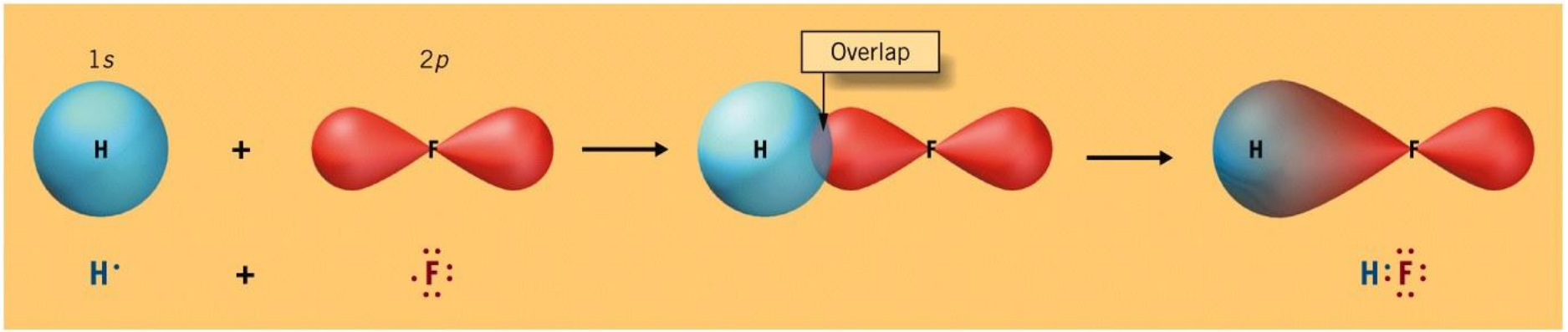
- F_2 bonds form because atomic valence orbitals overlap
- Here **2*p*** overlaps with **2*p***
- Same for all halogen, just different **n*p*** orbitals

Valence Bond Theory – HF

- HF involves overlaps between **1*s*** orbital on H and **2*p*** orbital of F



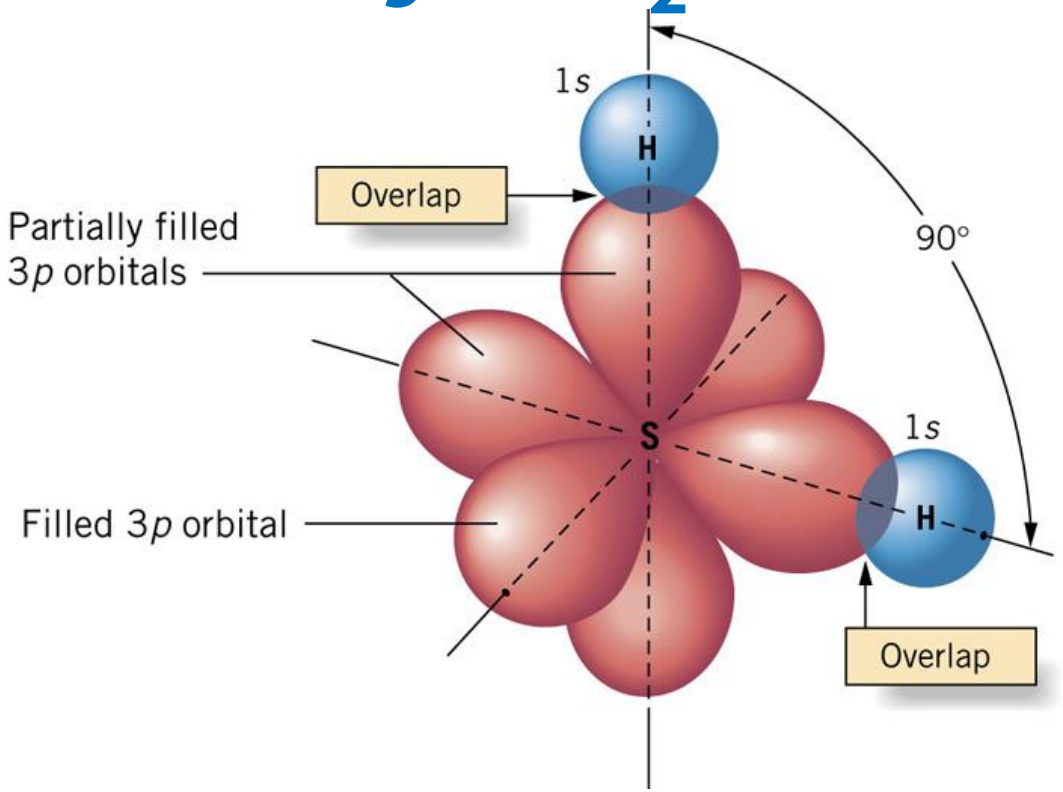
1*s* 2*p*



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Valence Bond Theory – H_2S

- Assume that unpaired **e**⁻s in S and H are free to form paired bond
- We may assume that H—S bond forms between ***s*** and ***p*** orbital



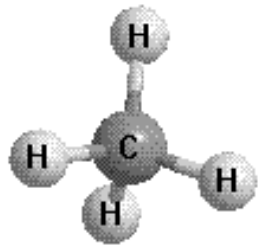
- Predicted 90º bond angle is very close to experimental value of 92º.

Difficulties with VB Theory so far

- Most experimental bond angles do not support those predicted by mere atomic orbital overlap

Ex. **C** $1s^2 2s^2 2p^2$ and **H** $1s^1$

- Experimental bond angles in methane
  - Are 109.5° and all are same
  - ***p*** orbitals are 90° apart
  - Not all valence $\mathbf{e^-}$ in C are in ***p*** orbitals
- How can multiple bonds form?



Hybridization

- Mixing of atomic orbitals to allow formation of bonds that have realistic bond angles.
  - Realistic description of bonds often requires combining or blending 2 or more atomic orbitals
- Hybridization just rearranging of e^- probabilities

Why do it?

- To get maximum possible overlap
- Best (strongest) bond formed

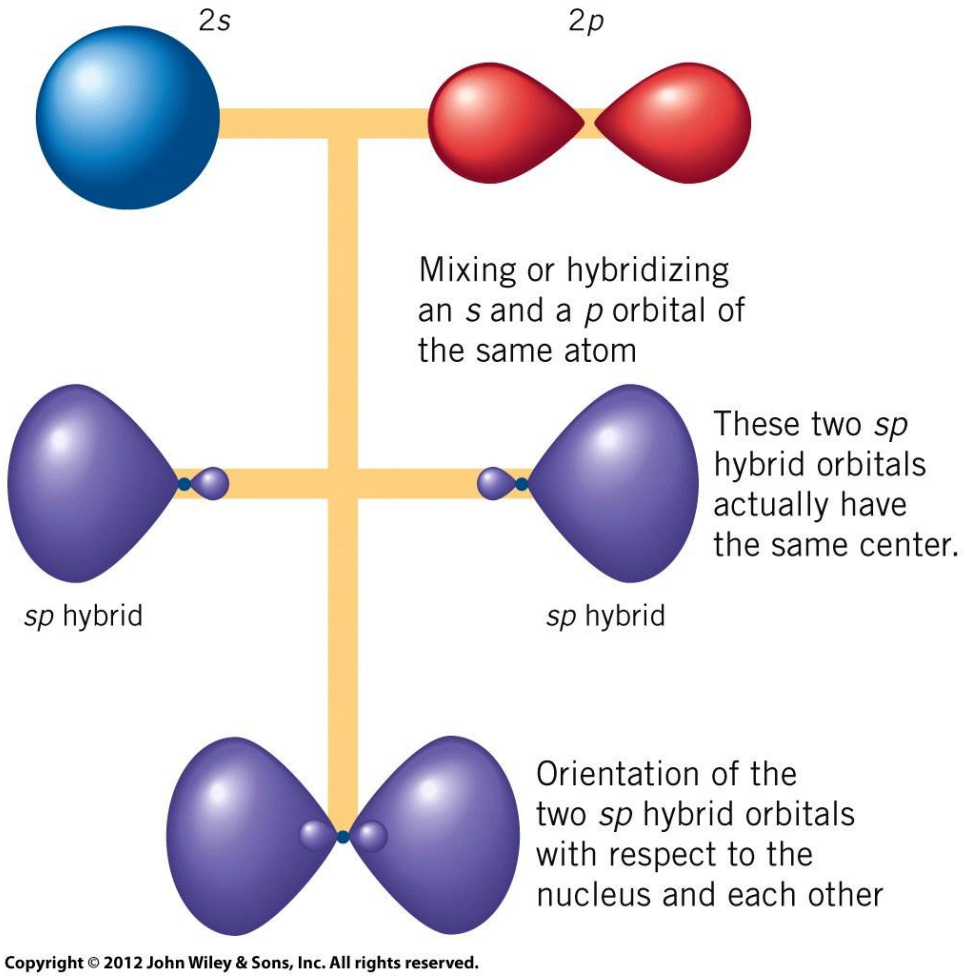
Hybrid Orbitals

- Blended orbitals that result from hybridization process
- Hybrid orbitals have
  - New shapes
  - New directional properties
  - Each HO combines properties of parent AOs
- ***Number of hybrid orbitals required = number of bonding domains + number of non-bonding domains*** on atom being hybridized

What Call These New Orbitals?

- Name hybrid as combination of orbitals used to form new hybrids
  - Use ***s*** + ***p*** form 2 ***sp*** hybrid orbitals
  - Use ***s*** + $\boldsymbol{p_x}$ + $\boldsymbol{p_y}$ form 3 $\boldsymbol{sp^2}$ hybrid orbitals, etc.
- One atomic orbital is used for every hybrid formed (orbitals are conserved)
- Sum of coefficients in hybrid orbital must add up to number of atomic orbitals used
- Number of atomic orbitals in = number of hybrid orbitals formed

Let's see How Hybridization Works

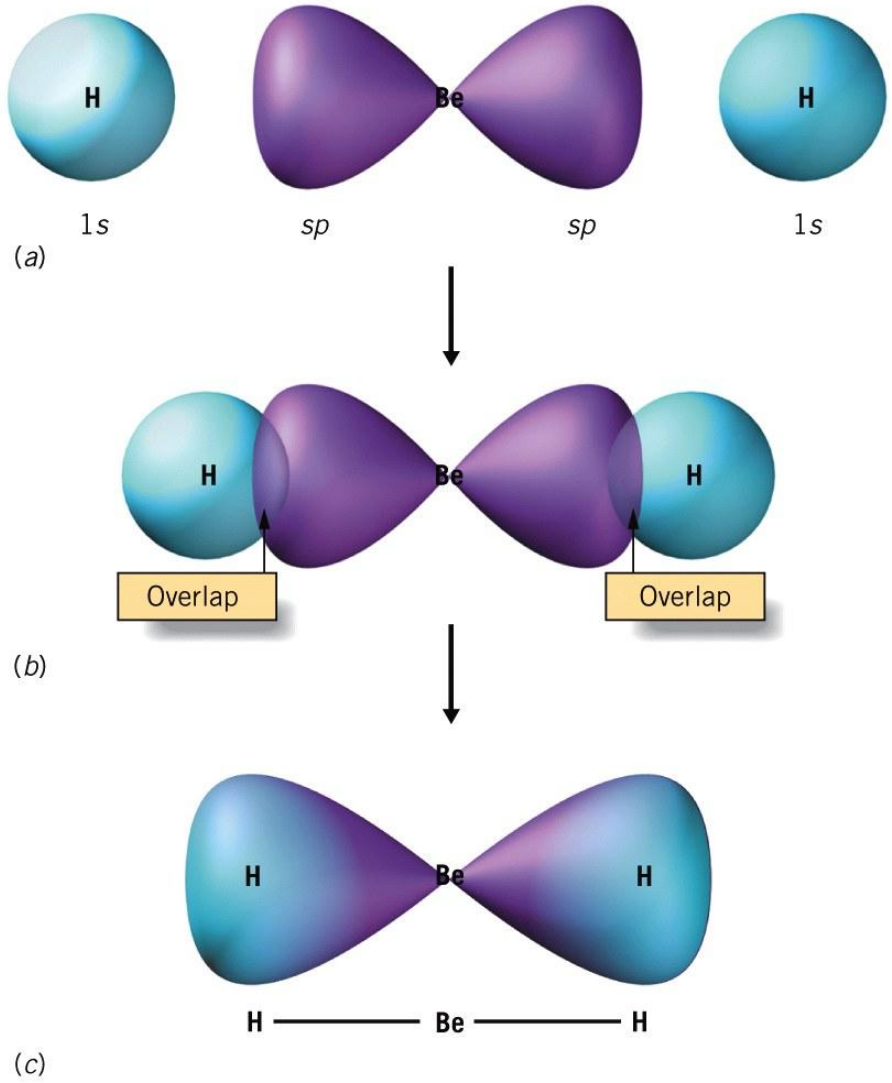


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- Mixing or hybridizing *s* and *p* orbital of same atom results in two *sp* hybrid orbitals
  - Two *sp* hybrid orbitals actually have same center
- Two *sp* hybrid orbitals point in opposite directions

Using *sp* Hybrids to Form Bonds

- Now have two ***sp*** hybrid orbitals
- Oriented in correct direction for bonding
- 180° bond angles
  - As VSEPR predicts and
  - Experiment verifies
- Bonding =
  - Overlap of H 1***s*** atomic orbitals with ***sp*** hybrid orbitals on Be

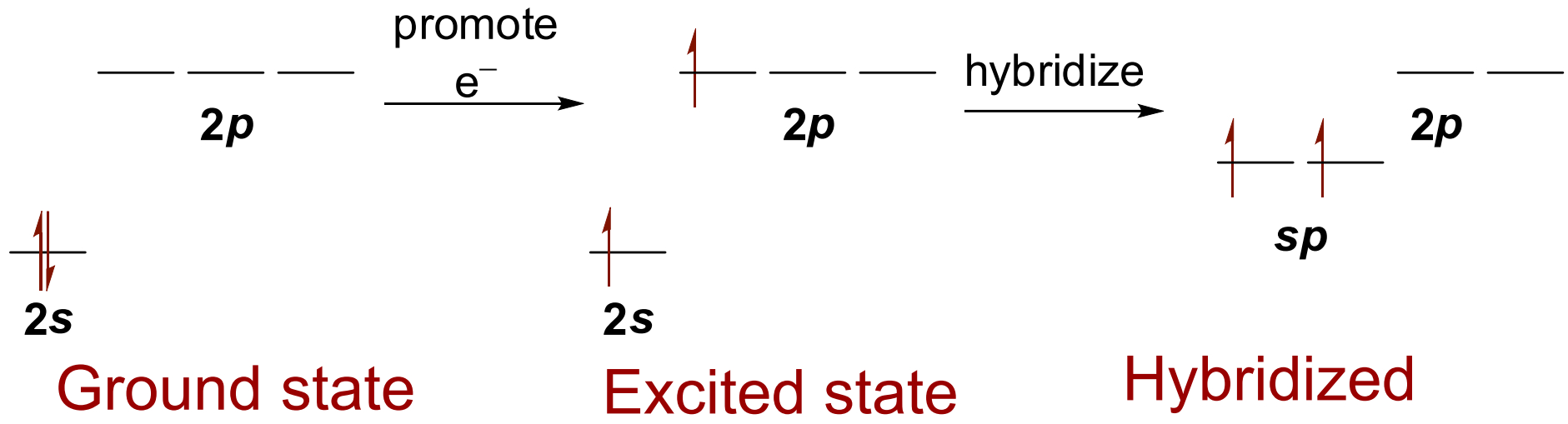


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What Do We Know?

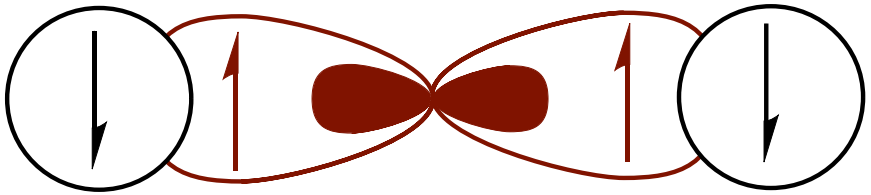
- Experiment and VSEPR show that
  - BeH_2 (*g*) is linear
  - 180º bond angle
- For Be to form these bonds it must have
  - 2 orbitals (HO) on BE that must point in opposite directions
  - Give correct bond angle
- Each Be orbital must contain $1e^-$ only
  - Each resulting bond with H contains only 2 e^-s
  - Each H supplies 1 e^-

Hybridization - Energetics

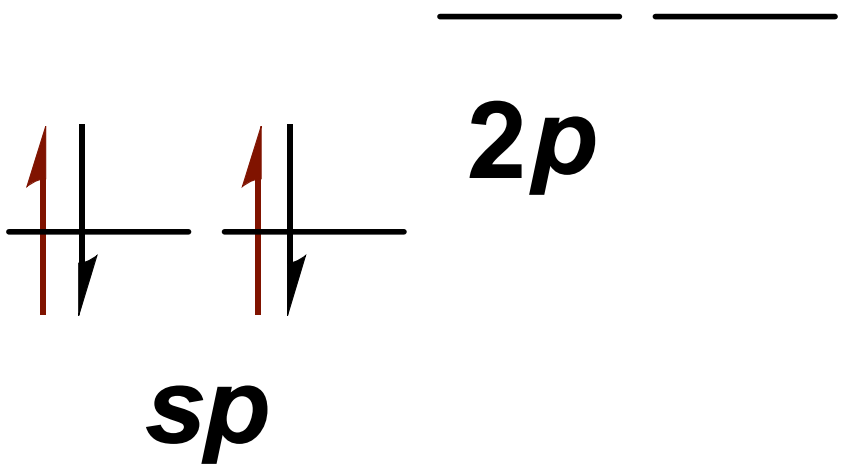


- By forming *sp* hybrids, Be satisfies these conditions
  - 2 HO's identical in shape, size and energy
  - Opposite in direction
  - ½ filled
  - Bonds equivalent

Hybridization – Bonding in BeH_2



- Now have overlap of 2 half-filled orbitals
- Form bond



Be in BeH_2

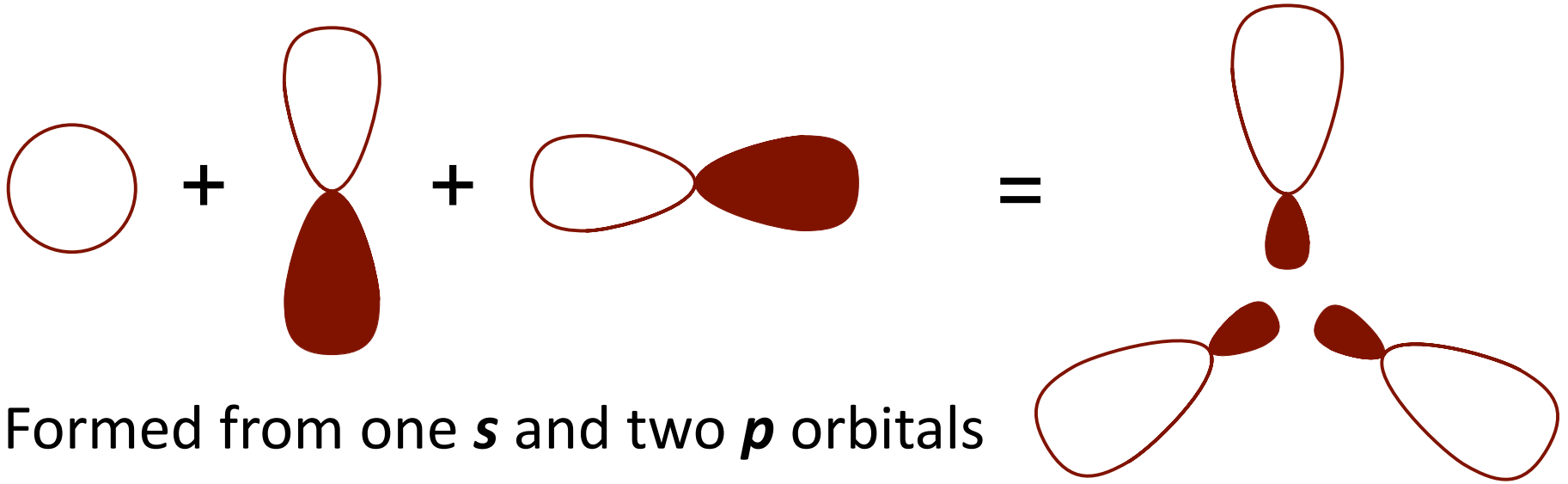
Hybrid Orbitals

Hybrid	Atomic Orbitals Used	Electron Geometry
sp	$s + p_x$	**Linear** Bond angles 180˚
sp^2	$s + p_x + p_y$	**Trigonal planar** Bond angles 120˚
sp^3	$s + p_x + p_y + p_z$	**Tetrahedral** Bond angles 109.5˚
sp^3d	$s + p_x + p_y + p_z + d_z2$	**Trigonal Bipyramidal** Bond Angles 90 and 120˚
sp^3d^2	$s + p_x + p_y + p_z + d_z2 + d_x2 - {}_y2$	**Octahedral** Bond Angles 90˚

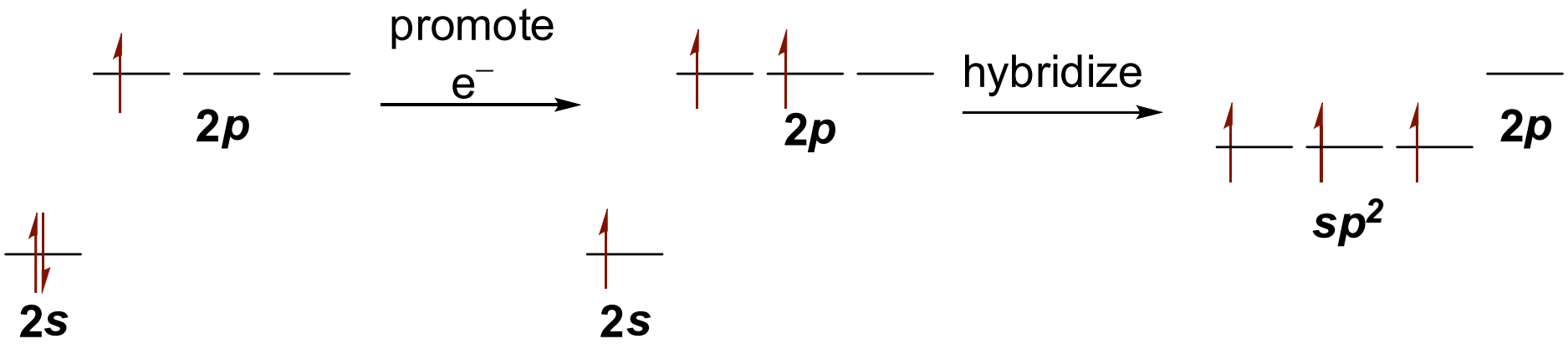
Common Procedure to All of These

1. Number of HO's formed = number of AOs mixed
2. Each HO has same shape, size, and energy, but point in different directions.
3. Large lobe extends farther from nucleus than either AO from which it was formed
   - More effective overlap with orbitals of other atom
   - Forms a stronger, more stable bond
4. Label HO's as $\boldsymbol{sp^n}$
   - Superscript after ***p*** orbital tells how many ***p*** orbitals are mixed with ***s*** orbital to make HOs

sp^2 Hybrid Orbitals

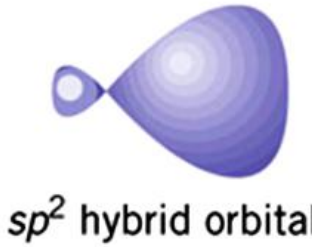


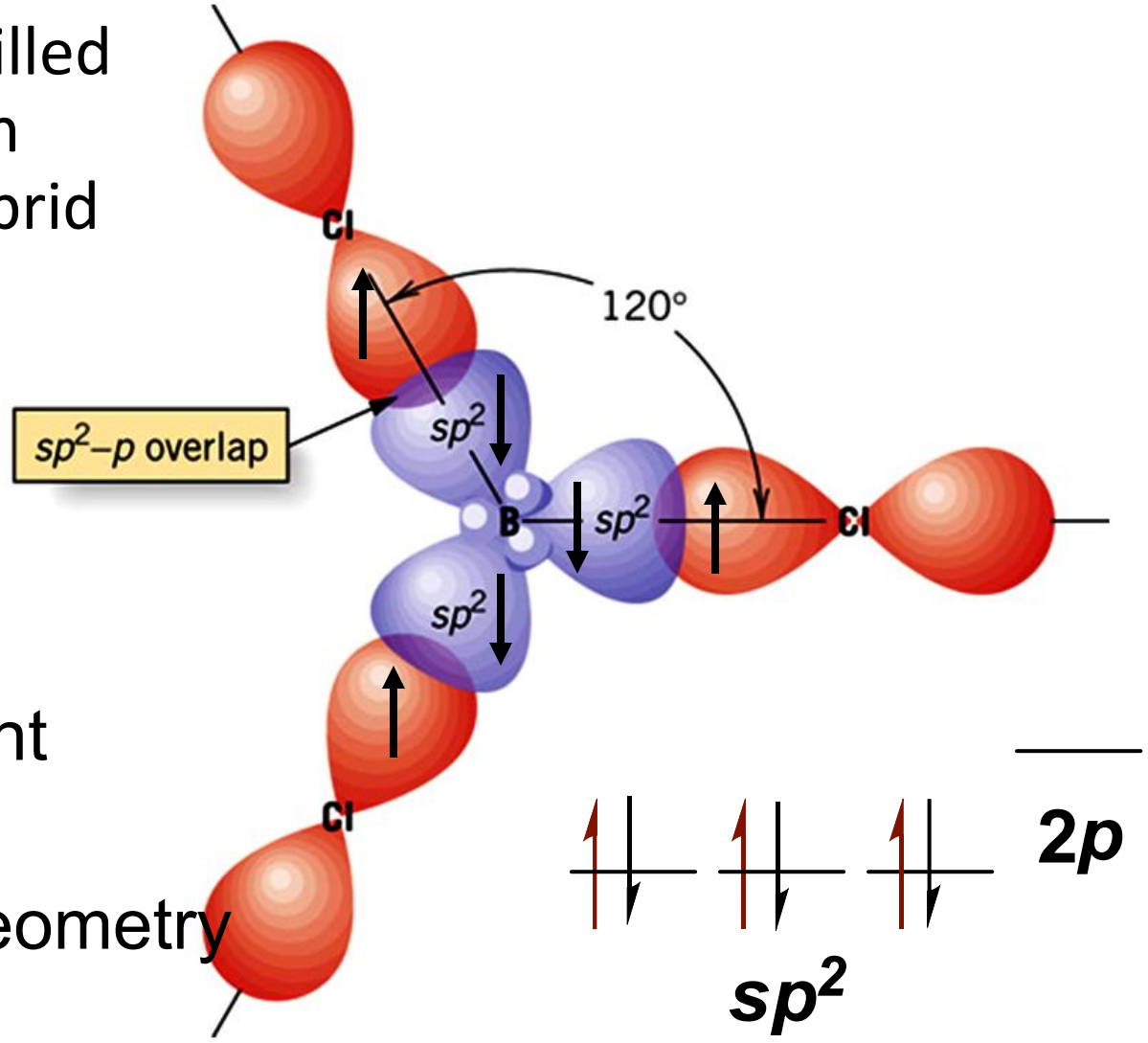
- Formed from one ***s*** and two ***p*** orbitals
- Lie in plane and point to corners of triangle



Bonding in BCl_3

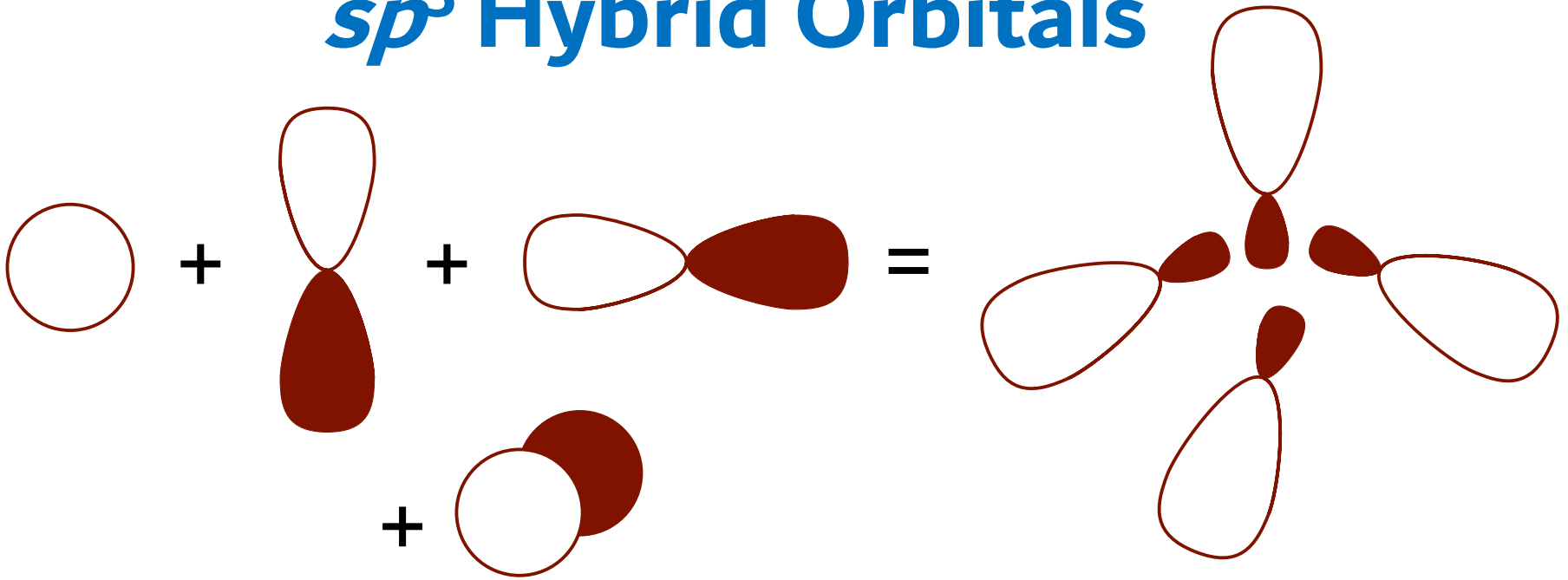
- Overlap of each ½ filled **3*p*** orbital on **Cl** with each ½ filled ***sp*²** hybrid on **B**



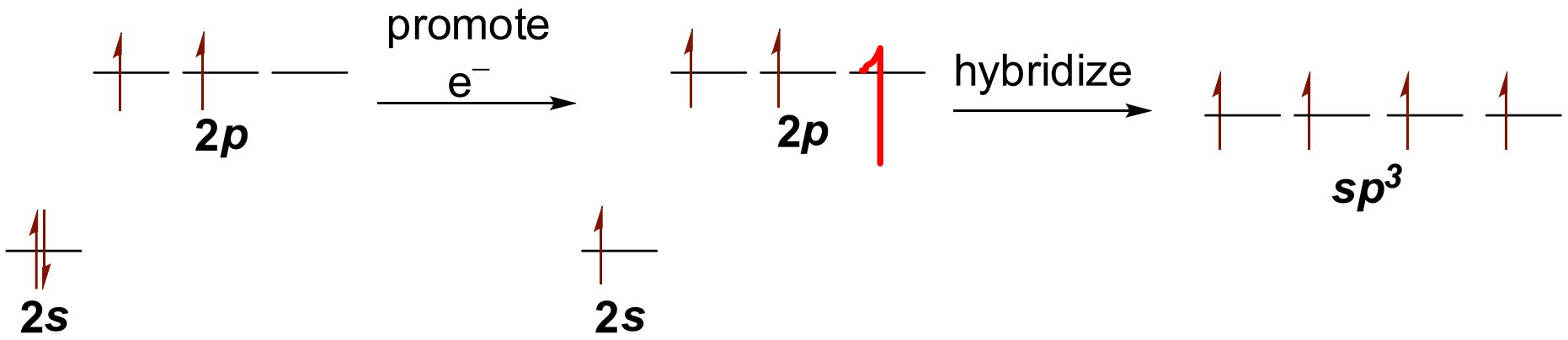


- Forms 3 equivalent bonds
- Trigonal planar geometry
- 120° bond angle

sp^3 Hybrid Orbitals

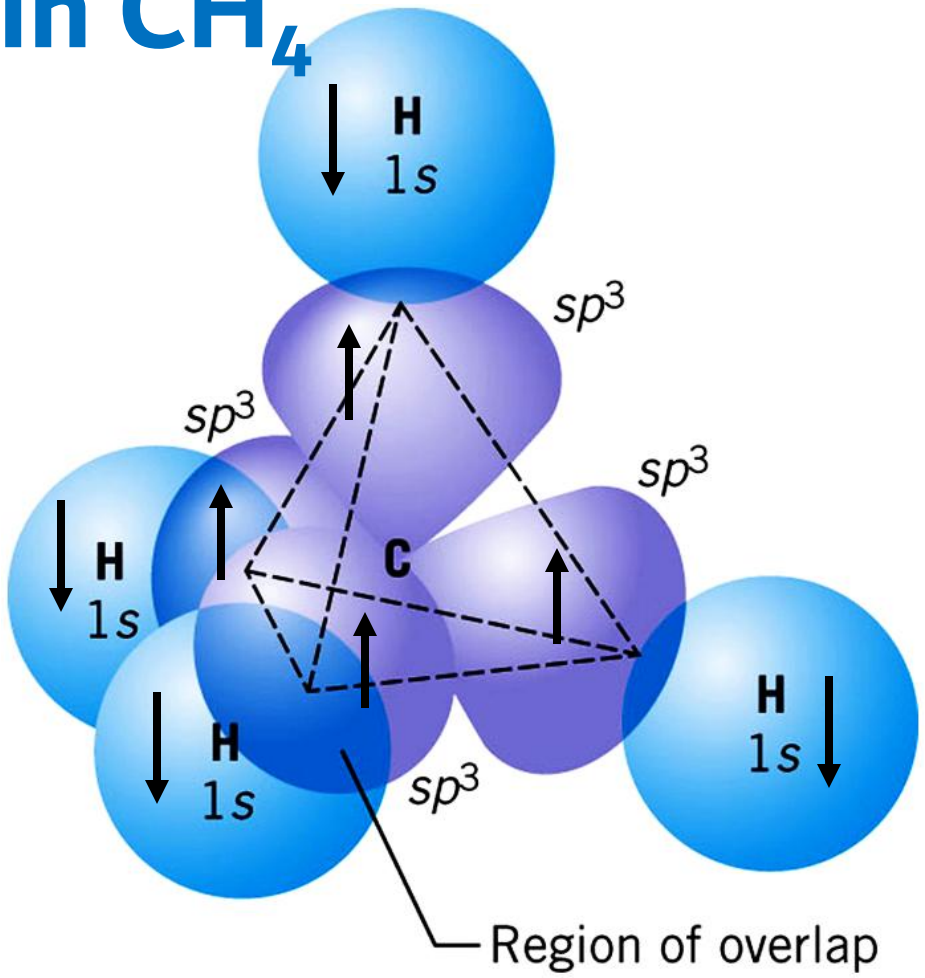


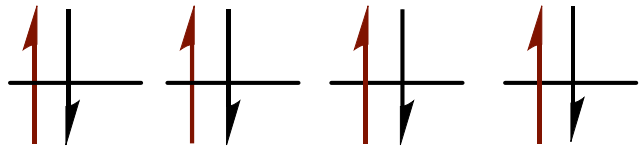
- Formed from one ***s*** and three ***p*** orbitals
- Point to corners of tetrahedron



Bonding in CH_4

- Overlap of each ½ filled 1*s* orbital on H with each ½ filled sp^3 hybrid on C
- Forms 4 equivalent bonds
- Tetrahedral geometry
- 109.5° bond angle

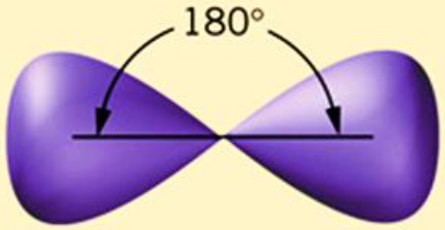




sp^3

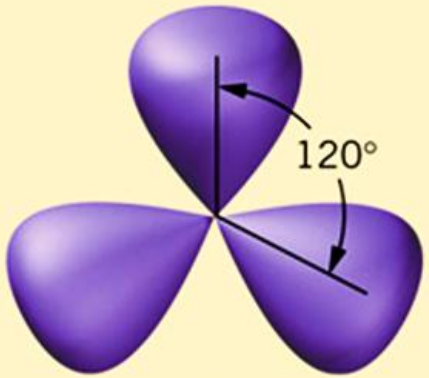
Hybrid Orbitals

Two ***sp*** hybrids



Linear

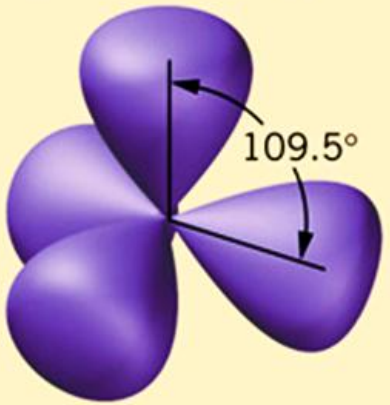
Three sp^2 hybrids



All angles 120°

Planar Triangular

Four sp^3 hybrids

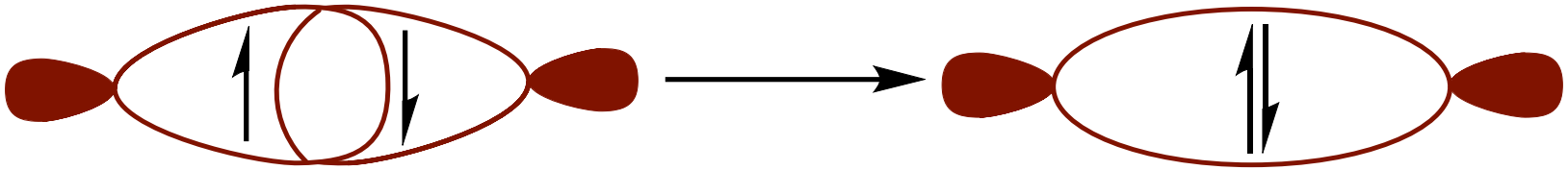


All angles 109.5°

Tetrahedral

Tetrahedral Carbon

- In ethane C_2H_6
- Each C—H bond
  - Overlap of one ***sp³*** HO on C with ***1s*** AO on H
- C—C bond
  - Overlap of one ***sp³*** HO on each C!



Your Turn!

What is oxygen's hybridization in OCl_2 ?

A. sp

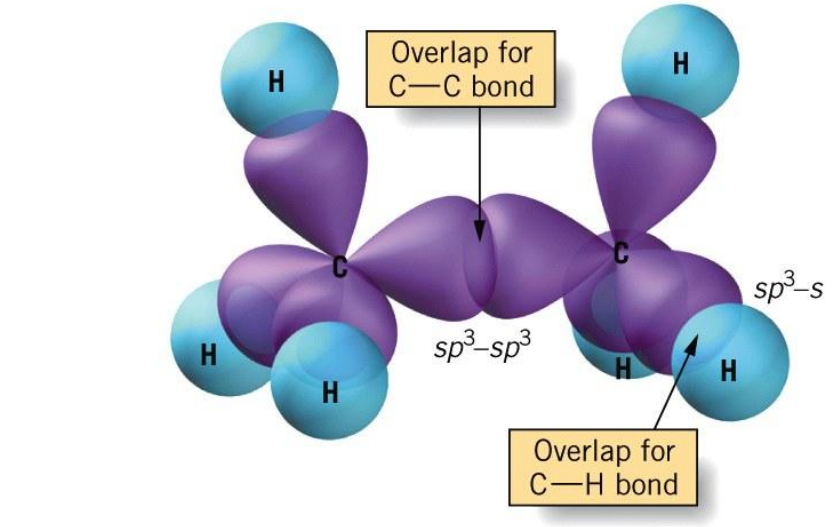
B. sp^3

C. sp^2

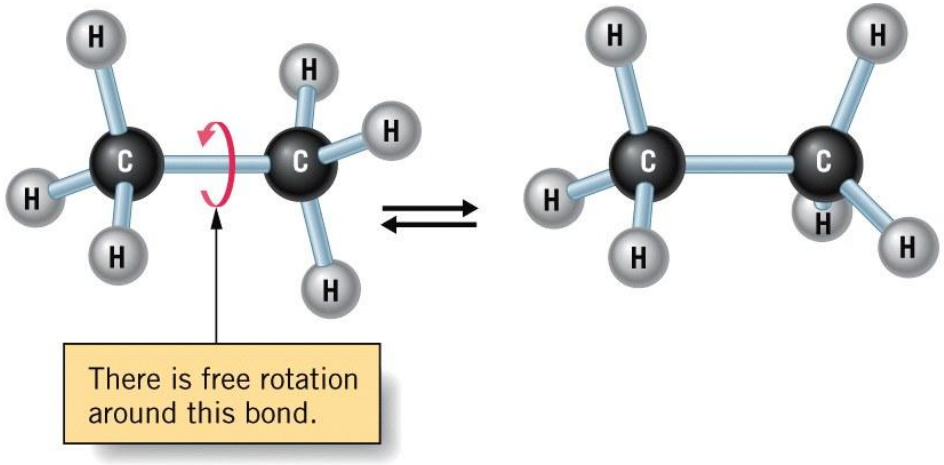
D. No hybridization

Conformations

- C—C bond unaffected by rotation around C—C bond
  - Due to cylindrical symmetry of bond
- **Conformations**
  - Different relative orientations on molecule upon rotation



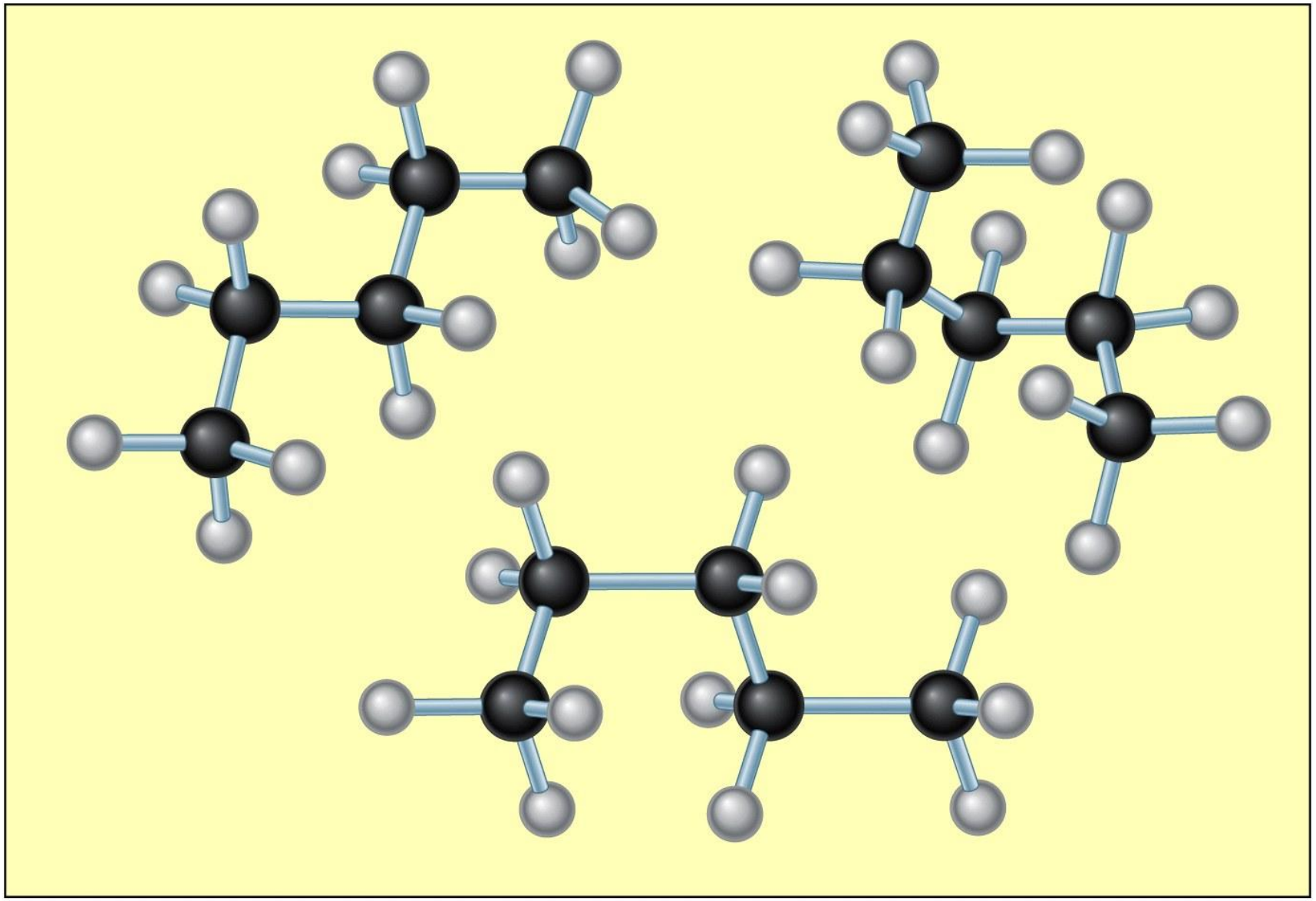
(*a*)



(*b*)

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Multiple Conformations of Pentane



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Expanded Octet Hybridization

Hybridization When Central Atom Has More Than Octet

- If there are more than 4 equivalent bonds on central atom, then must add ***d*** orbitals to make HO's

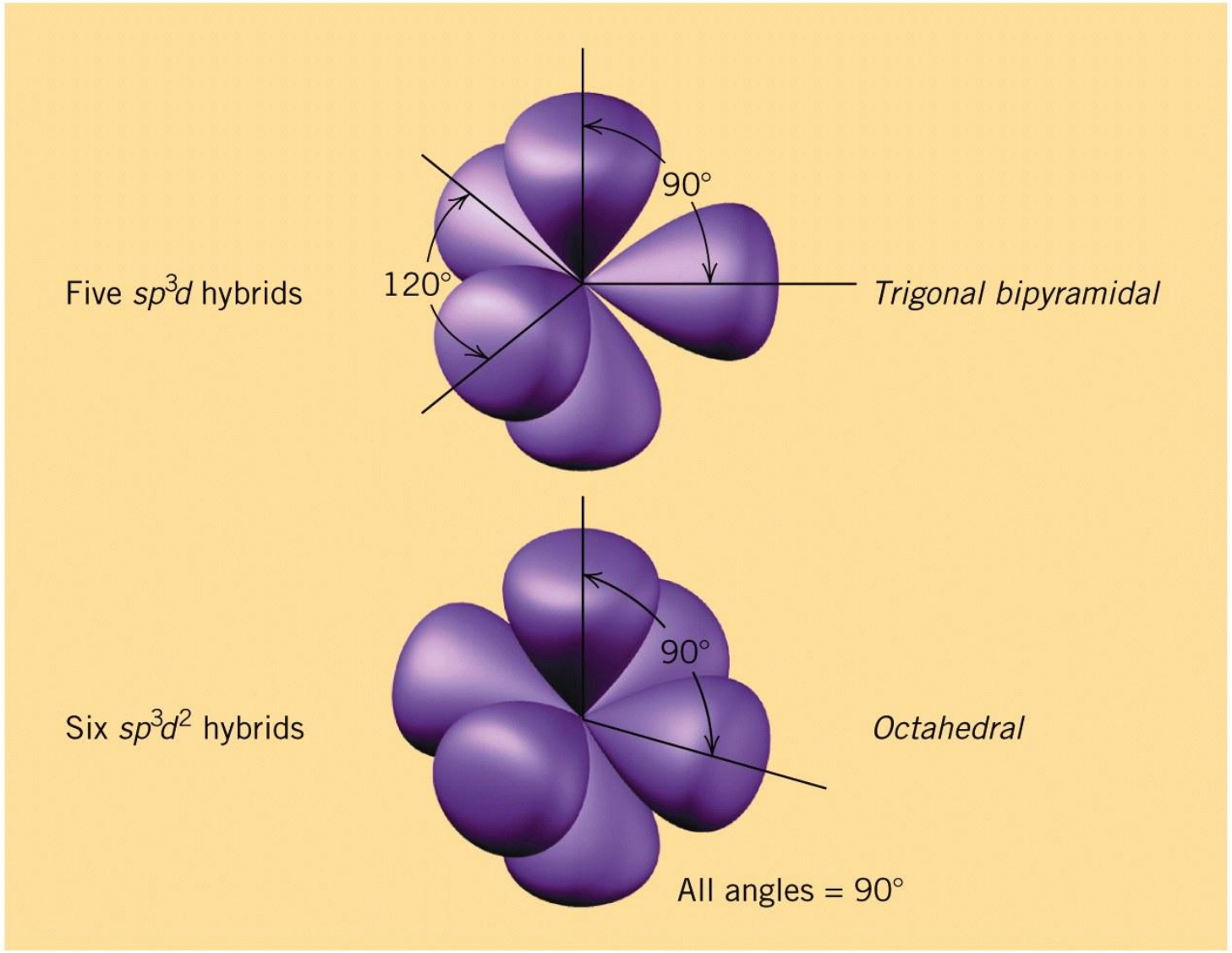
Why?

- One ***s*** and three ***p*** orbitals means that four equivalent orbitals is the most you can get using ***s*** and ***p***'s alone

Expanded Octet Hybridization

- So, only atoms in third row of the Periodic Table and below can exceed their octet
  - These are the only atoms that have empty ***d*** orbitals of same **n** level as ***s*** and ***p*** that can be used to form HO's
- One ***d*** orbital is added for each pair of electrons in excess of standard octet

Expanded Octet Hybrid Orbitals



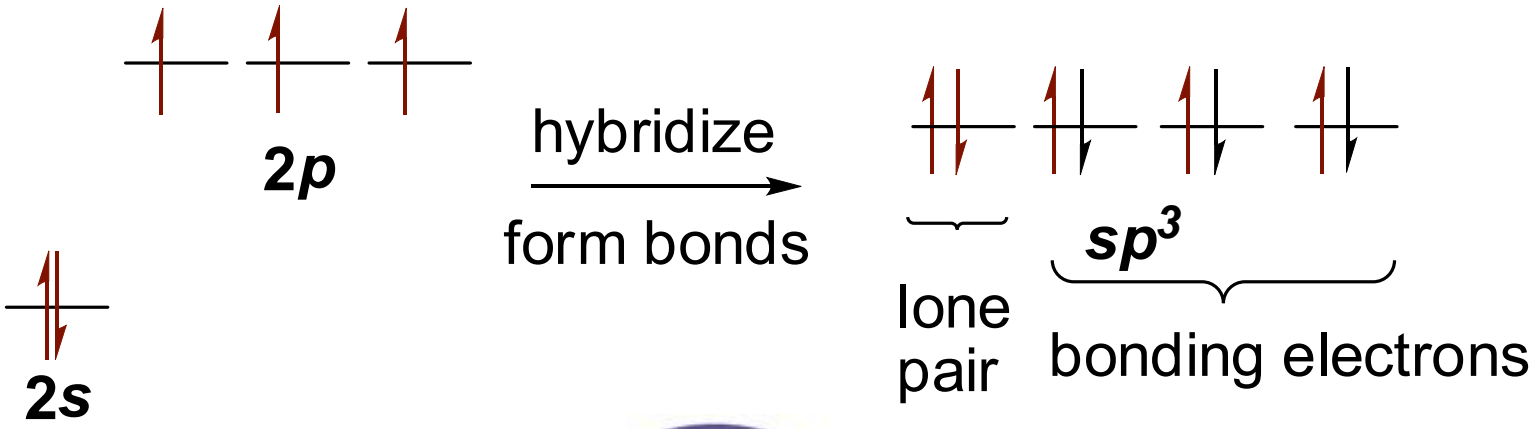
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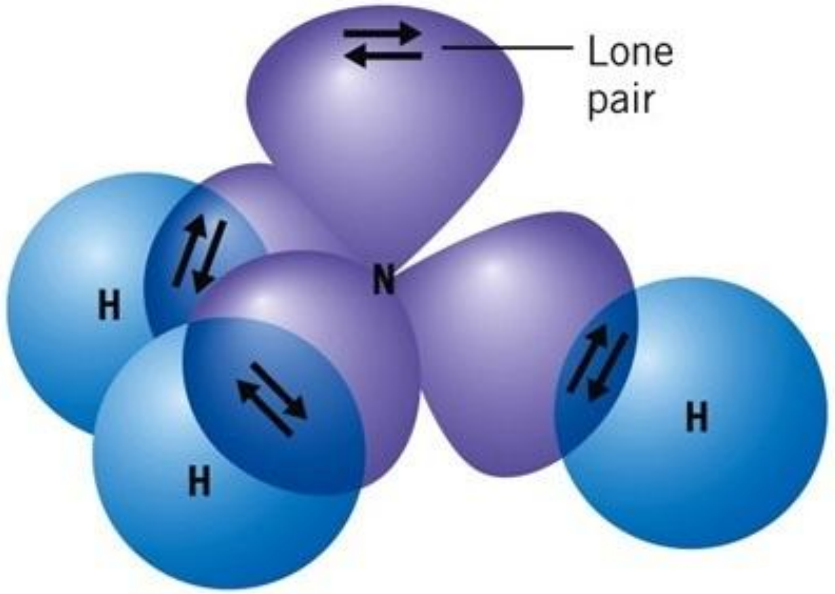
Hybridization in Molecules that have Lone Pair Electrons

CH_4	sp^3	tetrahedral geometry	109.5° bond angle
NH_3			107° bond angle
H_2O			104.5° bond angle

- Suggests that NH_3 and H_2O both use sp^3 HO's in bonding
- Not all HO's used for bonding e^-
- Lone pair e^-'s can reside here too
- Explains why in VSEPR you must count lone pairs to determine geometry
  - Lone pairs occupy hybrid orbitals

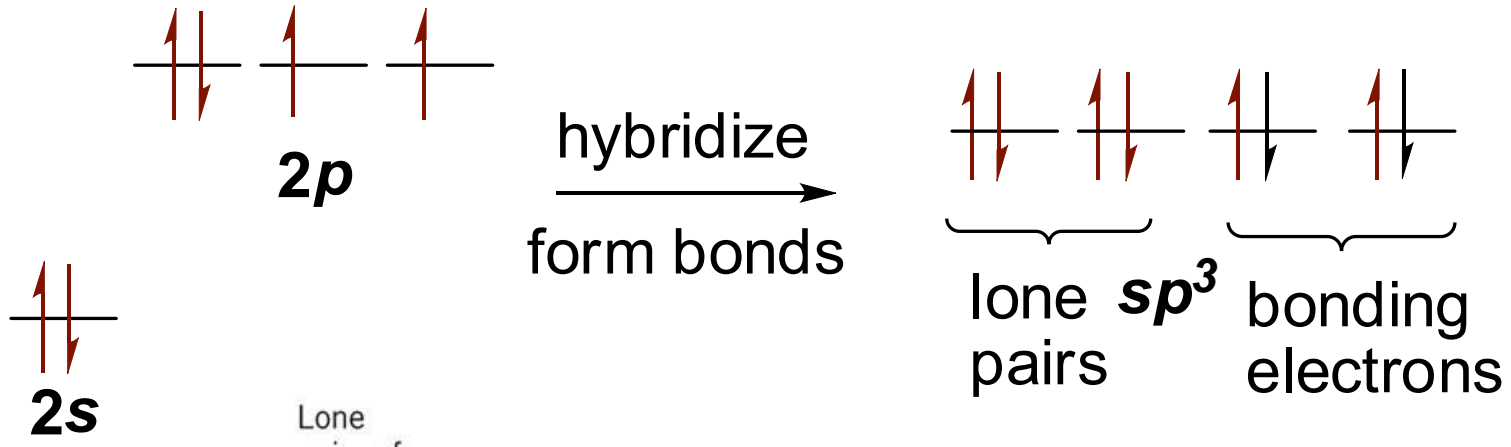
Hybridization in Molecules that have Lone Pair Electrons – NH_3

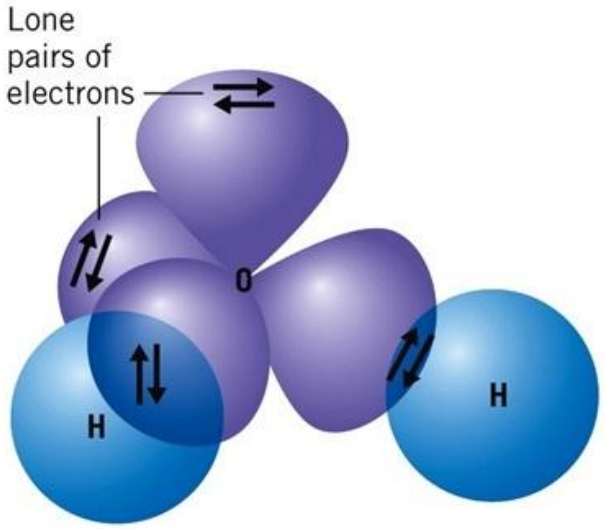




(b)

Hybridization in Molecules that have Lone Pair Electrons – H_2O

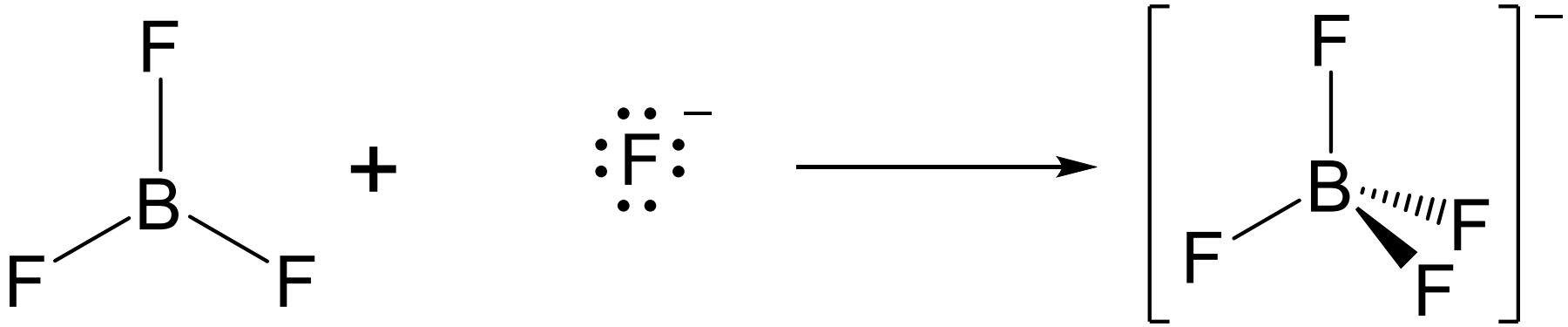




(a)

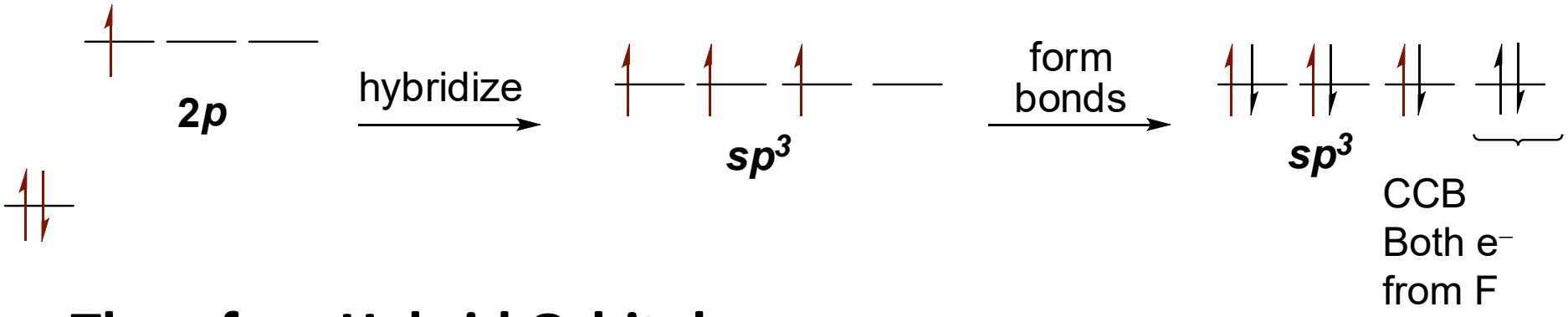
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Coordinate Covalent Bonds and HO‘s



- Both shared e⁻'s come from one atom
- Once formed, no different from any other covalent bond
- VBT requirements for bond formation
  - 2 overlapping orbitals sharing 2 paired e⁻s can be met two ways
  1. Overlapping of two ½ filled orbitals
  2. Overlapping of one full and one empty orbital

Coordinate Covalent Bonds and HO‘s



Therefore Hybrid Orbitals

- Determine molecular geometry by number of electron domains in HO's around central atom
- Directed as far apart as possible
- Contain
  1. All lone pairs of e⁻s on central atom
  2. All e⁻pairs forming single bonds (or σ bonds.)
  3. One and only 1 of e⁻ pairs in multiple bonds

Your Turn!

For the species ClF_2^+, determine the following:

1. *electron domain geometry*
2. *molecular geometry*
3. *hybridization around the central atom*
4. *polarity*

A. tetrahedron, trigonal planar, sp^3, polar
B. pentagon, tetrahedron, sp^3, non-polar
C. tetrahedron, bent, sp^3, polar
D. trigonal planar, bent, sp^2, non-polar

Your Turn!

For the species XeF_4O, determine the following:

1. *electron domain geometry*
2. *molecular geometry*
3. *hybridization around the central atom*
4. *polarity*

A. octahedron, square pyramid, sp^3d, polar

B octahedron, square pyramid, sp^3d^2, polar

C. square pyramid, octahedron, sp^3d^2, polar

D. trigonal bipyramid, planar, sp^3d, non-polar

Double and Triple Bonds

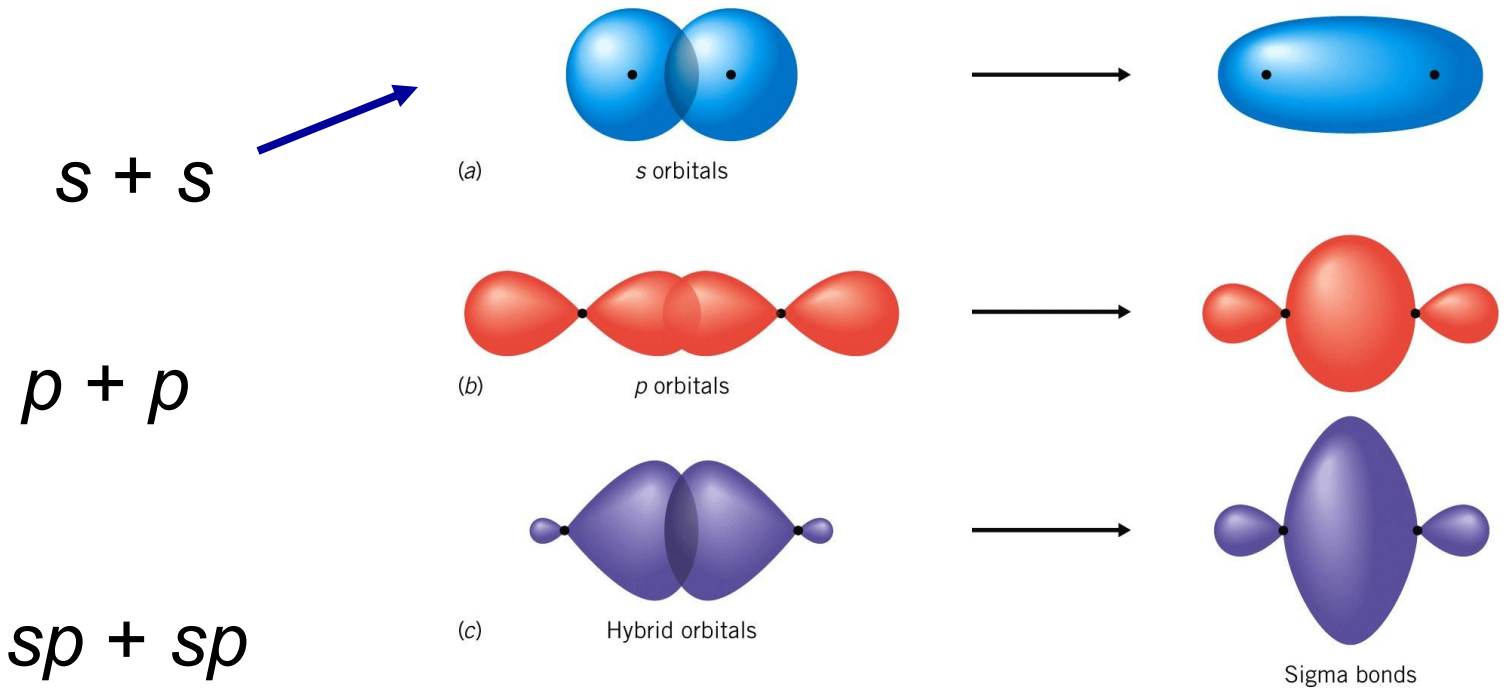
- So where do extra e^- pairs in multiple bonds go?
  - Not in Hybrid orbitals
  - Remember VSEPR, multiple bonds have no effect on geometry
- Why don't they effect geometry?

2 types of bonds result from orbital overlap

- **Sigma (σ) bond**
  - Accounts for first bond
- **Pi (π) bond**
  - Accounts for 2nd and 3rd bonds in multiple bonds

Sigma Bonds (σ)

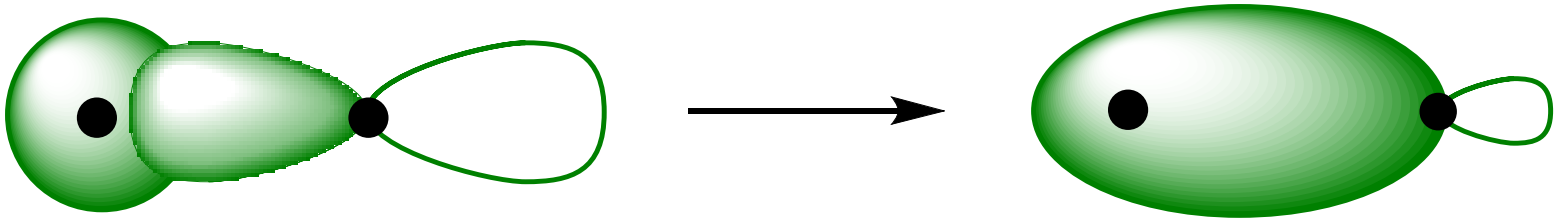
- Head on overlap of orbitals
- Concentrate electron density concentrated most heavily between nuclei of 2 atoms
- Lie along imaginary line joining their nuclei



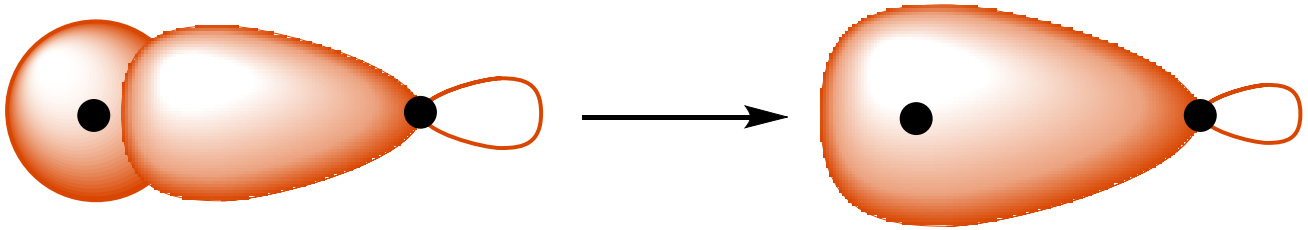
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Other Types of Sigma Bonds (σ)

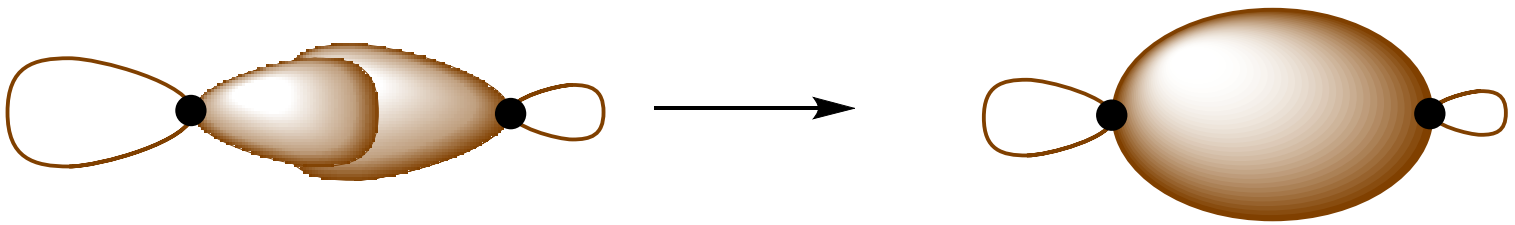
s + *p*



s + *sp*



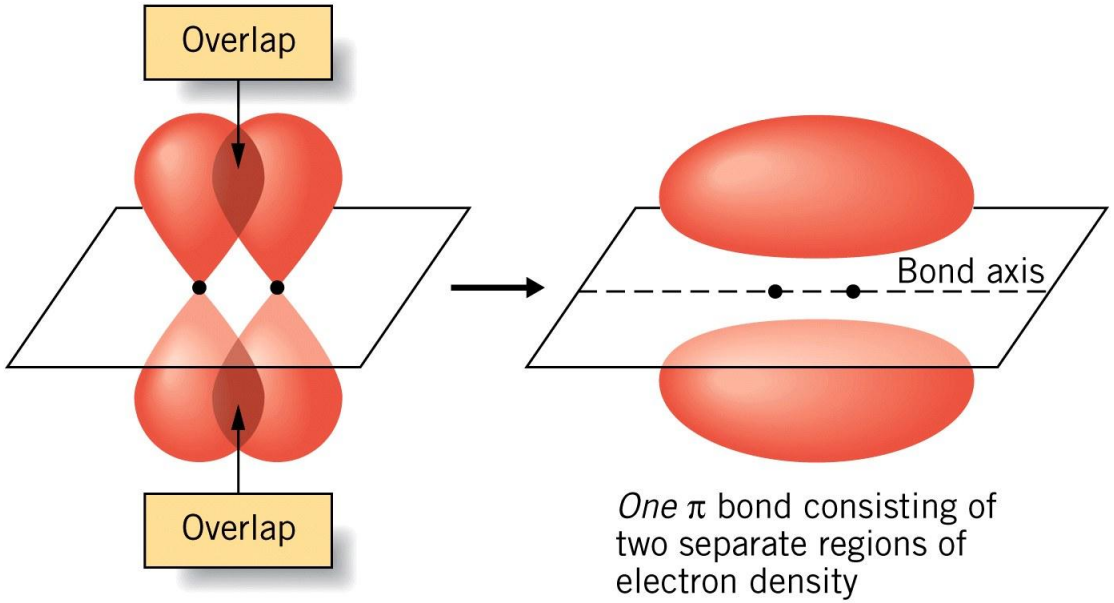
p + *sp*



Pi Bonds (π)

- Sideways overlap of unhybridized ***p*** orbitals
- e^- density divided into 2 regions (lobes)
  - Lie on opposite sides of imaginary line connecting 2 atoms
- e^- density above and below line of σ bond.

- No e^- density along σ bond axis
- π bond consists of both regions
- Both regions = 1 π bond



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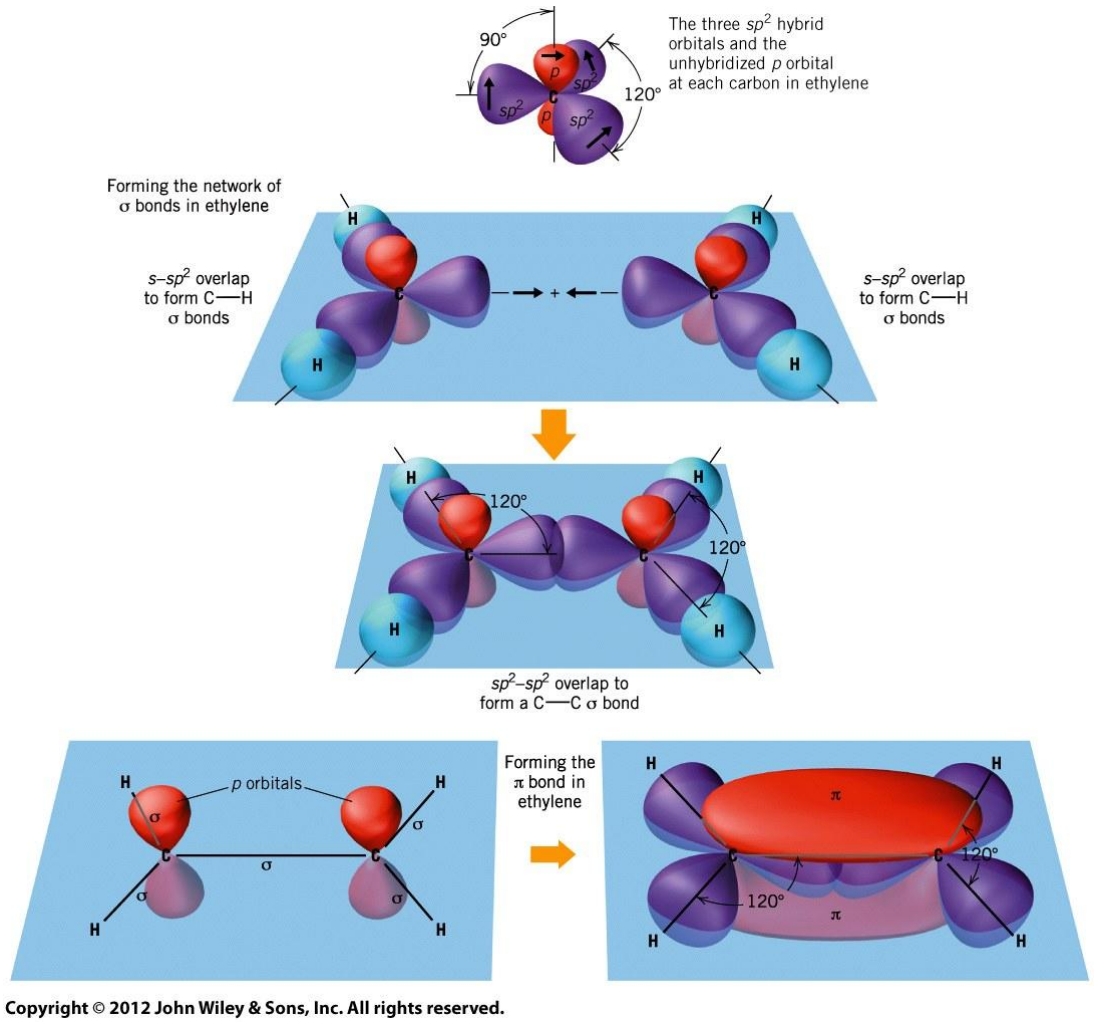
Pi Bonds (π)

- Can never occur alone
  - Must have σ bond
- Can form only if have unhybridized ***p*** orbitals remaining on atoms after form σ bonds
- π bonds allow atoms to form double and triple bonds

Bonding in Ethene (C_2H_4)

- Each **C** is
  - ***sp²*** hybridized (violet)
  - Has 1 unhybridized ***p*** orbital (red)
- C=C double bond is
  - 1 **σ** bond (***sp² – sp²***)
  - 1 **π** bond (***p – p***)

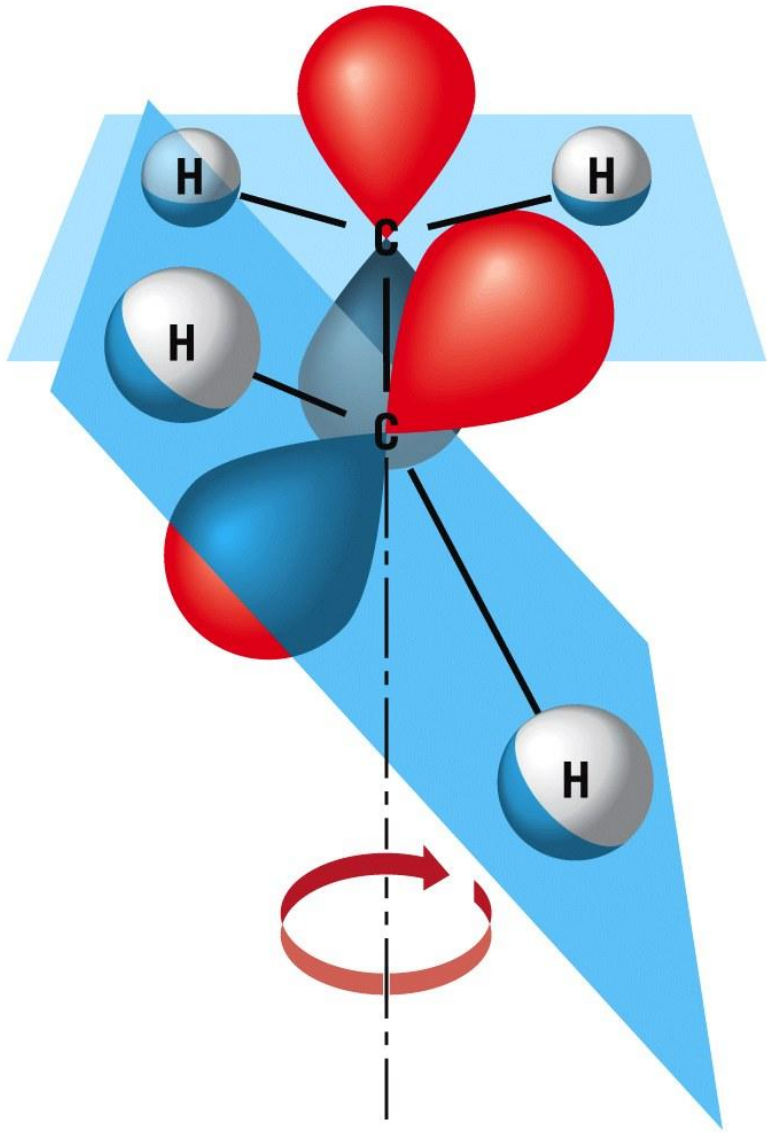
p—*p* overlap to form C—C π bond



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Properties of i Bonds π-Bonds

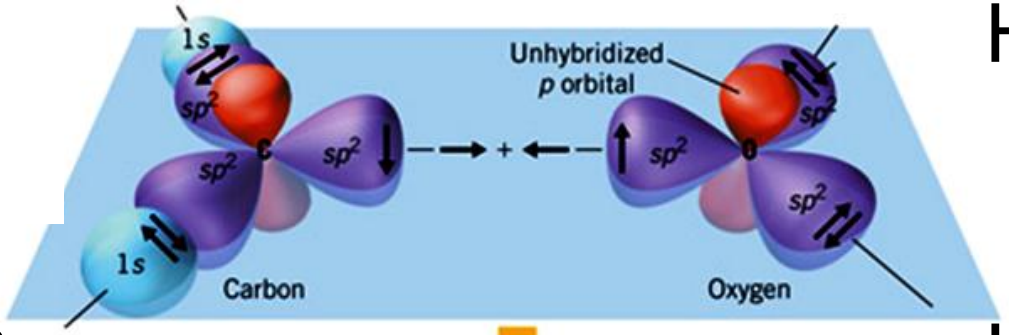
- Can't rotate about double bond
- π bond must first be broken before rotation can occur

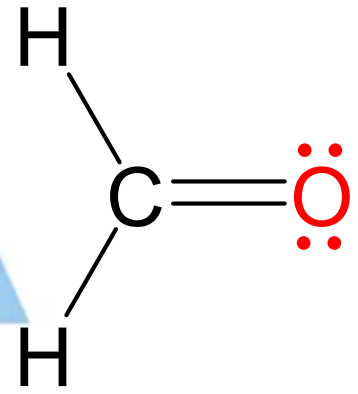


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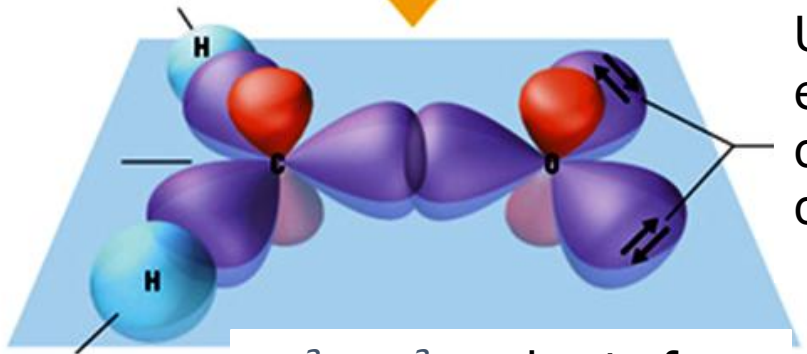
Bonding in Formaldehyde

- **C** and **O** each
  - ***sp²*** hybridized (violet)
  - Has 1 unhybridized ***p*** orbital (red)



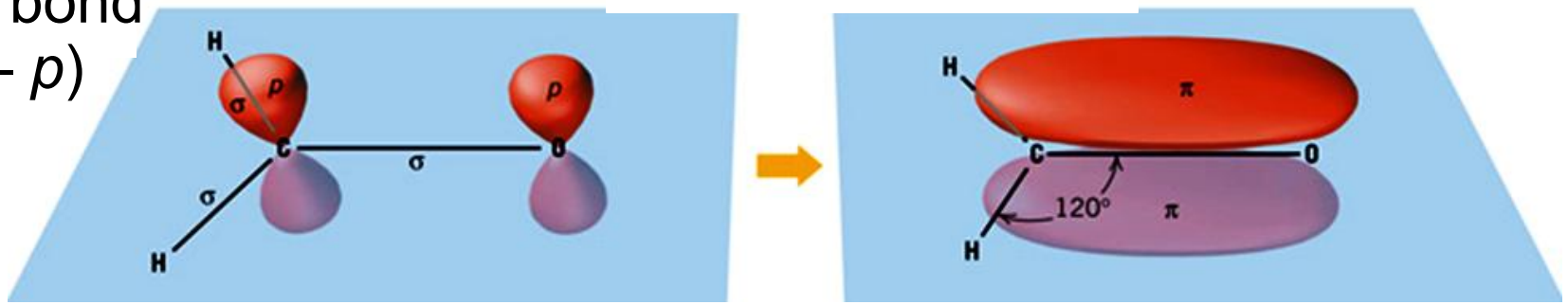


- C=O double bond is
  - 1 σ bond ($sp^2 – sp^2$)
  - 1 π bond ($p – p$)



Unshared pairs of electrons on oxygen in sp^2 orbitals

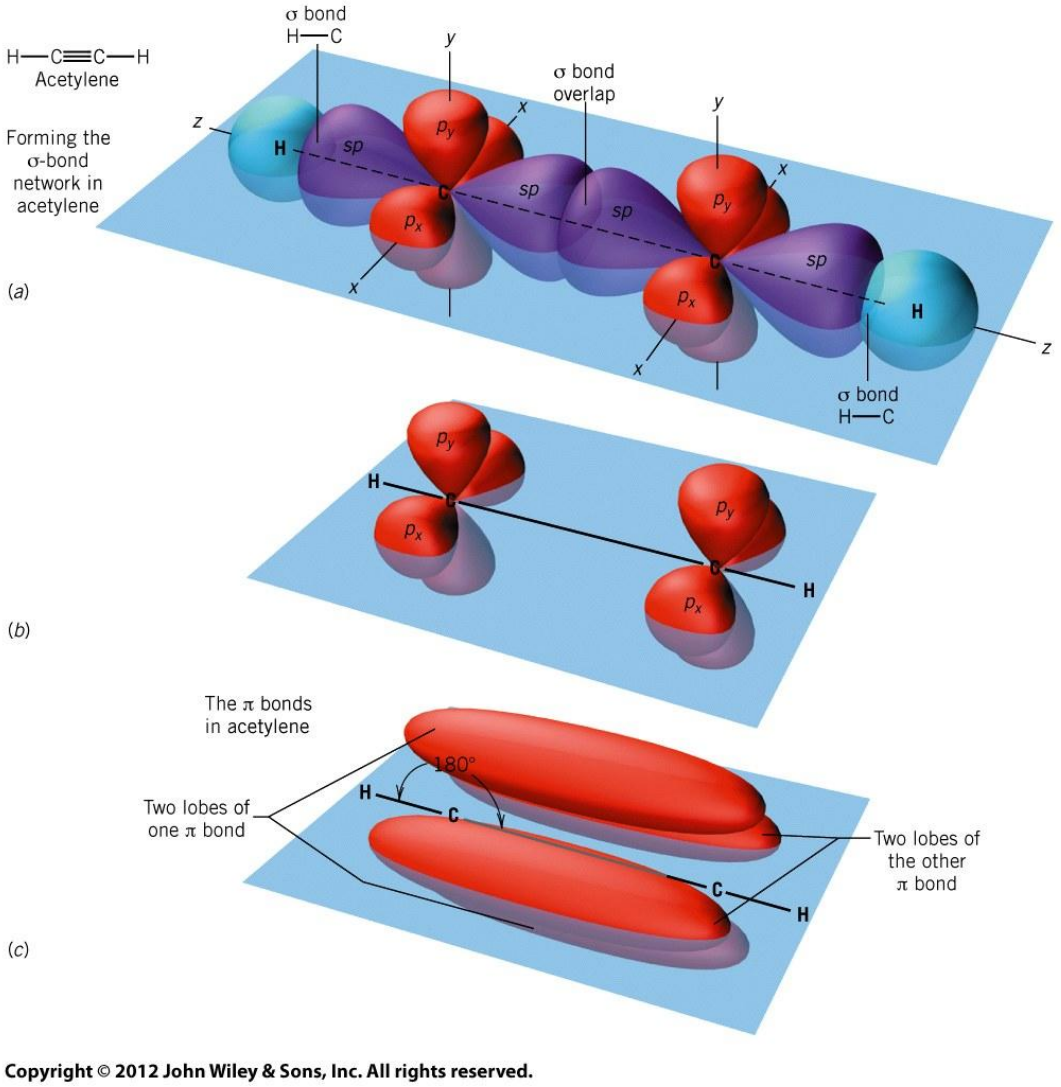
sp^2—sp^2 overlap to form C—O σ bond



Bonding in Ethyne or Acetylene

H—C≡C—H

- Each C
  - is *sp* hybridized (violet)
  - Has 2 unhybridized *p* orbitals, p_x and p_y (red)
- C≡C triple bond
  - 1 σ bond
    - *sp* – *sp*
  - 2 π bonds
    - $p_x - p_x$
    - $p_y - p_y$



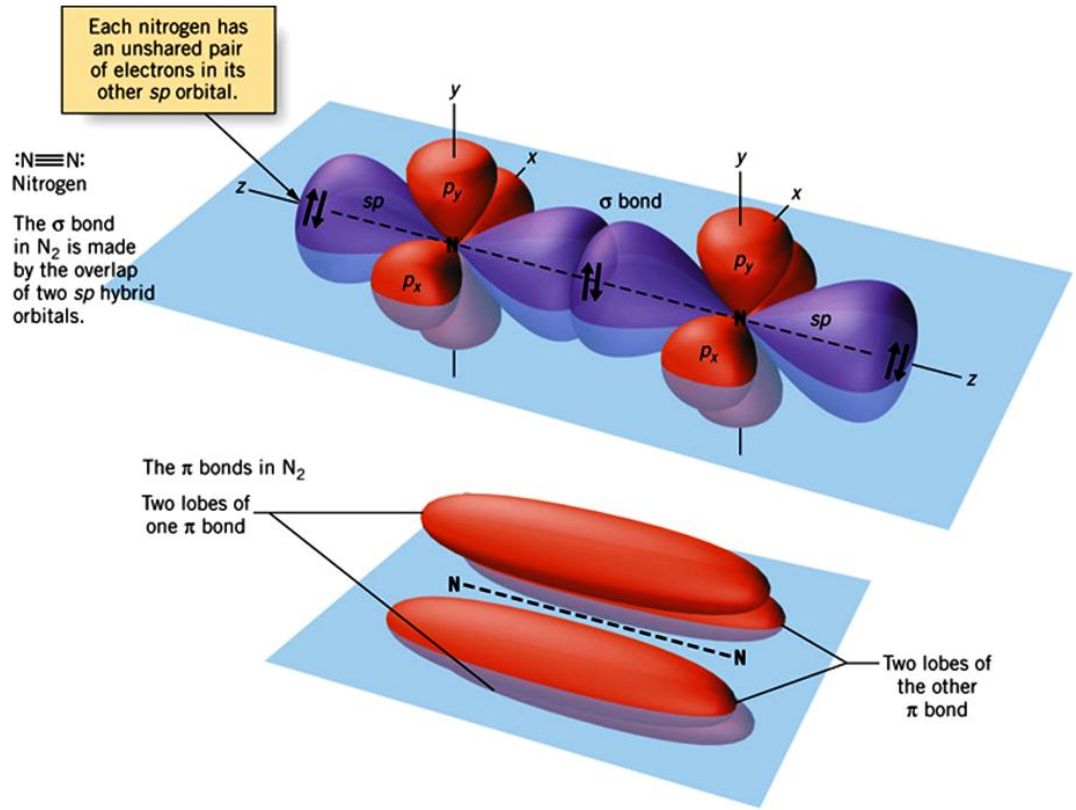
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Bonding in N_2

N≡N triple bond

- 1 σ bond
  - *sp* – *sp*
- 2 π bonds
  - $p_x - p_x$
  - $p_y - p_y$

- Each N
  - *sp* hybridized (violet)
  - Has 2 unhybridized *p* orbitals, p_x and p_y (red)



Your Turn!

How many σ and π bonds are there in $CH_2CHCHCH_2$ respectively, and what is the hybridization around the carbon atoms?

A. 7, 1, sp

B. 8, 2, sp^3

C. 9, 2, sp^2

D. 9, 3, sp^2

E. 8, 2, sp

Molecular Orbital Theory

1. Uses Molecular Bonding Orbital which results from interaction of Atomic orbitals (AOs) of bonded atoms
2. Molecular orbitals (MOs) are associated with entire molecule as opposed to 1 atom
3. Allows us to accurately predict magnetic properties of molecules
4. Shapes of MOs determined by combining e^- waves of AOs

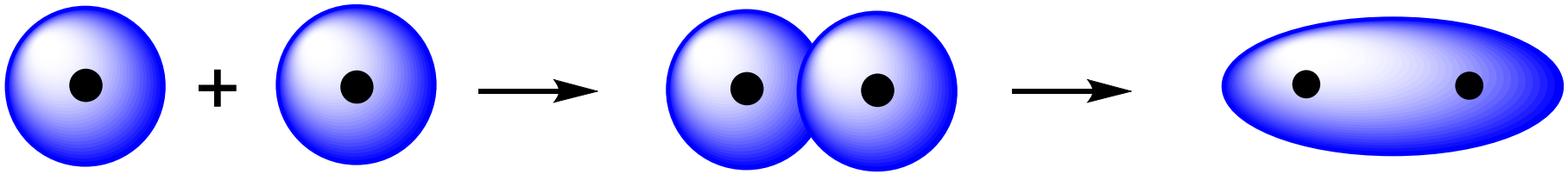
Molecular Orbital Theory – H_2

- Let's go back to H_2
- What forces are at work?
  - e^- – nuclear attractions
  - e^- – e^- repulsions
  - nuclear – nuclear repulsions
- Net attractions

Bonding Molecular Orbitals

- Come from various combinations of AOs
- For H_2, two **1*s*** orbitals, so two MOs
- Like hybridization but now AOs come from different atoms

$1s_A + 1s_B$ Overlapping ⟶ Bonding MO



Constructive interference of waves

σ Bonding Molecular Orbitals

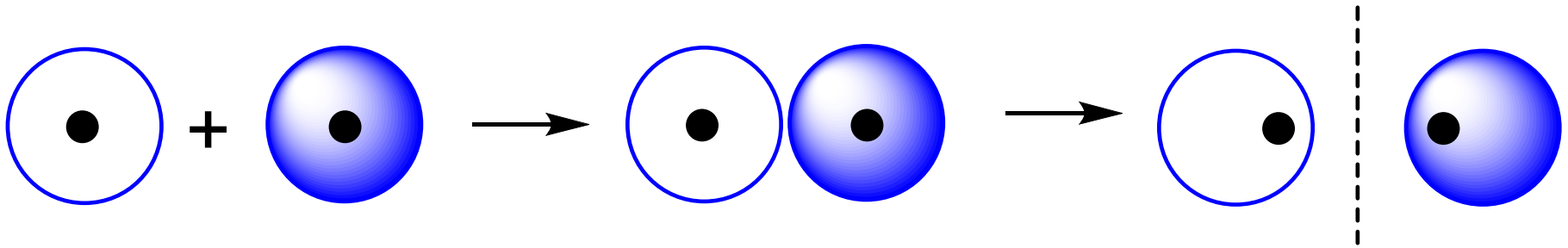
- **σ MO**
- Bonding MO
- Lower in energy than $\mathbf{1s_A}$ or $\mathbf{1s_B}$
- e^- density (Ψ^2) greatest where AOs overlapped
- e^- spends most of its time between nuclei
- Build up of e^- density
- Stabilized by ↓ P.E

H_2 : σ Bonding Molecular Orbitals

- When you put e^-'s into molecular orbital
  - ***Pauli Exclusion Principle*** still applies
  - 2 e^- per MO
  - Must have spins paired
- Energy of molecular orbital lower than energy of parent atomic orbitals
  - Motivation for forming bond
- Covalent bond
  - = shared pair of e^-'s
  - = atomic orbital overlap gives molecular orbital

Antibonding Molecular Orbitals

- number of atomic orbitals in must equal number of molecular orbitals out
- Other possible combination of two **1*s*** orbitals: $1s_A - 1s_B$



Destructive interference of waves

σ^* Antibonding Molecular Orbitals

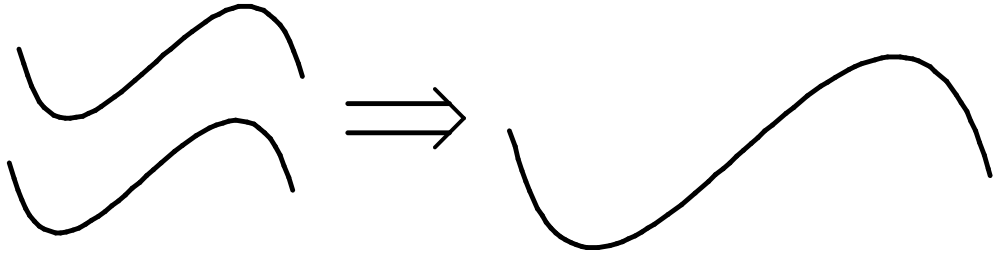
- e^- density (Ψ^2) subtracts between nuclei
  - Gives rise to node
- Cancellation of e^- waves
- **σ^* MO**
- This orbital keeps e^-'s away from region between nuclei
- $\therefore$ Nuclear – nuclear repulsions tend to push atoms apart
- Antibonding molecular orbital
- Higher in energy than $\mathbf{1s_A}$ or $\mathbf{1s_B}$
- Destabilized by $\uparrow$ **P.E.**

Molecular Orbitals

- When 2 AOs combine - forms 2 MOs
- In general **number of AOs in = number of MOs out**

Bonding Molecular Orbitals

- Lower in energy than atomic orbitals from which formed
- Greater stability
  - Wavefunctions constructively interfere

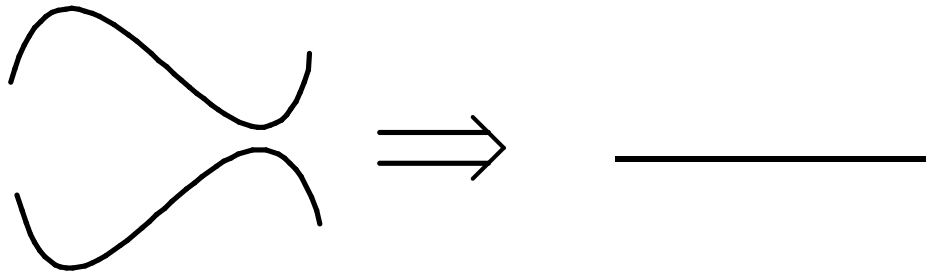


  - Bonding electrons stabilize molecule

Molecular Orbitals

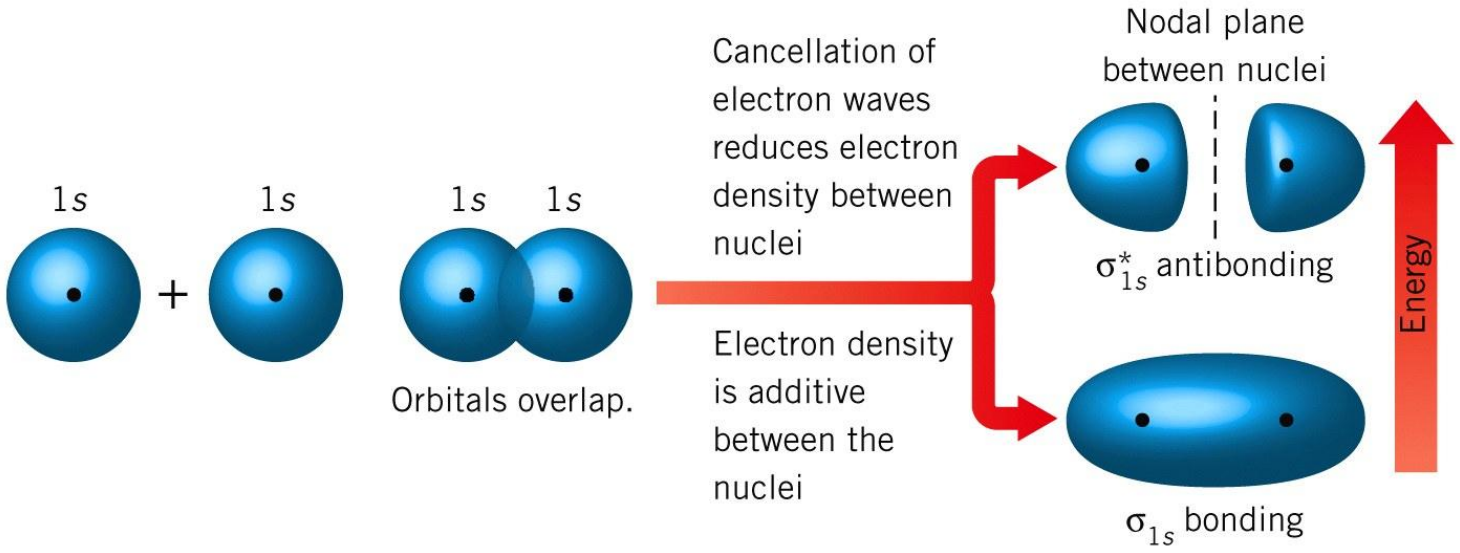
Antibonding Molecular Orbitals

- Higher in energy than AOs from which formed
- Lower stability
  - wavefunctions destructively interfere



- Antibonding electrons destabilize molecule
- Tend to cancel effects of bonding electrons

Summary of MO from 1s AO

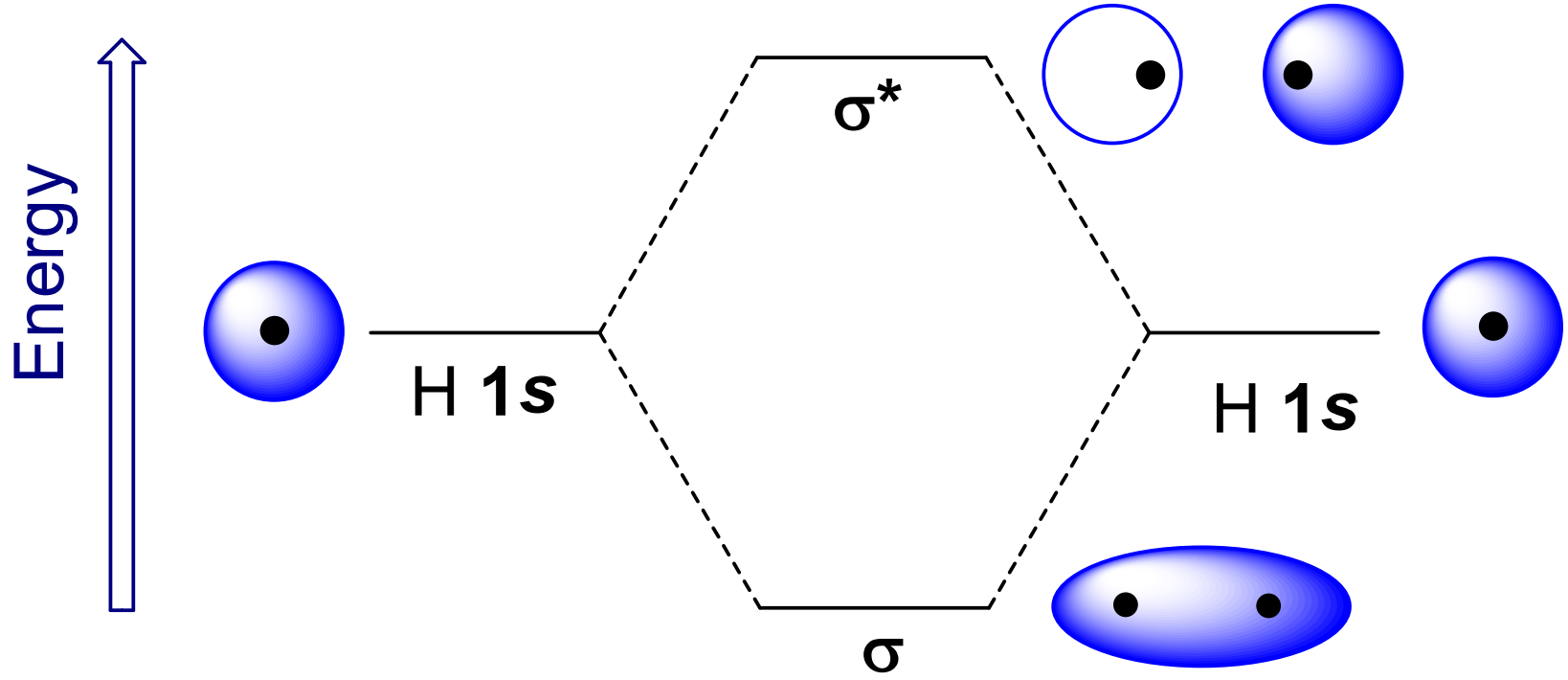


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- **Bonding MO**
  - Electron density builds up between nuclei
  - Electrons in bonding MOs tend to stabilize molecule
- **Antibonding MO**
  - Cancellation of electron waves reduces electron density between nuclei
  - Electrons in antibonding MOs tend to destabilize molecule

Molecular Energy Level Diagram

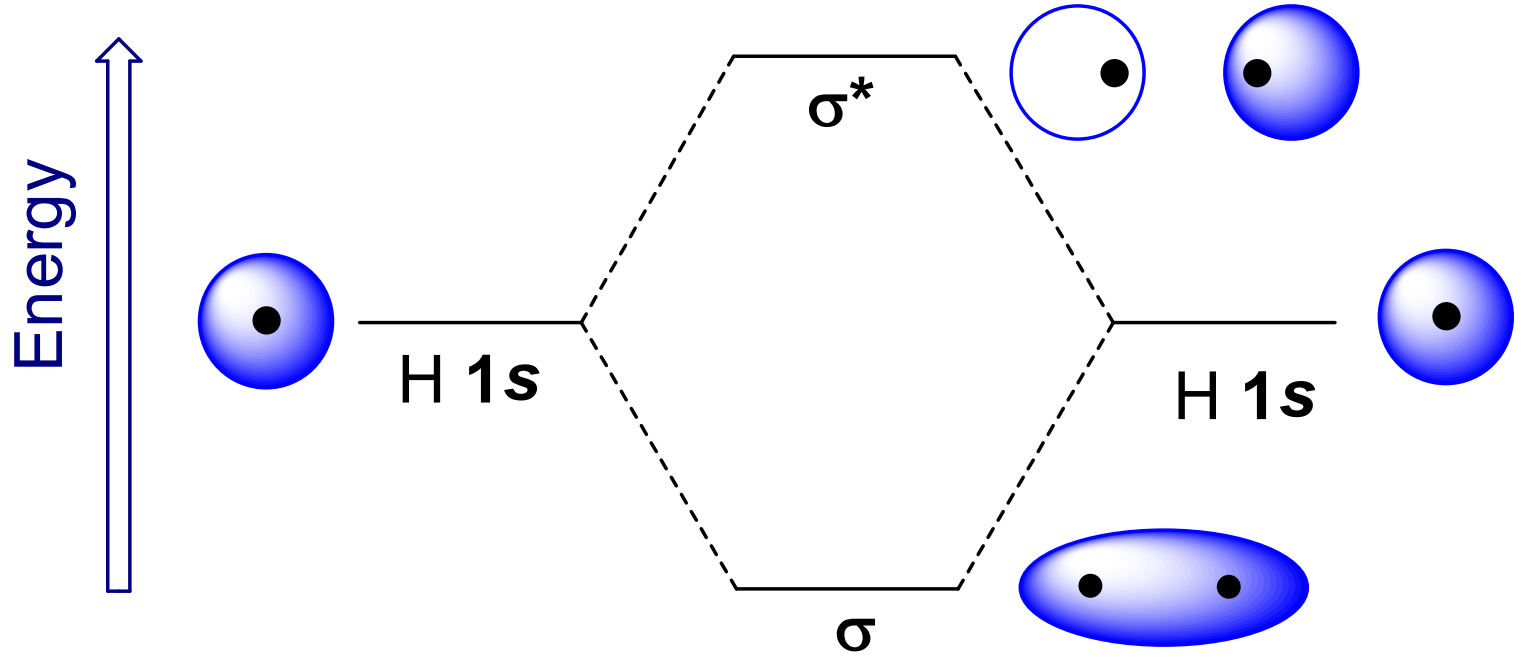
- Very useful pictures
- Can be used to account for existence of certain molecules and nonexistence of others
- Energy diagram for the two MOs formed by two **1*s*** orbitals



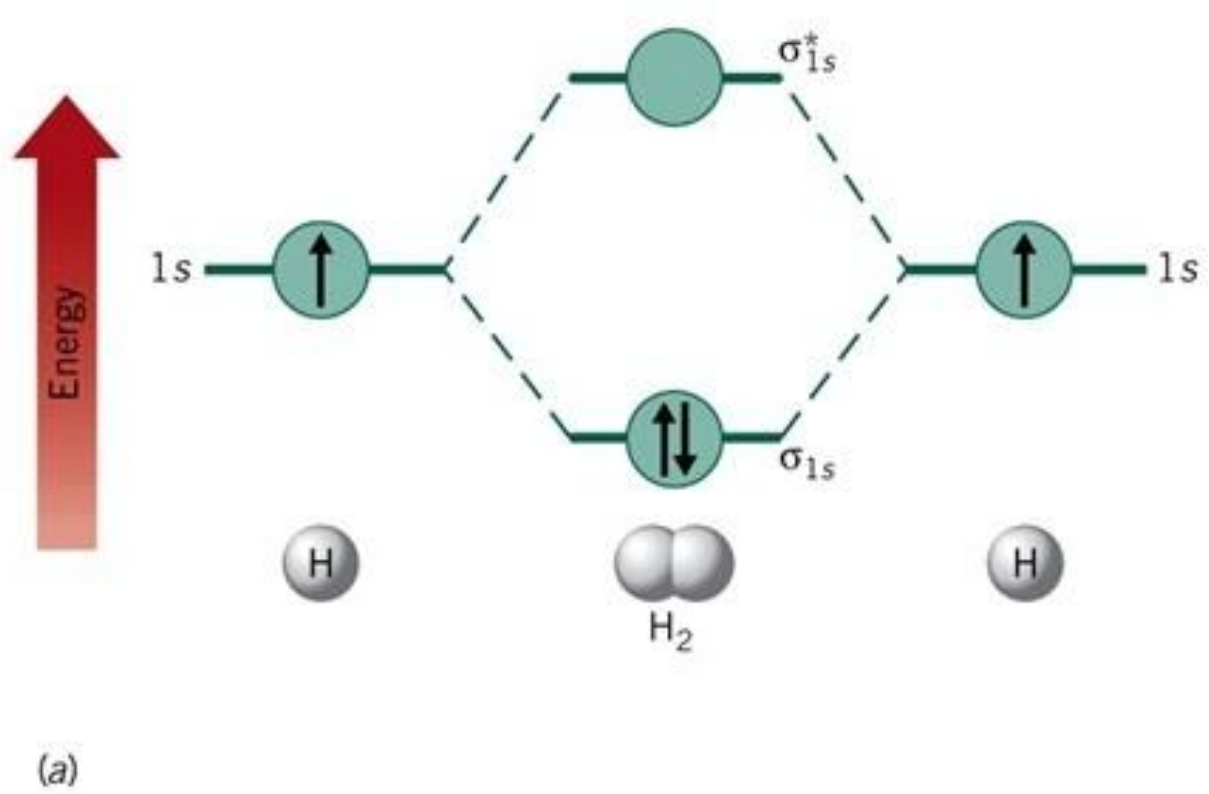
Molecular Energy Level Diagram

When you form MO s

- Bonding MO ***stabilized*** by same amount that antibonding MO ***destabilized***
- σ lowered by E_B in energy over $1s_A$ or $1s_B$
- σ* raised by E_B in energy over $1s_A$ or $1s_B$



Molecular Orbital Energy Diagram for H_2



(a)

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- H_2 is very stable molecule
- Has e^- configuration σ_{1s}^2

Rules for Filling Molecular Orbital Energy Diagrams

1. Electrons fill lowest-energy orbitals that are available
   - **Aufbau Principle applies**
2. No more than 2 electrons, with spin paired, can occupy any orbital
   - **Pauli Exclusion Principle applies**
3. Electrons spread out as much as possible, with spins unpaired, over orbitals of same energy
   - **Hund's Rules apply**

Bond Order

- Measure of number of electron pairs shared between 2 atoms

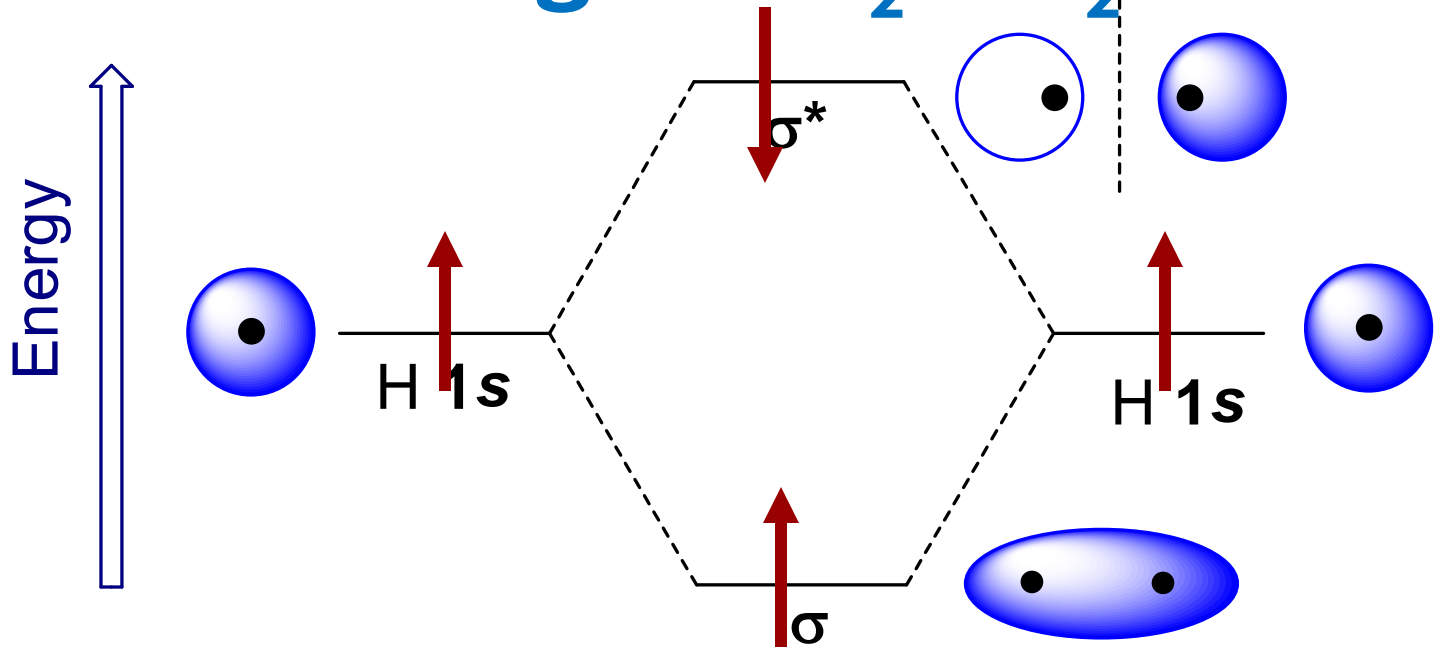
Bond Order = ½(#bonding e^- – #antibonding e^-)

- H_2 has B.O. = 1

What happens when we shine light on H_2?

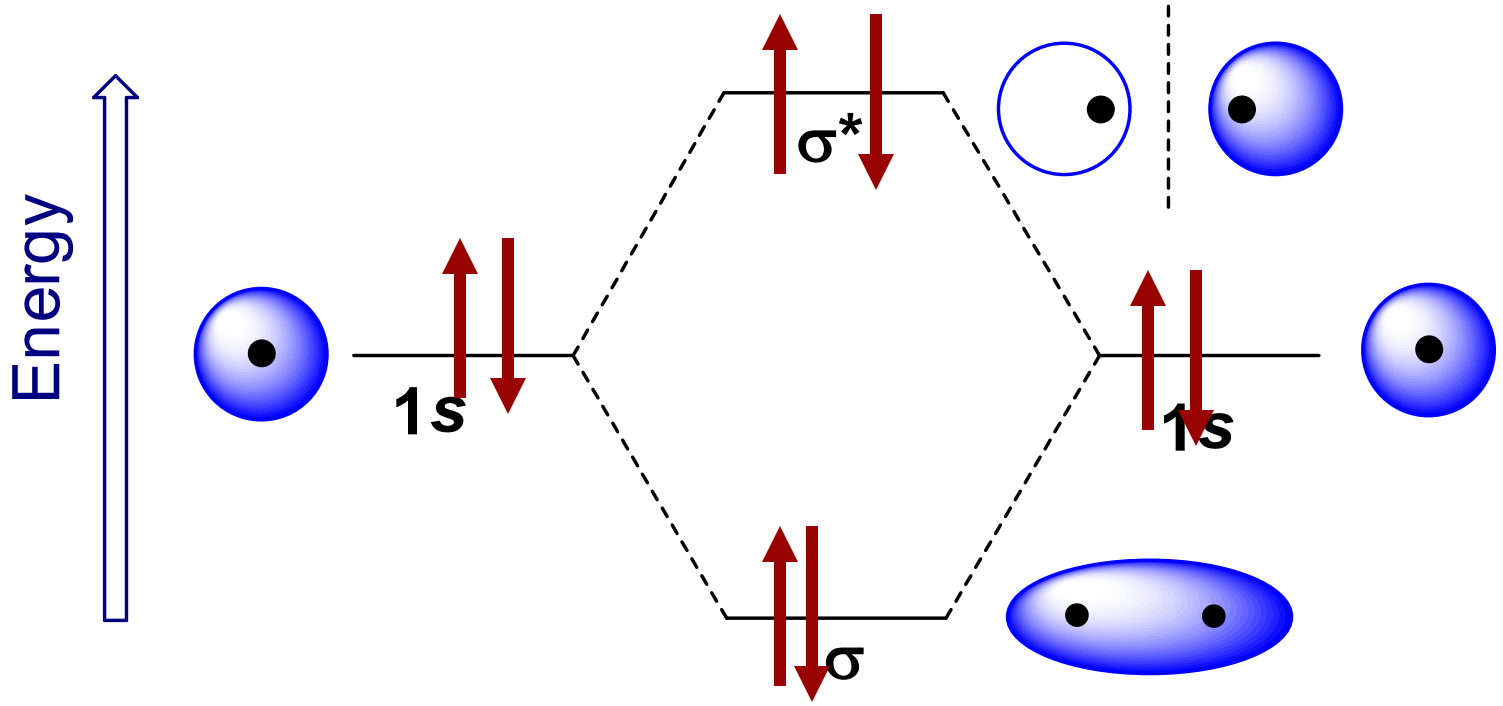
- Makes e^- jump from σ to σ^* MO

Excited State of Molecular Orbital Energy Level Diagram $H_2 = H_2$ *



- Get excited state of H_2 molecule!
- Molecular electron configuration = $\sigma^1(\sigma^*)^1$
- Now **no** benefit in bonding
- B.O. = 0
- So molecule dissociates

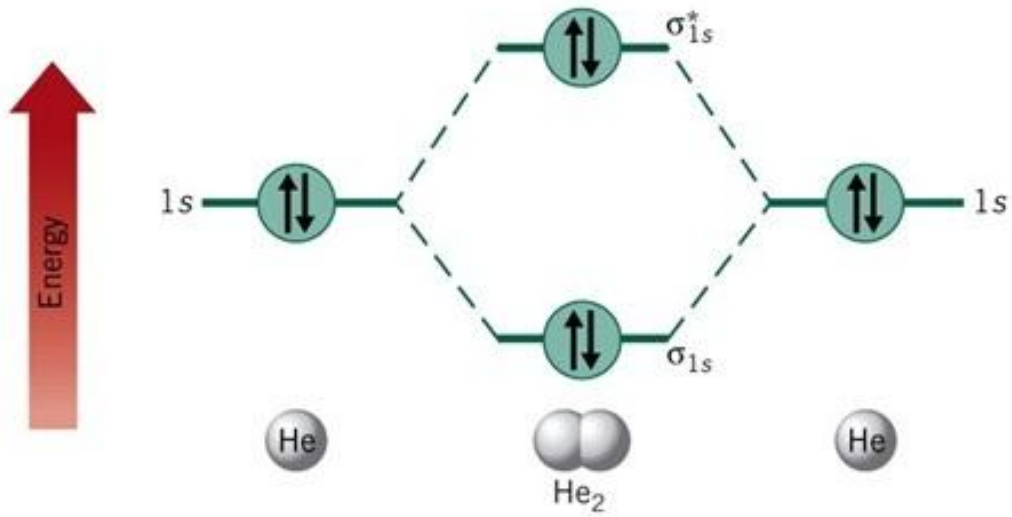
MO Energy Diagram for He_2



- Same picture can be used for predicting if He_2 is stable molecule
- Now 2 valence e^-s for each atom
- 2 e^-s in **σ** and 2 e^-s in **σ*** MO

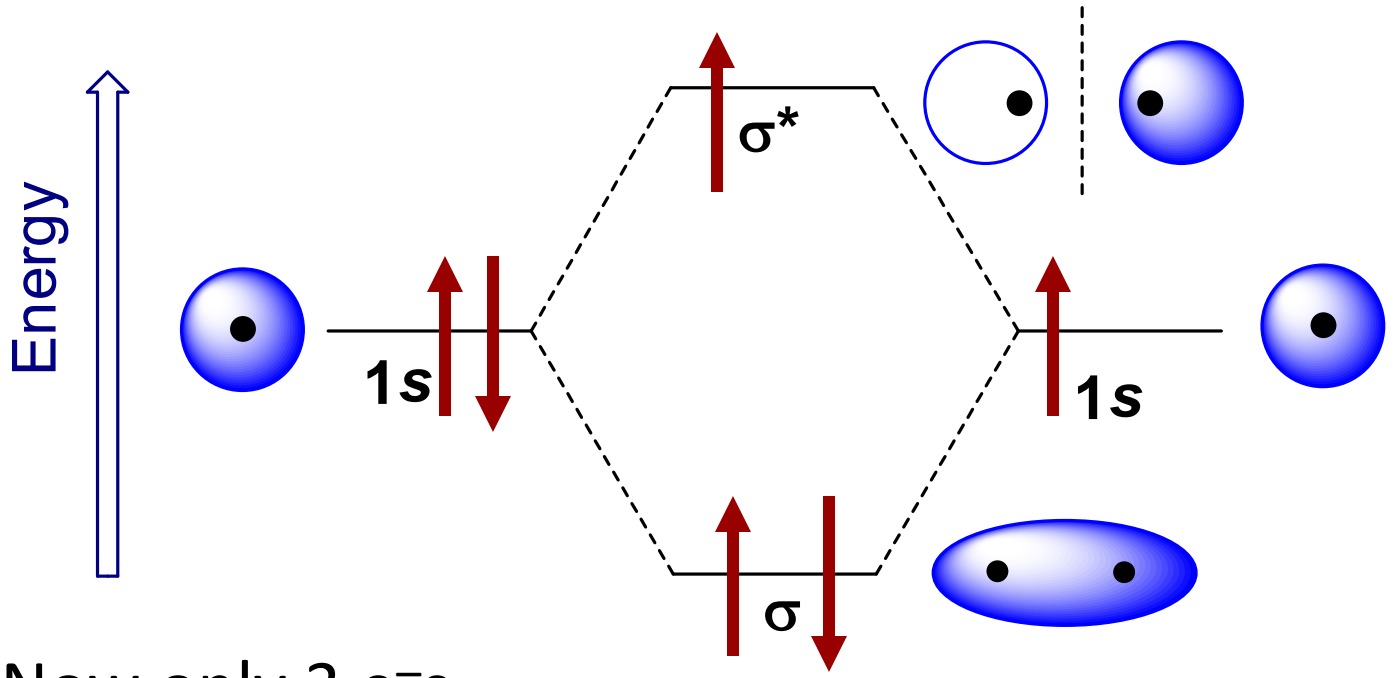
Molecular Orbital Energy Level Diagram for He_2

- 4 e^-s, so both σ and σ^* MO are filled
- Electron configuration = $\sigma_{1s}^2 (\sigma^*_{1s})^2$
- B.O. = ½(2 – 2) = 0
- ∴ no net bonding
- He_2 **Not** stable molecule



(b)

What about He_2^+ ?



- Now only 3 e⁻s
- 2 in σ; 1 in σ*
- Bond Order = ½(2 – 1) = ½
- ∴ net bonding
- He_2^+ is stable

Your Turn!

What is the bond order of H_2^- ?

A. 1

B. 0

C. ½

D. 1 ½

MOT Deals with Odd e^- species

- Bond order does NOT have to be whole number

Diatomics of all the second row elements

- Same principle: combine AOs to form bonding and antibonding MOs
- What combinations can we make for 2nd row?

2s

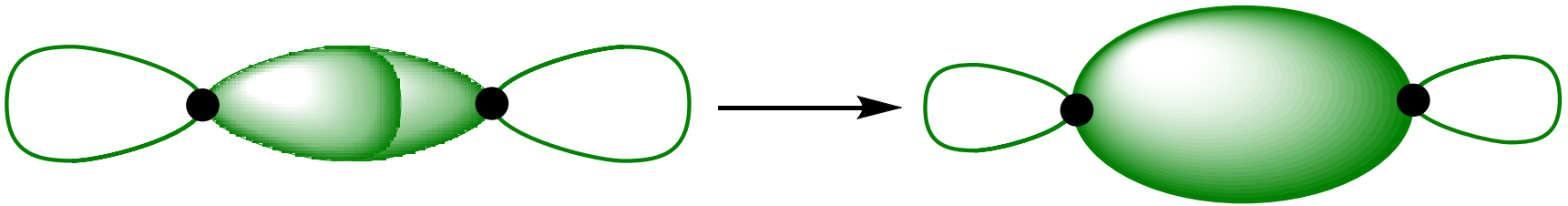
- Give rise to σ and σ^* MOs as before

2p 2 possibilities

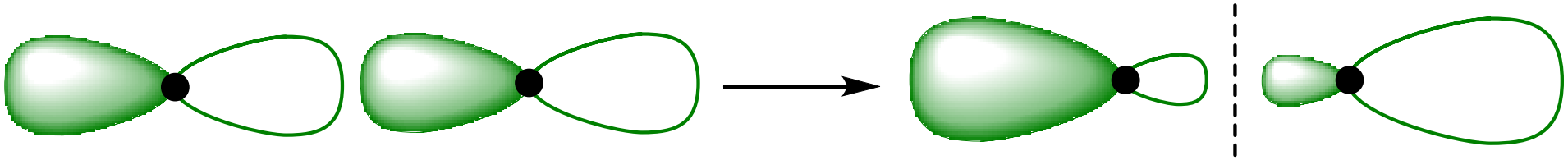
1. Head-on overlap gives rise to σ_{2p} and σ_{2p}^* MO s
2. Sideways overlap gives rise to π_{2p} and π_{2p}^* MO s

2p Orbitals – Head-on Overlap

Gives rise to σ_{2p} and σ_{2p}^* MOs



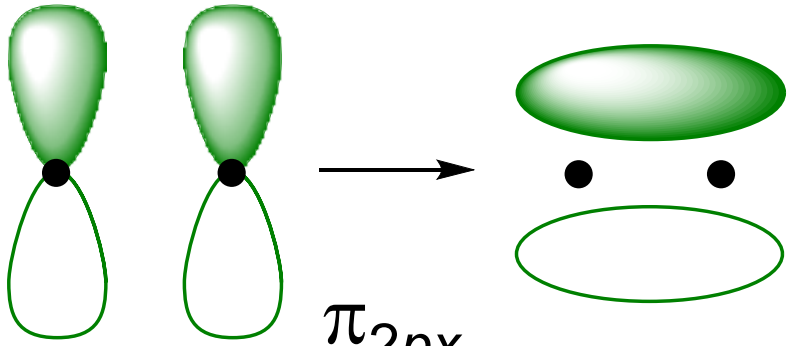
Bonding σ_{2p} MO



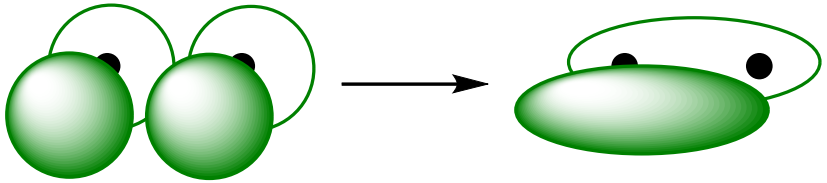
Antibonding σ_{2p}^* MO

π_{2p} Molecular Orbitals

Bonding combinations

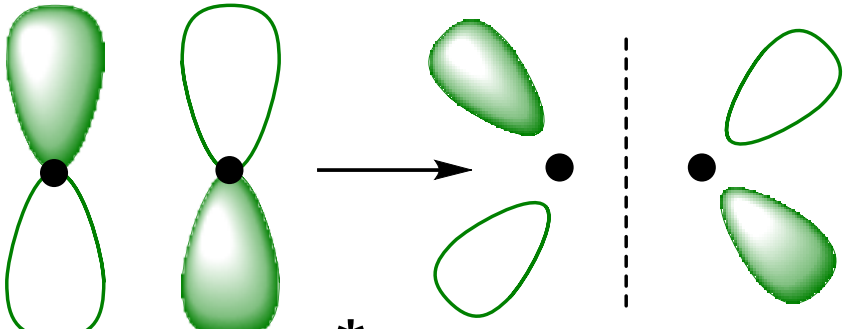


π_{2px}

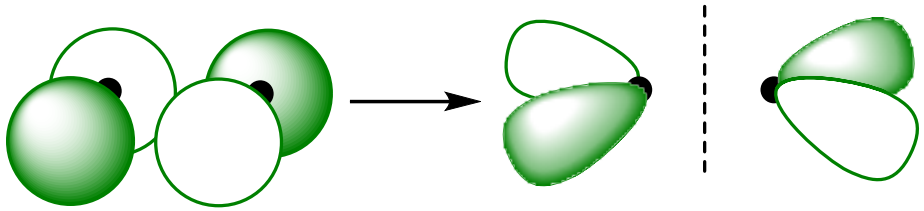


π_{2py}

Antibonding combinations

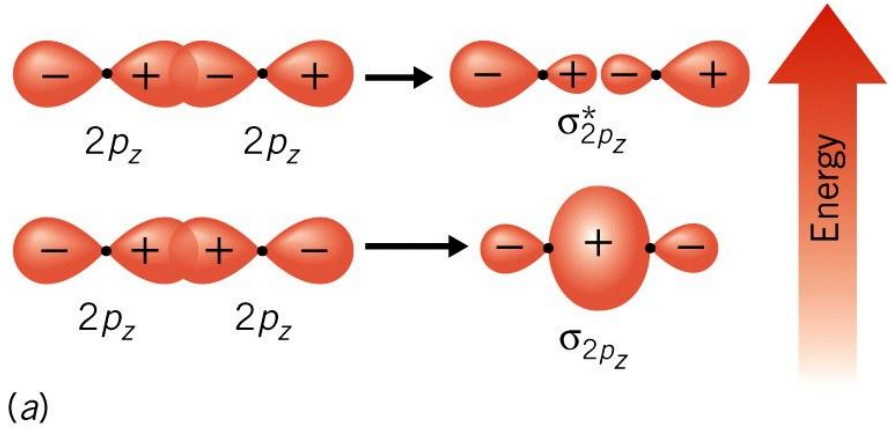


π^*_{2px}

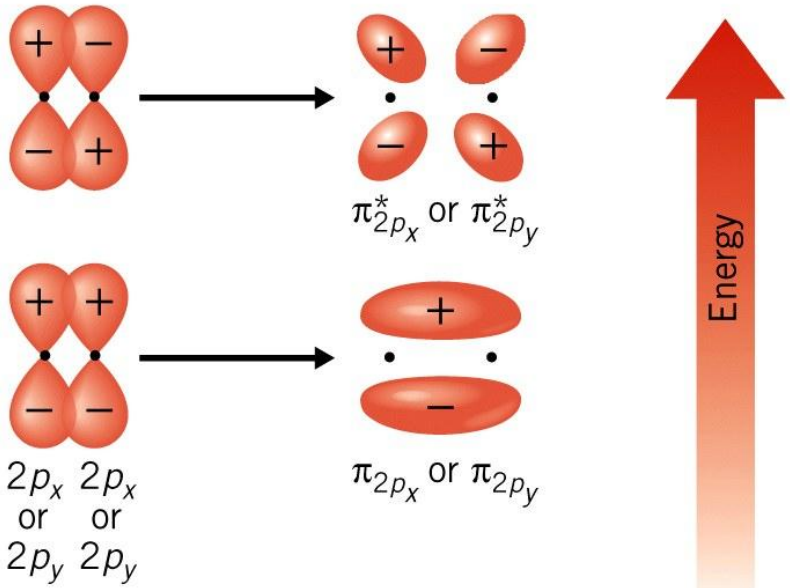


π^*_{2py}

Summary of 2p Orbital Overlaps



(*a*)



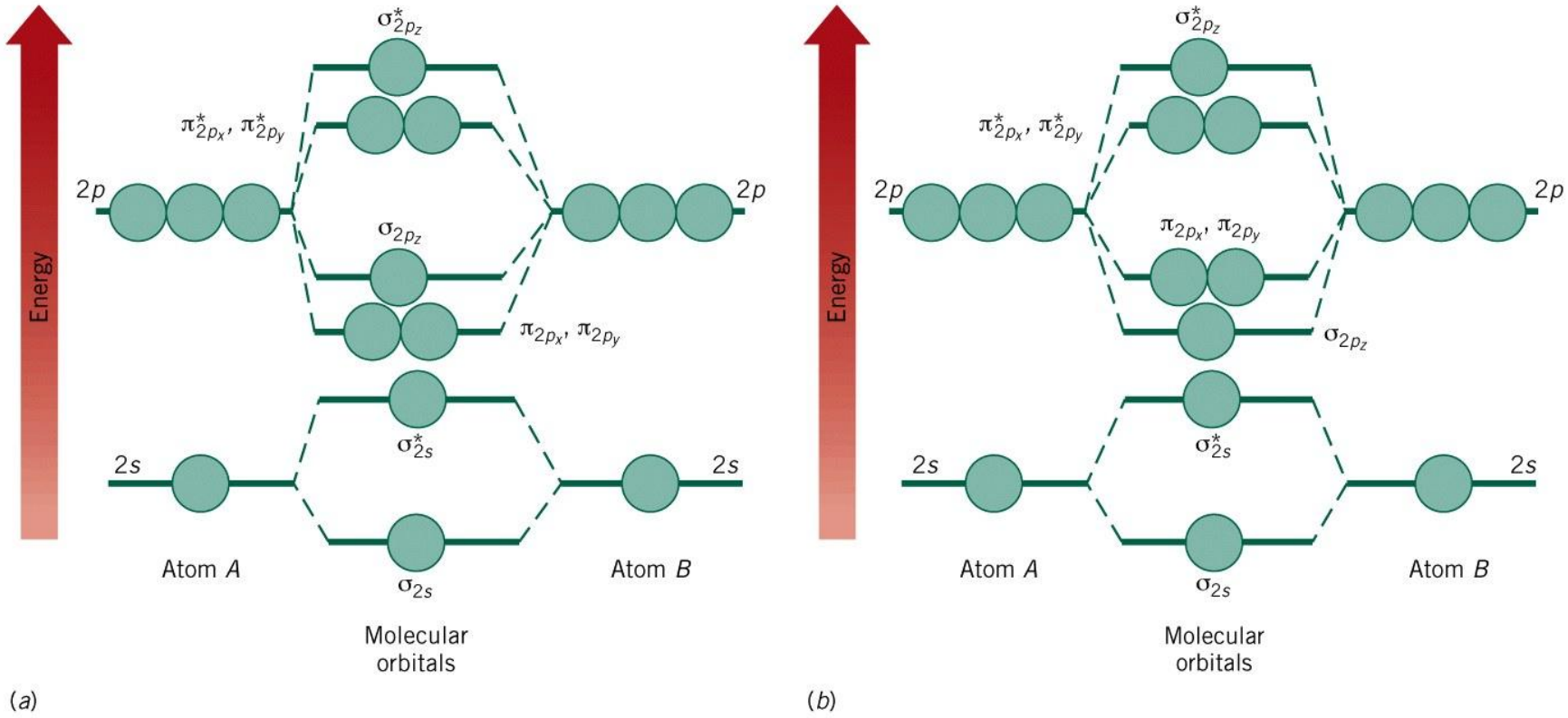
(*b*)

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How are π_{2p} Ordered in Energy?

- **2*s*** always lower in energy than **2*p***
- So energy of σ_{2s} lower than energy of MOs arising from **2*p***.
- Ordering of MOs arising from **2*p*** orbitals is subtle
- Get different situations depending on Atomic Number (Z)
- $Li_2 \longrightarrow N_2$ π_{2p} lower in E than σ_{2p}
- O_2, F_2 and above σ_{2p} lower in E than π_{2p}

MO Energy Diagrams for 2nd Row of Periodic Table



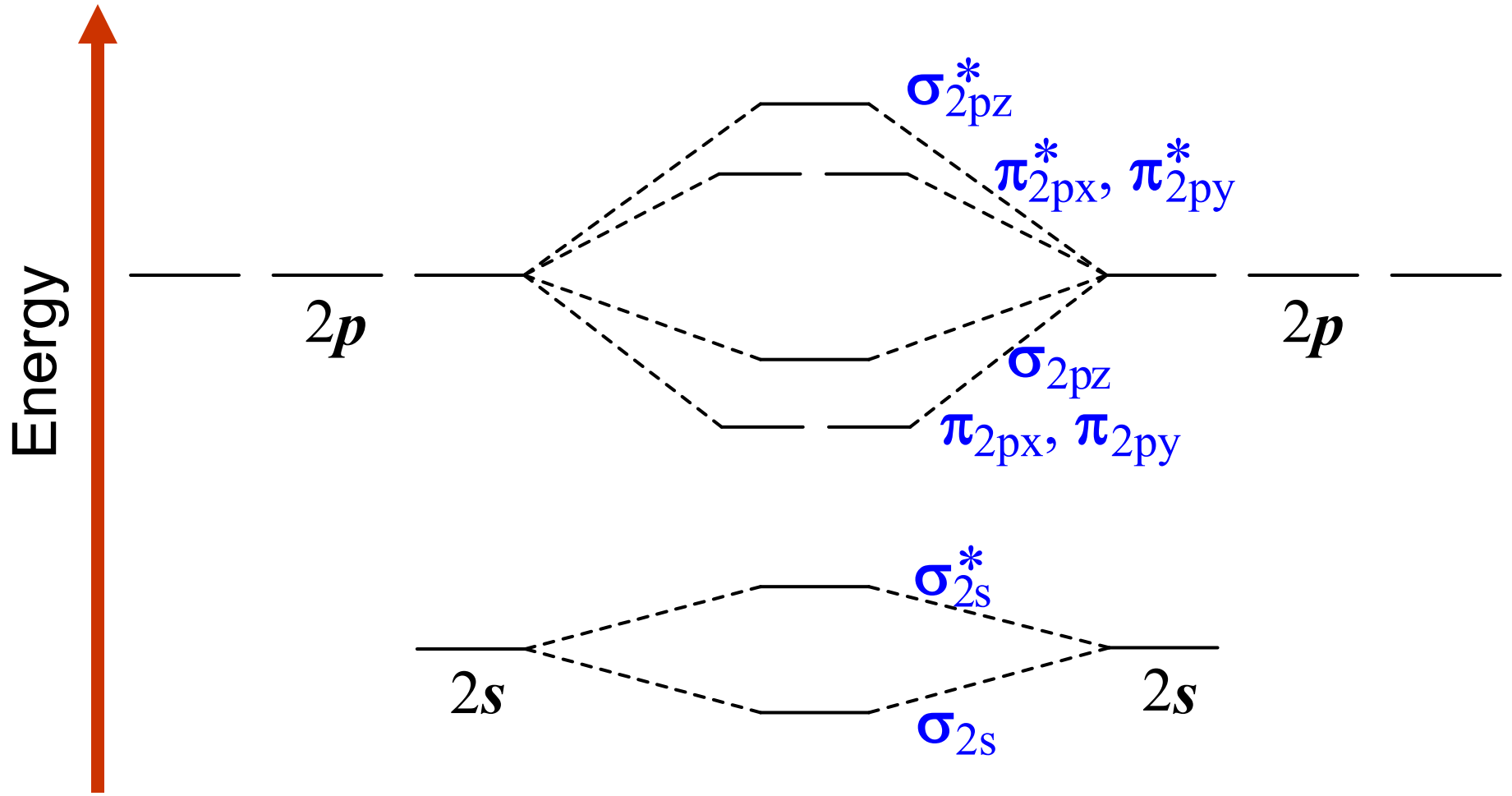
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$Li_2 \rightarrow N_2$

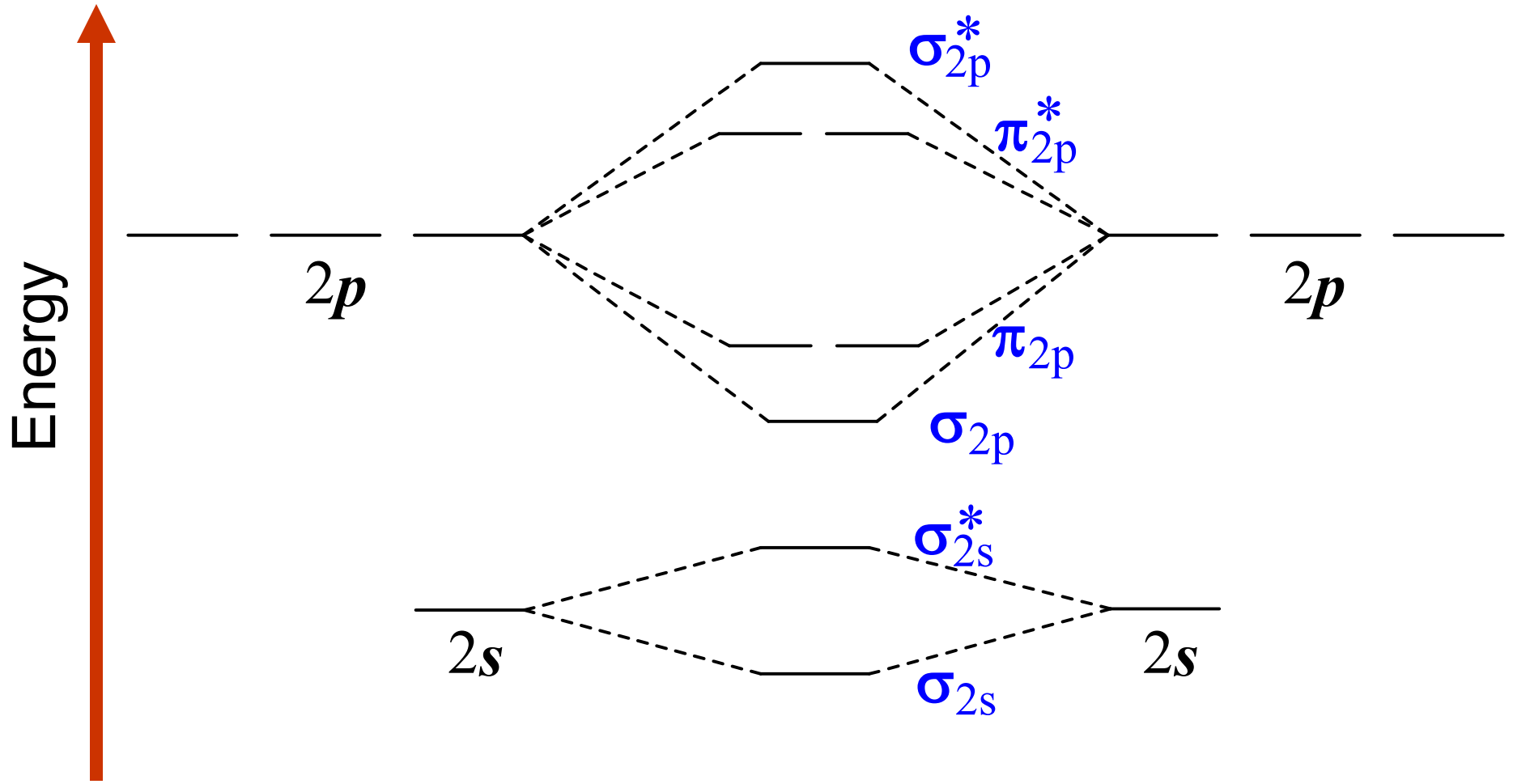
π_{2p} *Lower* in energy than σ_{2p}

O_2, F_2 and Higher σ_{2p} *Lower* in energy than π_{2p}

MO Energy Diagram for $Li_2 \rightarrow N_2$
π_{2p} Lower in Energy than σ_{2p}



MO Energy Diagram for O_2, F_2 and Ne
σ_{2p} Lower in Energy than π_{2p}



Let's Look at Diatomic of 2nd Row Elements

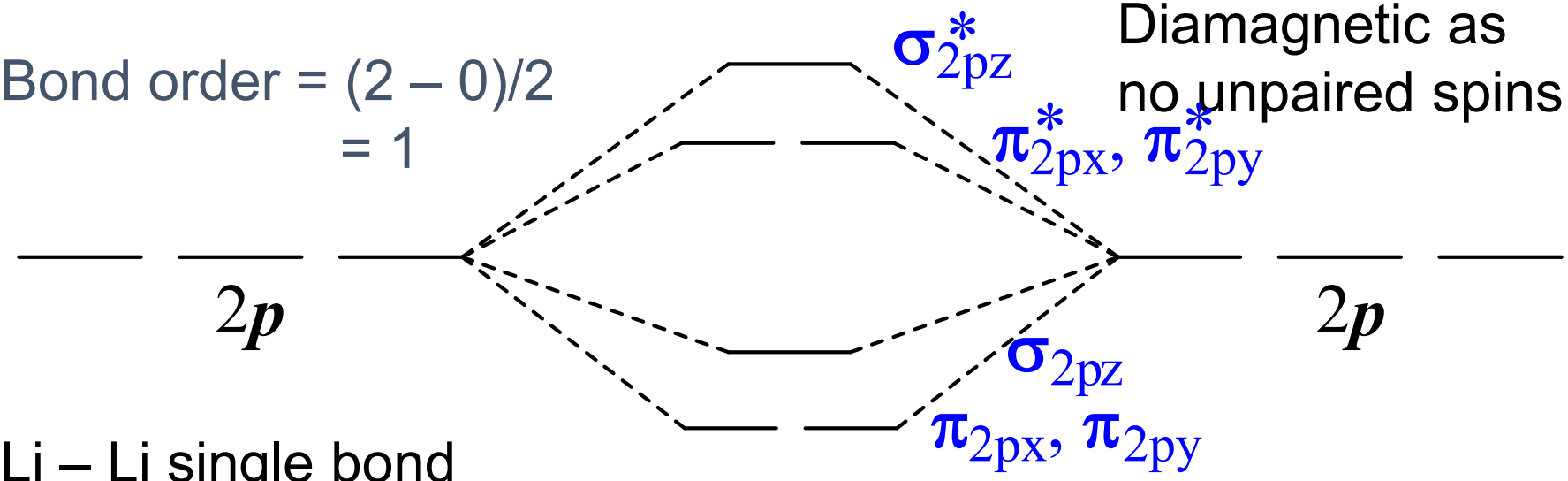
Li_2

- **1s** orbital smaller than **2s**
- From **Li** on overlap of n = 2 orbitals will be much more than **1s**
- Also **1s** orbitals both completely filled
- So both σ and σ^* MOs formed from these are filled
  - Therefore no net bonding
  - Can ignore **1s**
  - Can focus on valence e^-s and orbitals

MO Energy Diagram for Li_2
Lower in Energy than σ_{2p}

Li electron configuration = [He]$2s^1$

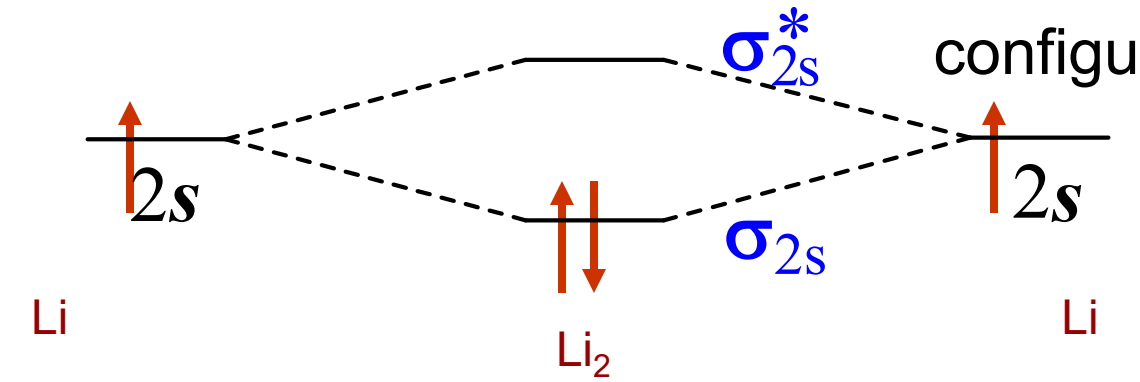
Bond order = (2 – 0)/2
= 1



Diamagnetic as no unpaired spins

Li – Li single bond
∴ stable molecule

Molecular electron configuration = σ_{2s}^2

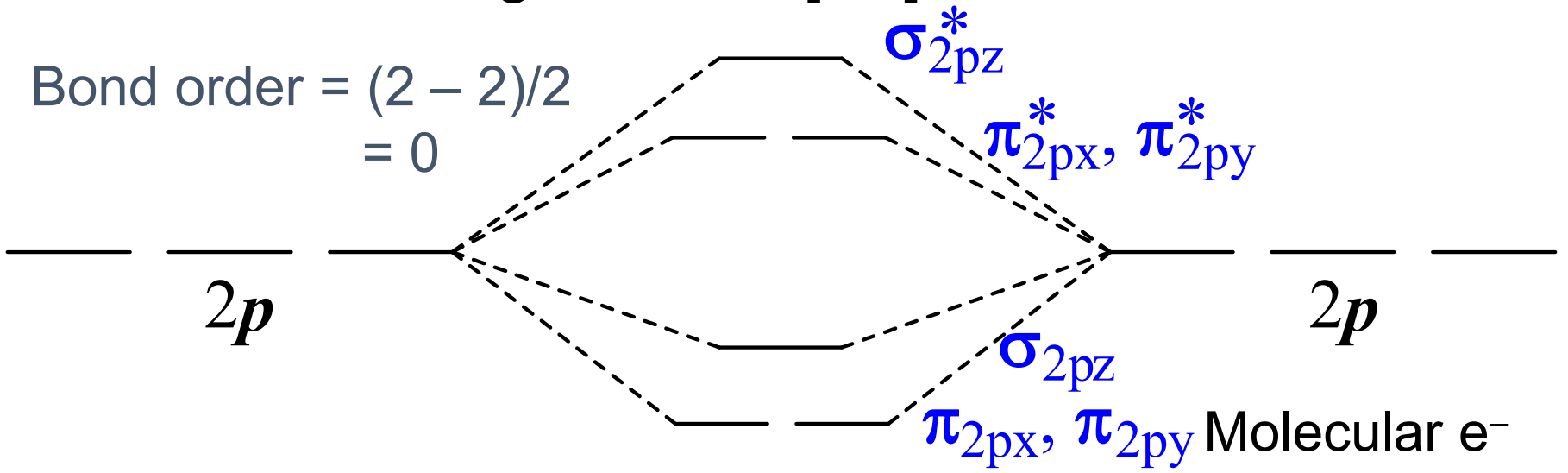


MO Energy Diagram for Be_2

π_{2p} Lower in Energy than σ_{2p}

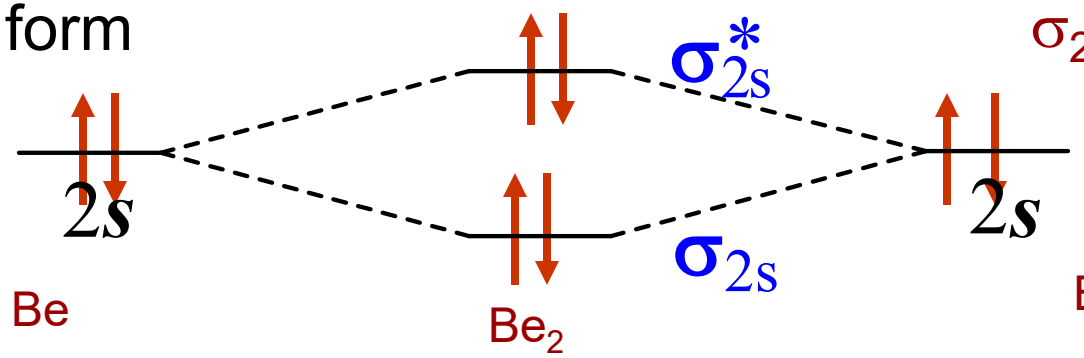
Be electron configuration = [He]$2s^2$

Bond order = (2 – 2)/2
= 0



Be – Be no net bond
∴ does not form

Molecular e^- configuration = $\sigma_{2s}^2 (\sigma^*_{2s})^2$



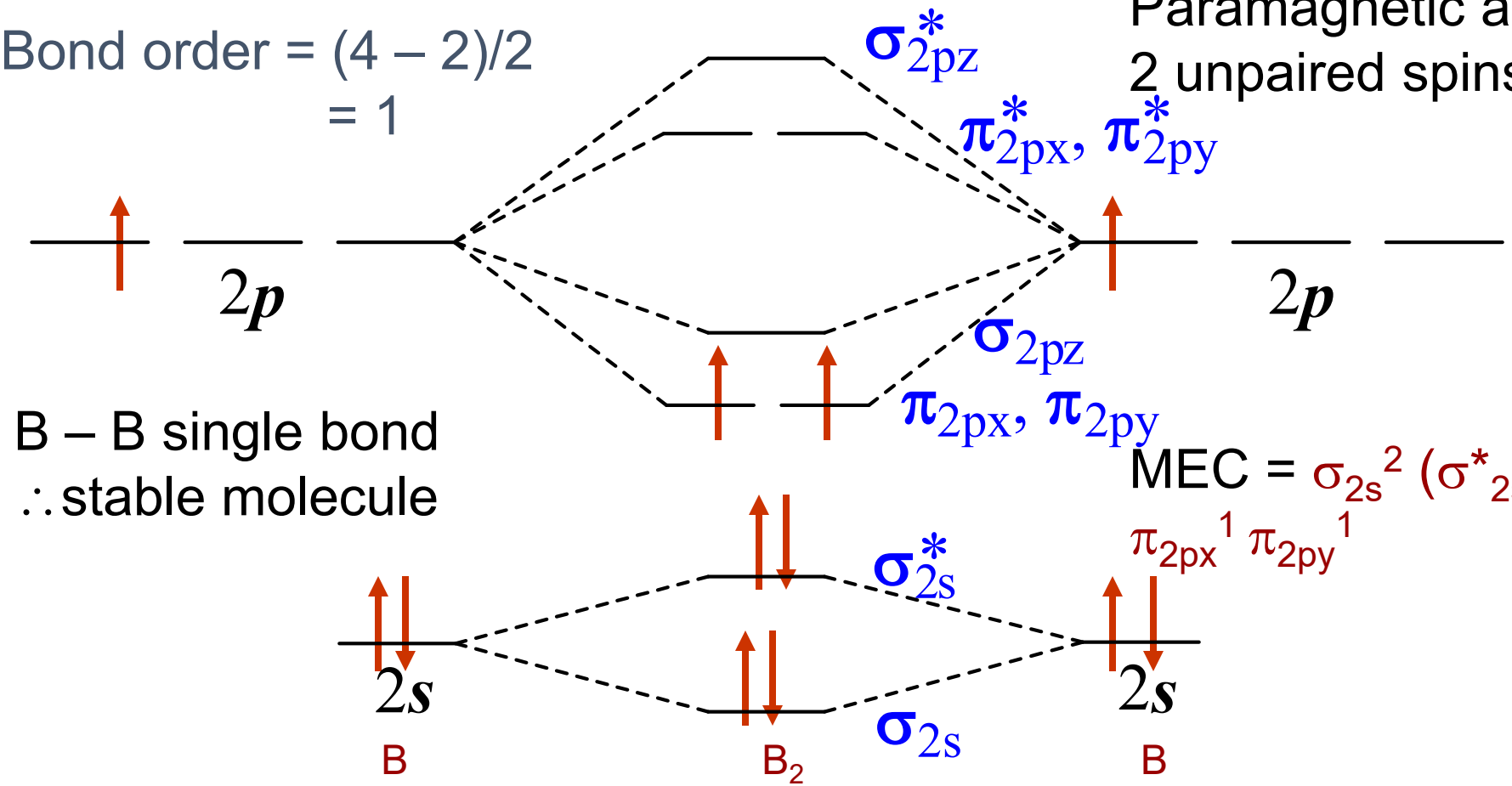
MO Energy Diagram for B_2

π_{2p} *Lower* in Energy than σ_{2p}

B electron configuration = [He]$2s^2 2p^1$

Bond order = (4 – 2)/2
= 1

Paramagnetic as
2 unpaired spins



B – B single bond
∴ stable molecule

MEC = $\sigma_{2s}^2 (\sigma^*_{2s})^2$ $\pi_{2px}^1 \pi_{2py}^1$

This is how we know that π_{2p} is lower in energy than σ_{2p}

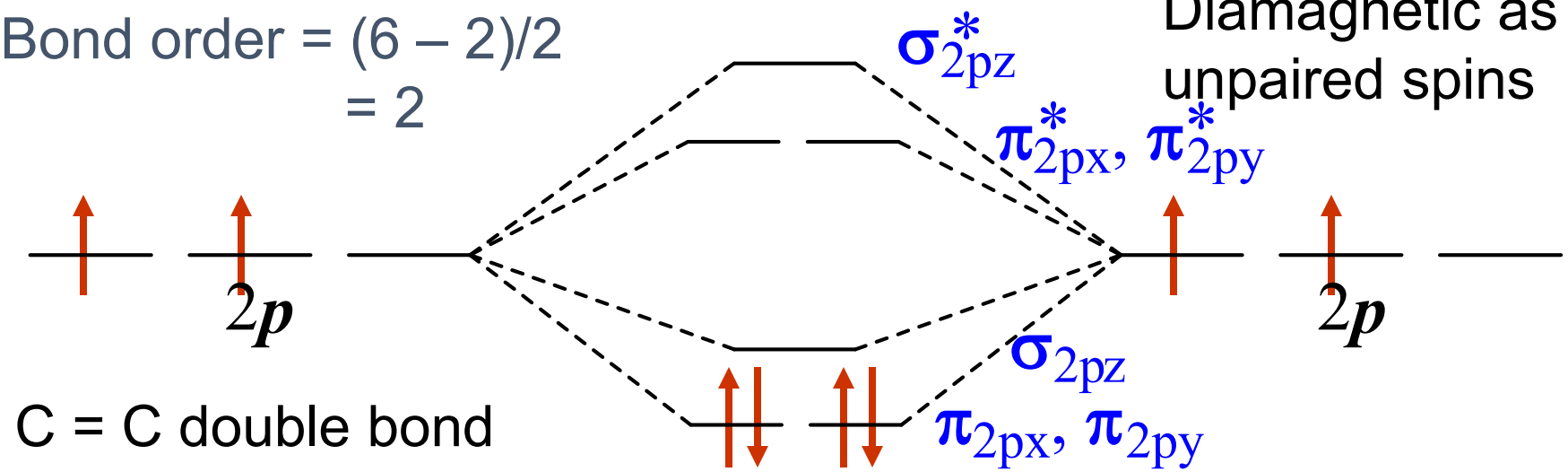
MO Energy Diagram for C_2

π_{2p} Lower in Energy than σ_{2p}

C electron configuration = [He]$2s^2 2p^2$

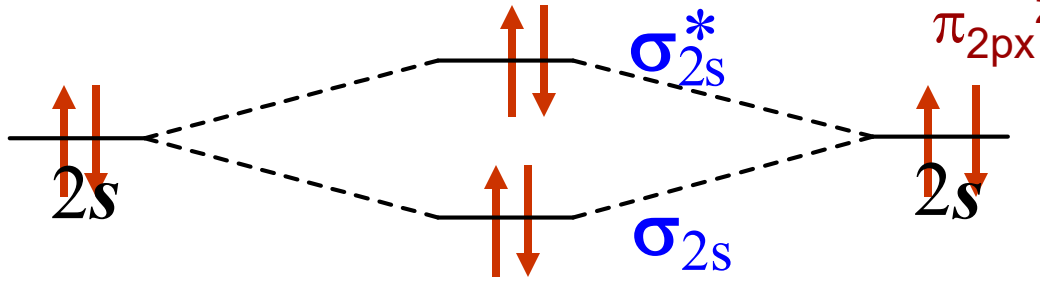
Bond order = (6 – 2)/2
= 2

Diamagnetic as no unpaired spins



C = C double bond
∴ stable molecule

MEC = $\sigma_{2s}^2 (\sigma^*_{2s})^2 \pi_{2px}^2 \pi_{2py}^2$



C C_2 C

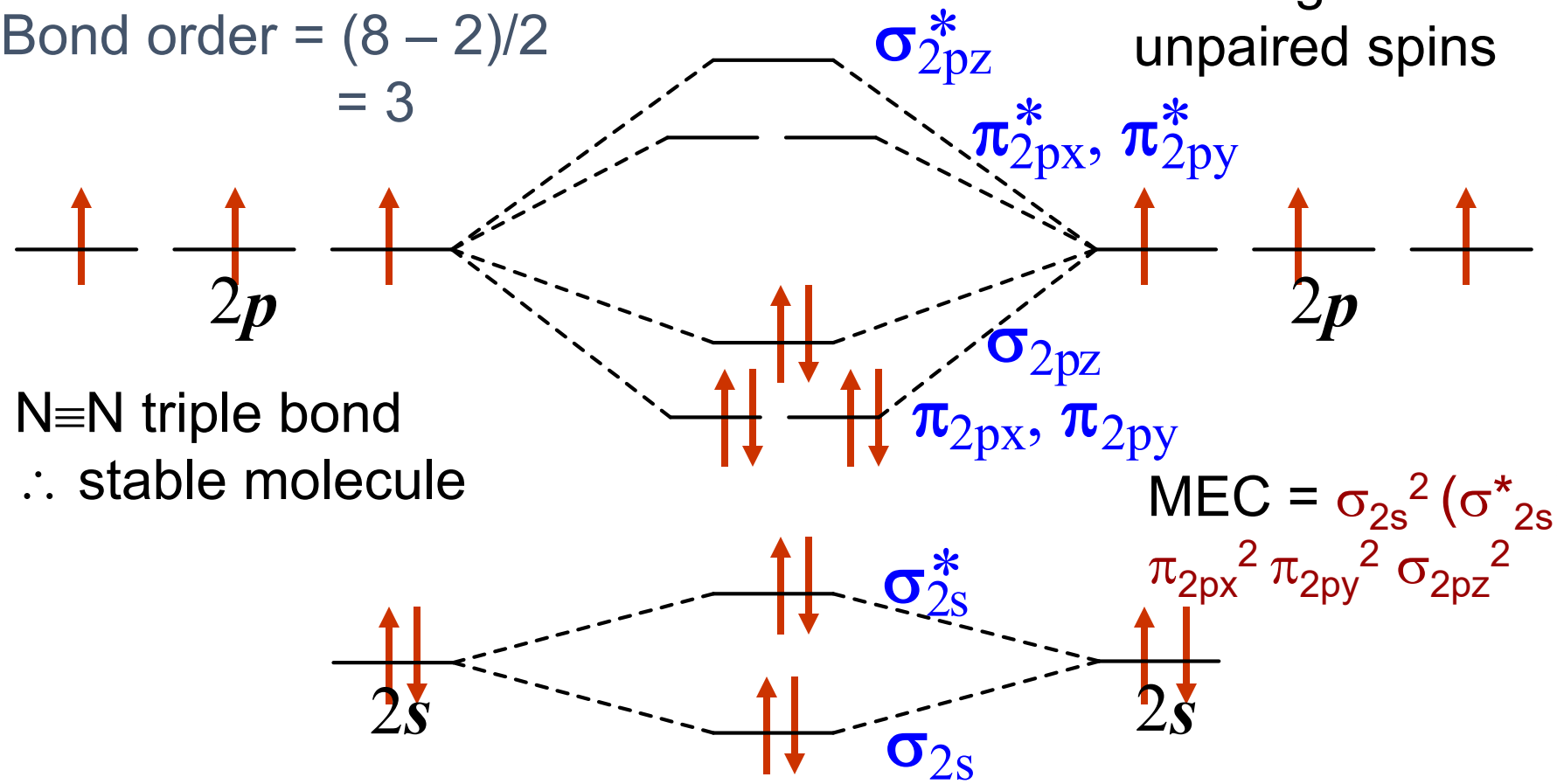
MO Energy Diagram for N_2

π_{2p} Lower in Energy than σ_{2p}

N electron configuration = [He]$2s^2 2p^3$

Bond order = (8 – 2)/2
= 3

Diamagnetic as no unpaired spins



N≡N triple bond
∴ stable molecule

MEC = $\sigma_{2s}^2 (\sigma^*_{2s})^2$
$\pi_{2px}^2 \pi_{2py}^2 \sigma_{2pz}^2$

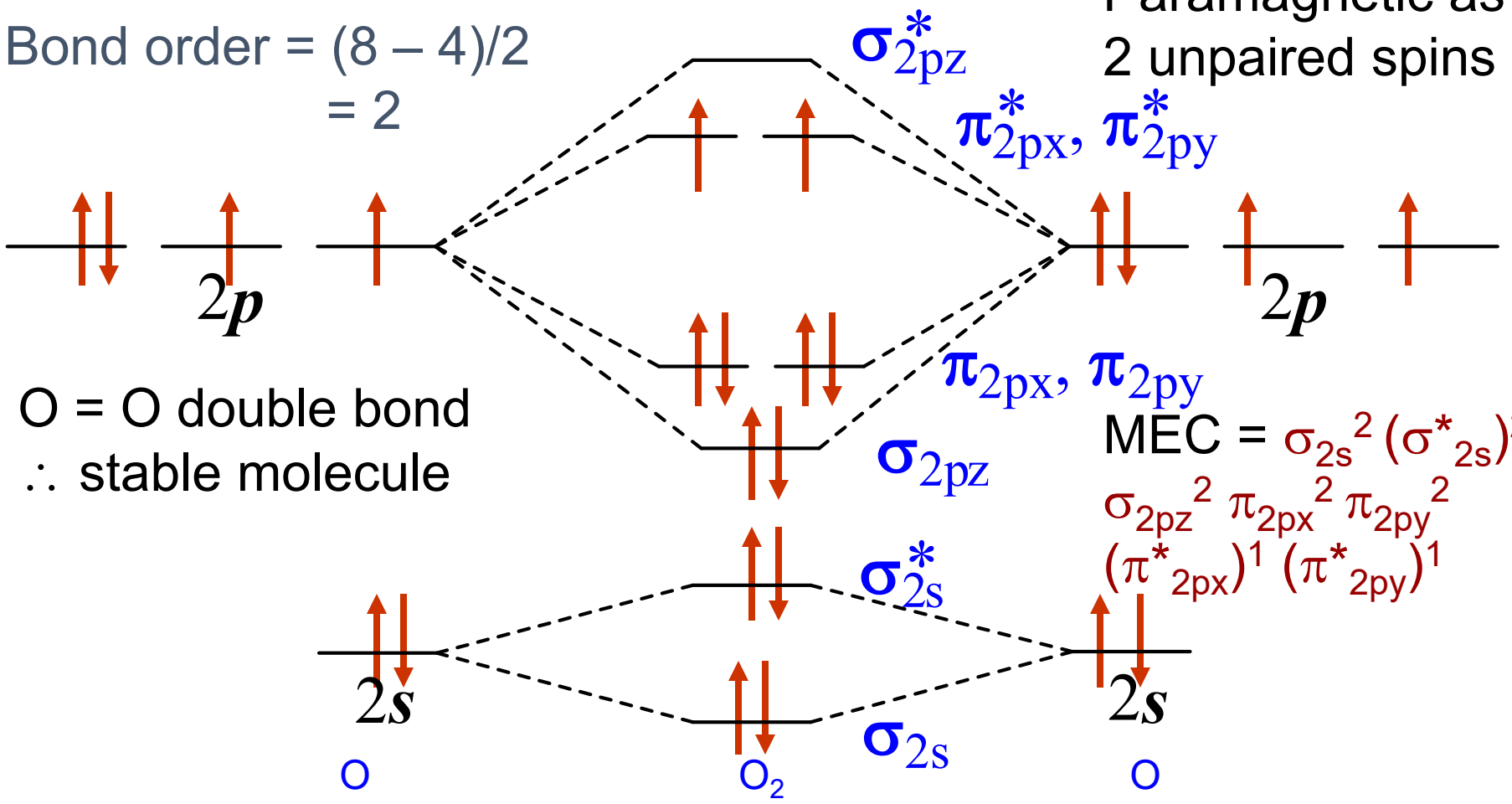
MO Energy Diagram for O_2

σ_{2p} Lower in Energy than π_{2p}

O electron configuration = [He]$2s^2 2p^4$

Bond order = (8 – 4)/2
= 2

Paramagnetic as
2 unpaired spins



O = O double bond
∴ stable molecule

MEC = $\sigma_{2s}^2 (\sigma^*_{2s})^2$
$\sigma_{2pz}^2 \pi_{2px}^2 \pi_{2py}^2$
$(\pi^*_{2px})^1 (\pi^*_{2py})^1$

Lewis Structure Can't Tell us this!!

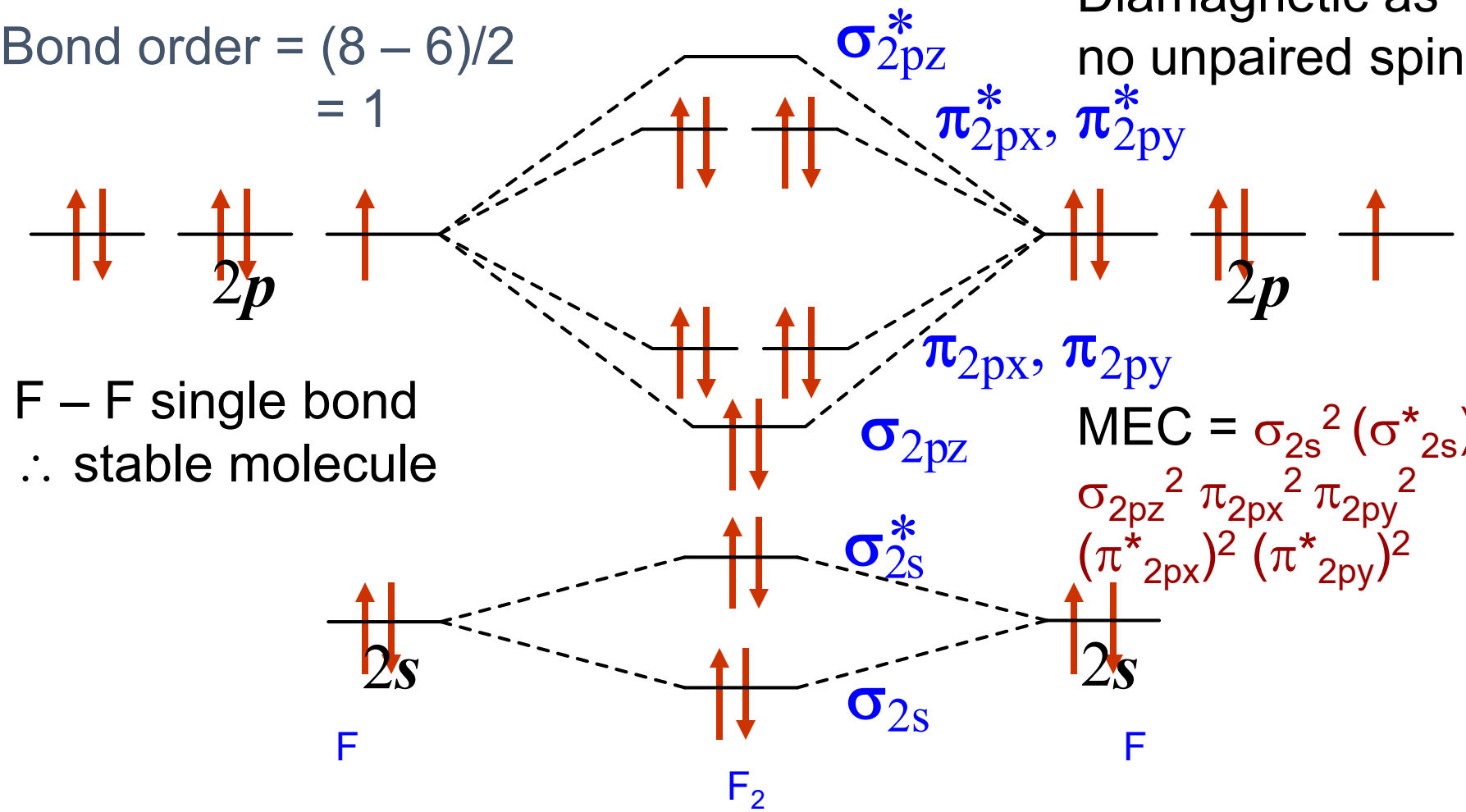
MO Energy Diagram for F_2

σ_{2p} Lower in Energy than π_{2p}

F electron configuration = [He]$2s^2 2p^5$

Bond order = (8 – 6)/2
= 1

Diamagnetic as
no unpaired spins



F – F single bond
∴ stable molecule

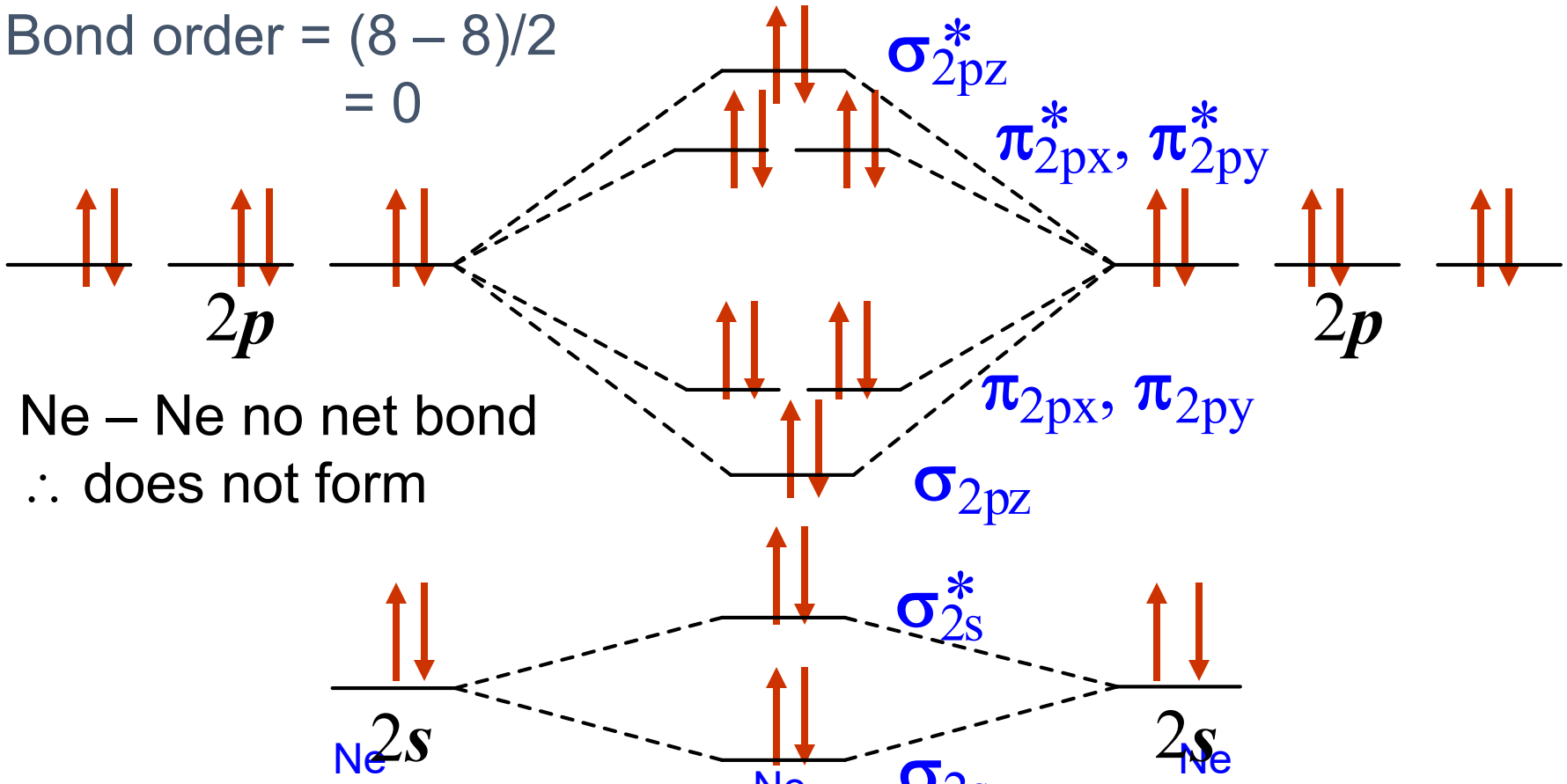
MEC = $\sigma_{2s}^2 (\sigma^*_{2s})^2$
$\sigma_{2pz}^2 \pi_{2px}^2 \pi_{2py}^2$
$(\pi^*_{2px})^2 (\pi^*_{2py})^2$

MO Energy Diagram for Ne_2

σ_{2p} Lower in Energy than π_{2p}

Ne electron configuration = [He]$2s^2 2p^6$

Bond order = (8 – 8)/2
= 0



Ne – Ne no net bond
∴ does not form

MEC = $\sigma_{2s}^2 (\sigma^*_{2s})^2 \sigma_{2pz}^2 \pi_{2px}^2 \pi_{2py}^2 (\pi^*_{2px})^2 (\pi^*_{2py})^2 (\sigma^*_{2pz})^2$

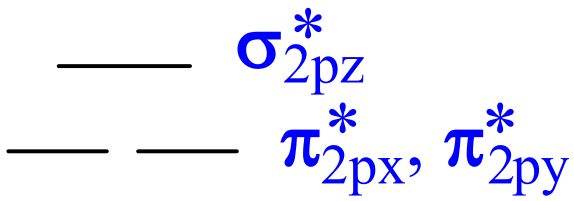
What about Heteronuclear Diatomic Molecules ?

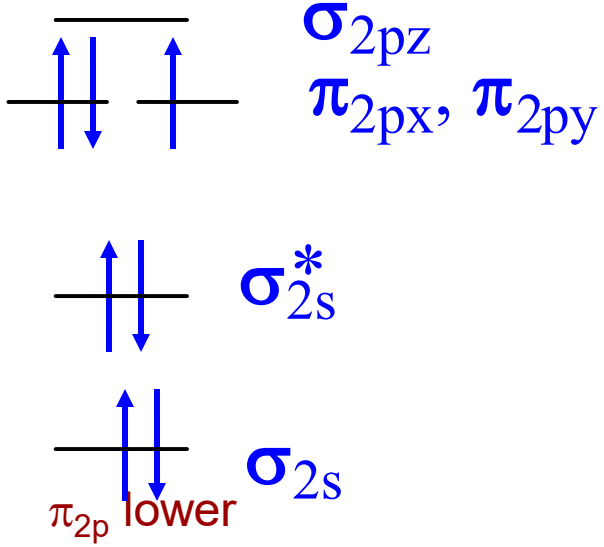
- If **Li** through N π_{2p} below σ_{2p}
- If **O**, **F** and higher atomic number, then σ_{2p} below π_{2p}

Ex.

- **BC** both are to left of **N**
  - so π_{2p} below σ_{2p}
- **OF** both are to right of **N**
  - so σ_{2p} below π_{2p}
- What about **NF**?
  - Each one away from **O** so average is **O** and σ_{2p} below π_{2p}

What is Bond order of NF and BC ?

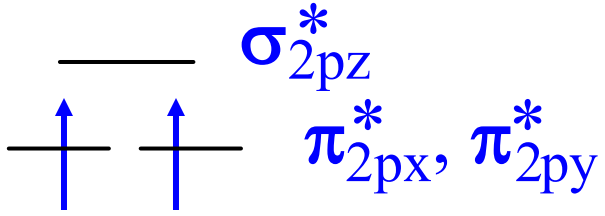


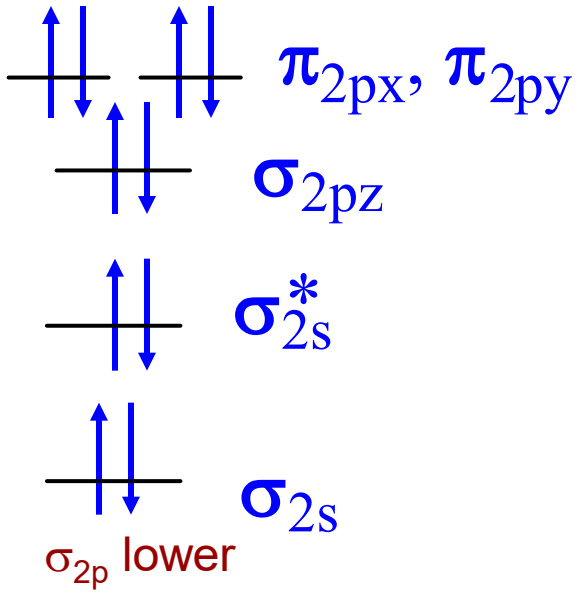


BC

Number of valence e^- = 3 + 4 = 7

Bond Order = (5 – 2)/2 = 1.5





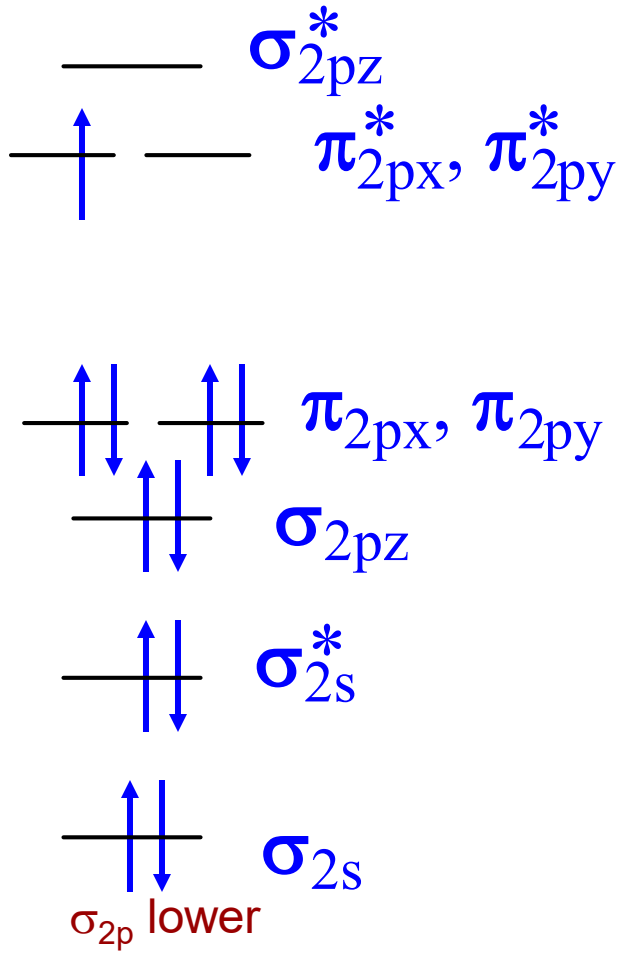
NF

Number of valence e^- = 5 + 7 = 12

Bond Order = (8 – 4)/2 = 2

What is Bond Order of NO ?

- Tricky
- **N** predicts π_{2p} lower
- **O** predicts σ_{2p} lower
- Have to look at experiment
- Shows that σ_{2p} is lower

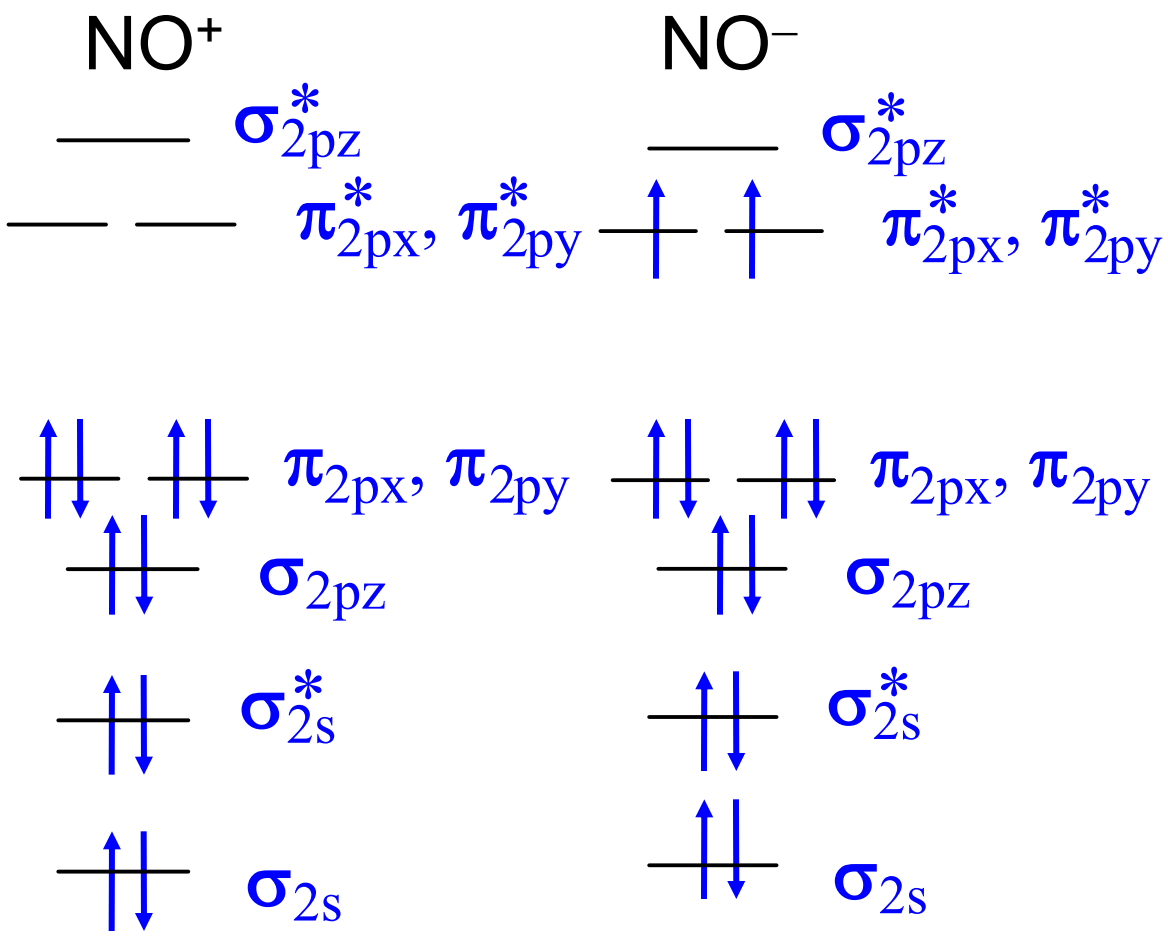


Number of valence e^- = 5 + 6 = 11

Bond Order = (8 – 3)/2 = 2.5

What is Bond Order of NO^+ and NO^- ?

- Same diagram
- Different number of **e⁻**
- **NO^+** has
  - **11 – 1 = 10 valence e⁻**
  - **Bond order = (8 – 2)/2 = 3**
- **NO^-** has
  - **11 + 1 = 12 valence e⁻**
  - **Bond order = (8 – 4)/2 = 2**



Compare Relative Stability of NO, NO^+ and NO^-

- Recall that as **bond order** ↑, **bond length** ↓, and **bond energy** ↑

Molecule or ion	Bond Order	Bond Length (pm)	Bond Energy (kJ/mol)
NO^+	3	106	1025
NO	2.5	115	630
NO^-	2	130	400

- So NO^+ is most stable form
  - Highest bond order, shortest and strongest bond

Your Turn!

What is the bond order for each species, and how many unpaired electrons are there in

O_2, O_2^+, and O_2^- ?

A. 2½, 2, 1½ ; 2, 1, 1

B. 2, 1, 1½ ; 1, 2, 1

C. 2, 1½ , 1½ ; 2, 1, 1

D. 2, 1½ , 2½ ; 1, 2, 1

E. 2, 2½ , 1½ ; 2, 1, 1

VBT vs MOT

- Neither VBT or MOT is entirely correct
  - Neither explains all aspects of bonding
  - Each has its strengths and weaknesses
- MOT correctly predicts unpaired e^- in O_2 while VBT does not
- MOT handles easily things that VBT has trouble with
- MOT is a bit difficult because even simple molecules require extensive calculations
- MOT describes “resonance” very efficiently

Successes of MO Theory

- MO theory is particularly successful in explaining electronic structure of O_2 and B_2
- Predicts paramagnetism as 2 electrons
  - 1 each in π_{2px} and π_{2py} (for B) or
  - 1 each in π^*_{2px} and π^*_{2py} (for O)
- Also in explaining why noble gases don't react

Your Turn!

Which of the following species is paramagnetic?

A. N_2

B. F

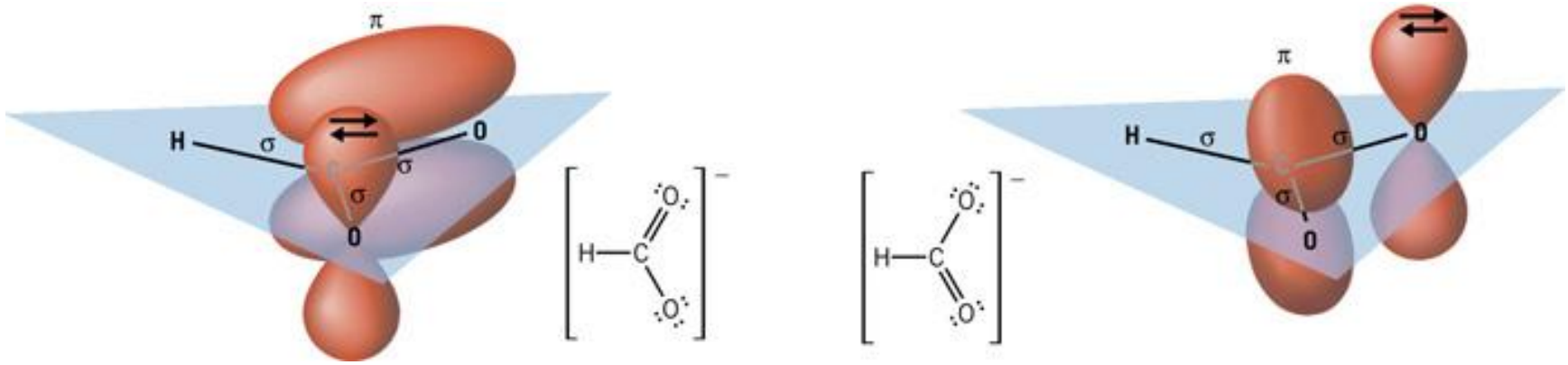
C. O_2^{2-}

D. N_2^{2+}

E. O_2^{2+}

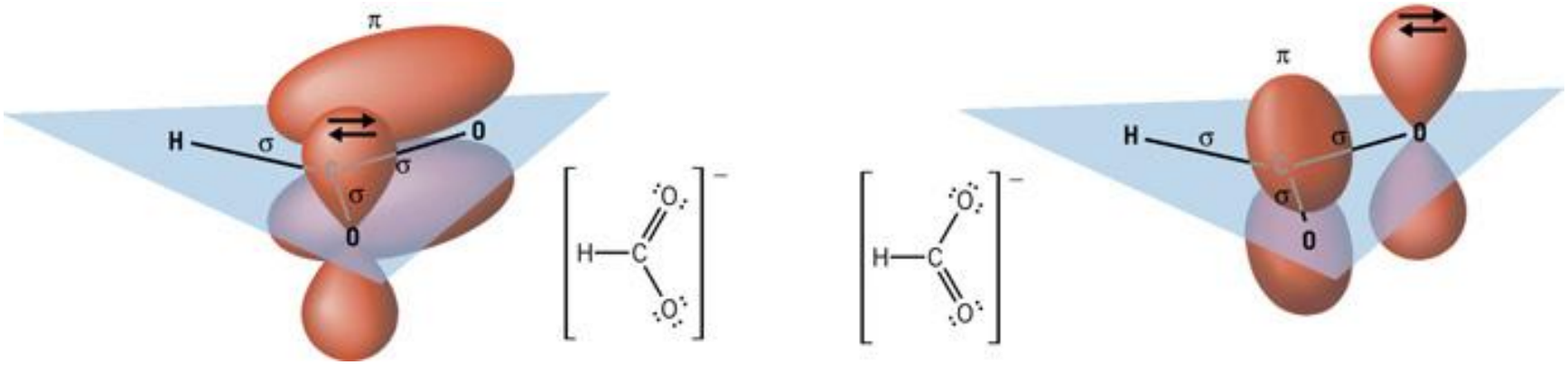
How Does MO Theory Deal with Resonance Structures ?

- Formate anion, $HCOO^-$
- C has 3 electron domains (all bonding pairs) so
  - ***sp²*** hybridized; trigonal planar
- Each O has 3 electron domains (1 bonding pair and 2 lone pairs)
  - so ***sp²*** hybridized; trigonal planar



Resonance Structures of Formate Anion, $HCOO^-$?

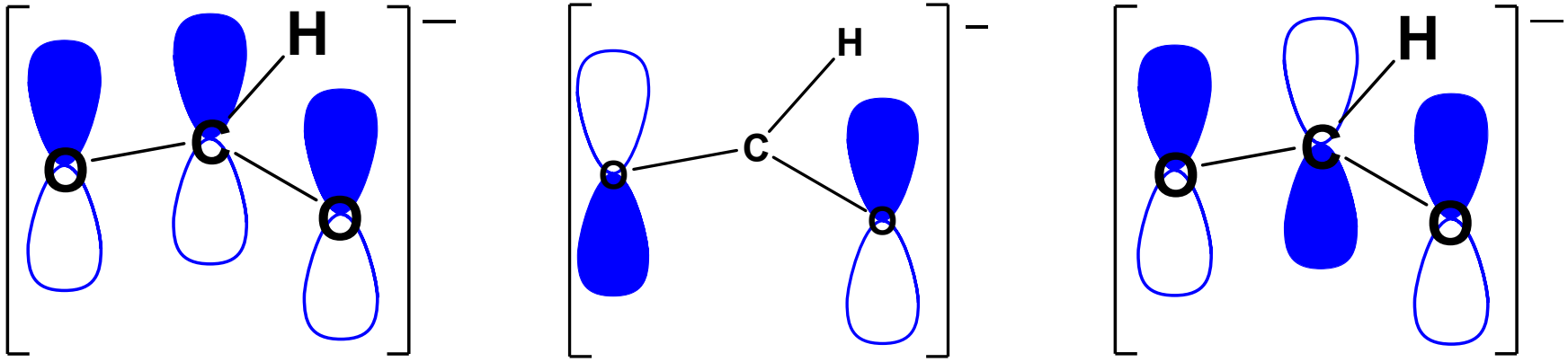
- Have 2 resonance structures
- Have lone pair on each O atom in unhybridized ***p*** orbitals as well as empty ***p*** orbital on C
- Lewis theory says
  - Lone pair on 1 O
  - Use lone pair of other O to form π (pi) bond
  - Must have 2 Lewis structures



Delocalized Molecular Orbitals

MO Theory

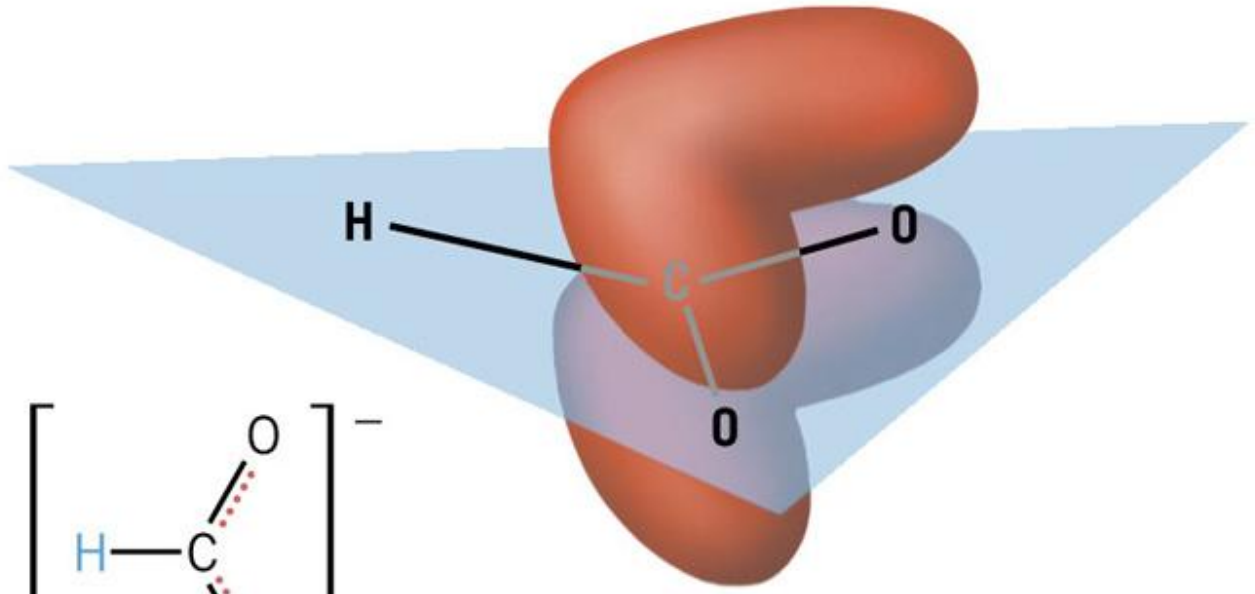
- Since have 3 unhybridized ***p*** orbitals
- Can form 3 π molecular orbitals
  - 1 bonding, 1 nonbonding, and 1 antibonding
  - Have 4 electrons that go into these MOs
  - 2 **e**⁻ go into π bonding MO *and*
  - 2 **e**⁻ go into π nonbonding MO

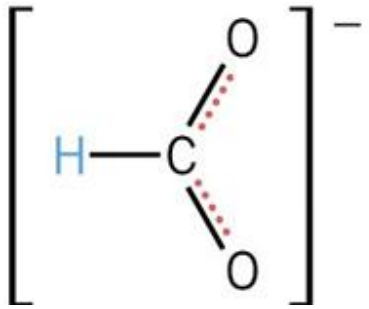


Bonding π MO Nonbonding π MO Antibonding π MO

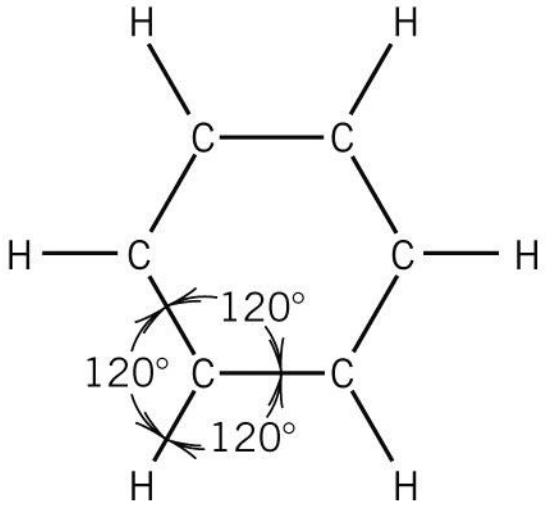
Delocalized Molecular Orbitals

- Bonding MO delocalized over all 3 atoms
- Gives our resonance hybrid picture, which is truer representation of what actually occurs

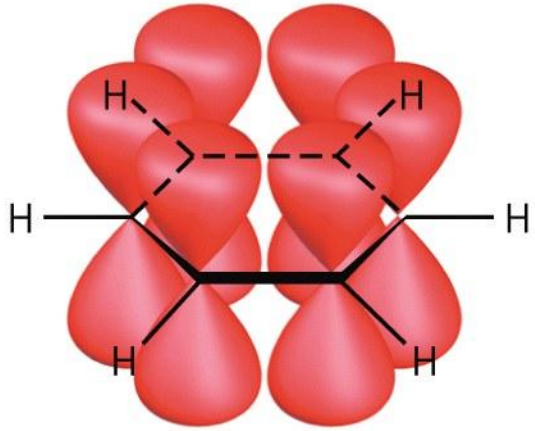




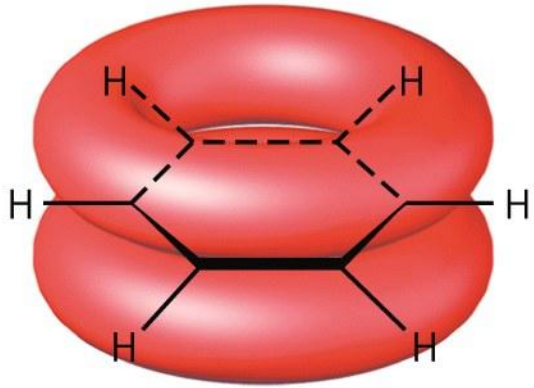
Benzene, in MO terms



σ-bond network in benzene (*a*)



p orbitals ready to overlap (*b*)

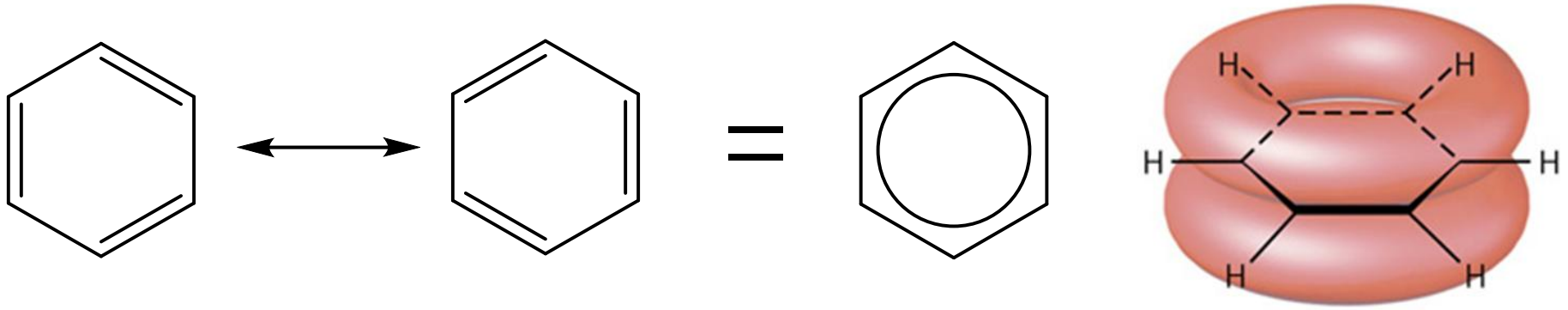


Delocalized π electrons (*c*)

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- 6 C atoms, each ***sp*²** hybridized (3 **σ** bonds)
- Each C also have 1 unhybridized ***p*** orbital (6 total)
- So 6 **π** MOs, 3 bonding and 3 antibonding
- So 3 **π** bonds

Benzene, in MO terms



- Can write benzene as 2 resonance structures
- But actual structure is composite of these two
- Electrons are **delocalized**
- Have 3 pairs of electrons delocalized over 6 C atoms
- Extra stability is **delocalization energy**
- Functionally, resonance and delocalization energy are "same"

Your Turn!

Which of the following species exhibits resonance ?

A. H_2O

B. SO_2

C. CH_2CH_2

D. CH_2CHCH_3

E. C_6H_{12} cyclohexane